

Janne's Guide to

Russian Ground Equipment

Field reference for recognition and unit association

54 systems: fires, air defence, radars, electronic warfare, command and signals, and engineers. What each looks like, which units hold it, how far it reaches.

F

FIRES

D

AIR DEFENCE

R

RADARS

E

ELECTRONIC
WARFARE

G

ENGINEERS

L

SUPPORT
VEHICLES

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About this guide

Status of this proof. This proof holds 54 entries, all checked claim by claim against their sources. Figures may still be wrong. Pictures are placeholders. Do not use this proof operationally.

How to read an entry. The black box at top right is the mobility code. *Operated by* lists unit types that hold the equipment. Each row ends with a tag of three parts, such as `[D 2016 H]`: basis, date and confidence. Basis is D for doctrinal tables of organisation, O for observed with an identified unit, R for reported or claimed. Confidence is H, M or L. *Recognise* bullets follow a fixed order and start with the first letter of the feature. Unknown values are printed INA.

Caution. Since 2022, army and district assets are often parcelled out to groupings of forces, new formations are under strength, and legacy or civilian vehicles appear in any unit. Equipment indicates a unit type. It does not prove it.

TR	Tracked, cross-country	TR-A	Tracked, swims without preparation
WH-AT	Wheeled all-terrain	WH-AT-A	Wheeled all-terrain, swims
WH-LT	Wheeled, limited off-road	RD	Roadbound, hard surface only
TW	Towed	MP	Man-portable
AIR-RW	Helicopter	AIR-FW	Fixed-wing aircraft

BM-21 Grad 122 mm MRL 5M-21 «Град»

NATO: M1964

WH-AT

Truck-mounted 40-tube 122 mm area-fire rocket launcher (divisional and brigade MRL) In service 1963

OPERATED BY motor rifle brigade, MRL battalion; 6 launchers per battery, 18 per battalion [\[D 2016 M\]](#)

INDICATES A group of about 18 launchers points to a brigade MRL battalion; 6 launchers to one battery. A few launchers inside a battalion tactical group suggest an attached MRL element.

RECOGNISE Cab Unarmoured Ural 6x6 truck cab (Ural-375D; Ural-4320 on BM-21-1). The 40-tube pack can be turned away from the cab.

Body Rear pack of 40 tubes in four rows of ten, each tube 3 m long, on a welded cradle; elevates 0 to +55 degrees. Traverse 102 left, 70 right of the vehicle.

Axles Three axles, 6x6 truck chassis, wheeled; not amphibious.

NOT TO BE CONFUSED WITH BM-21V Grad-V (9P125) 12 tubes on a GAZ-66B 4x4 truck, airborne version
9P138 Grad-1 36 tubes on a ZIL-131 6x6 truck

MOBILITY Unarmoured truck chassis on wheels with central tyre inflation. Fords 1.5 m without preparation. Not amphibious.

Weight, combat 13.7 t BM-21 on Ural-375D, loaded with crew
L × W × H 7.35 × 2.4 × 3.09 m travelling
Road speed 75 km/h maximum, hard road, loaded
Road range 750 km

Range, 9M22U HE-frag 20.4 km maximum, 9M22U (M-210F) HE-fragmentation rocket

Minimum range, 9M22U HE-frag 5 km 9M22U (M-210F) HE-fragmentation rocket

Range, 9M28F HE-frag 15 km maximum, 9M28F HE-fragmentation rocket

Minimum range, 9M28F HE-frag 1.5 km minimum effective, 9M28F HE-fragmentation rocket

SEEN WITH Resupply truck 9T254 (60 rockets); IV110 Bereza battery command vehicle (GAZ-66 chassis); Ural-375D / Ural-4320 truck (1 per launcher)

SOURCES ATP CNA IISS25 MISS6 RUSIPL RWW WEG13 WEG2015 WP-EN13 WP-RU15

Tornado-G 9K51M «Торнадо-Г»

WH-AT

122 mm 40-tube truck MRL; automated Grad upgrade for brigade and division artillery In service 2012

OPERATED BY motor rifle division, artillery regiment, MRL battalion (replacing Grad battalion) [\[D 2019 H\]](#)

tank division, artillery regiment, MRL battalion [\[D 2019 H\]](#)

motor rifle brigade, brigade artillery, MRL battalion, INA for Tornado-G; it replaces the BM-21, 18 per brigade MRL battalion (RWW p.100); 18 delivered to one brigade 2012 (RBASE) [\[D 2019 M\]](#)

motor rifle brigade, 18 [\[O 2012-04 M\]](#) seen with 20th Guards Separate Motor Rifle Brigade

INDICATES Tornado-G replaces the Grad in the MRL battalion of motor rifle and tank brigades and of division artillery regiments. A group of them suggests a manoeuvre brigade or division artillery group nearby.

RECOGNISE Cab Ural-4320 6x6 truck; cab seats driver and two. Launcher is laid and returned to travel position from inside the cab.

Body Single pack of 40 122 mm tubes, 10 × 4, on the rear body; traverses 102 deg left and 70 deg right, elevates 0 to 55 deg.

Axles Three axles, all driven (6x6); Ural-4320 has central tyre pressure regulation.

Details Rebuilt BM-21 Grad on Ural-4320 (about 20 rebuilt in 2019); additions are GLONASS navigation, a ballistic computer and a video monitor in the cab.

NOT TO BE CONFUSED WITH BM-21-1 Grad (Ural-4320) p. 4
Same truck and 40-tube pack. Tornado-G lays from the cab with no crew at the launcher; no reliable external mark found in text sources.
9A51 Prima 50 tubes instead of 40 on the same Ural-4320 chassis.
9K515 Tornado-S p. 4 12 tubes of 300 mm on an 8x8 MAZ-543M chassis, against 40 small tubes on a 6x6 Ural.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY 6x6 truck; WEG claims small-arms protection, WEG 2015 BM-21 page gives none. Fords about 1.5 to 1.75 m. Can fire from an unsurveyed position and leave straight after firing.

Weight, combat 14 t
L × W × H 7.35 × 2.5 × 3.1 m travelling
Road speed 85 km/h maximum
Road range 650 km cruising range

Range, 9M521 HE-frag 40 km maximum, 9M521 extended-range HE-frag rocket; attribution to 9M521 from RUWIKI (Karpenko), WEG, ZR and WW1 give 40 km as system maximum

Range, 9M522 HE-frag 37.5 km maximum, 9M522 rocket, separable HE-frag warhead

Range, 9M538 HE-frag 20 km maximum, 9M538 rocket, enhanced monobloc HE-frag warhead, claim

Range, 9M539 HE-frag 20 km maximum, 9M539 rocket, separable HE-frag warhead; Tecmash statement at IDEX 2019, claim

SOURCES CNA D24 EDR GS16 IISS25 JF18 MAILRU2 ORYX RBASE RDP5 RWW TASS12 WEG41 WEG2015 WMC16 WP-RU58 WWW13 ZR

BM-27 Uragan 9П140 «Ураган»

NATO: M1977

WH-AT

220 mm 16-tube MRL on ZIL-135LM 8x8 for area fire and remote minelaying at 10–35 km In service 1975

OPERATED BY rocket artillery regiment, three rocket launcher battalions, 54 per regiment (3 battalions of 18) [\[D 1991 M\]](#) rocket artillery brigade, rocket launcher battalions, 54 per brigade (3 battalions of 18) [\[D 1991 M\]](#) self-propelled artillery regiment, Uragan rocket artillery battalion, 8 per battalion [\[R 2017-12 M\]](#)

INDICATES A 9P140 points to army-level rocket or artillery brigade fires, or to a Uragan battalion (8 launchers) in a division artillery regiment; the MoD also planned such battalions for combined arms brigades. In the Airborne Forces it indicates the artillery brigade.

RECOGNISE Cab ZIL-135LM cab with a blast shield that is raised during firing.

Body 16 tubes in three banks at the rear: two lower rows of six, upper row of four. In travel the pack faces rearward; muzzles covered by a sheet-metal cover when loaded.

Axles Four axles, 8x8; only the front and rear axles steer. Two petrol engines, each driving the wheels on one side.

Details Two manually emplaced hydraulic jacks at the rear of the chassis are lowered for firing. Newer launchers sit on a 6x6 BAZ-69092 chassis.

NOT TO BE CONFUSED WITH 9T452 transloader Same ZIL-135 chassis but a flatbed with 16 rockets in two stacks either side and a loading device between them; no tube pack.

BM-21 Grad p. 4 40 tubes on a Ural-375 truck, against 16 larger tubes on an 8x8 ZIL-135.

9A52 Smerch 12 tubes of 300 mm on a MAZ-543A chassis, against 16 tubes on a ZIL-135.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Unarmoured 8x8 petrol truck; rear jacks must be lowered to fire; fords 1.2 m unprepared; 60% gradient.

Weight, combat	20 t
L × W × H	9.63 × 2.8 × 3.23 m travel position
Road speed	65 km/h maximum
Road range	500 km cruising

Range, 9M27F HE-frag	35 km maximum, 9M27F HE-frag rocket
Minimum range, 9M27F HE-frag	10 km 9M27F HE-frag rocket

Range, 9M27K cluster	35 km maximum, 9M27K rocket, 30 9N210 HE-frag submunitions
Minimum range, 9M27K cluster	10 km 9M27K rocket, 30 9N210 HE-frag submunitions

Range, 9M27K2 AT mines	35 km maximum, 9M27K2 rocket, 24 PTM-1 anti-tank mines
Minimum range, 9M27K2 AT mines	10 km 9M27K2 rocket, 24 PTM-1 anti-tank mines

SEEN WITH 9T452 transloader; ZIL-135LM (1)

VARIANTS 9K57-1 Uragan-1 Launcher on 6x6 BAZ-69092 chassis; some seen with net screens over cab and tubes.

SOURCES CNA D242 FAS4 GS19 IISS25 IZ17 MIL31 MIL32 MIL110 MFER ORYX RUSIPL RWW W1R WEG46 WEG2015 WP-EN49 WP-EN50 WP-RU62

Uragan-1M 9K512 «Ураган-1М»

WH-AT

Heavy 220/300 mm MRL with swappable rocket pods; successor to 9K57 Uragan In service 2007

OPERATED BY self-propelled artillery regiment, rocket artillery battalion (planned replacement for 9K57 Uragan, MoD statement; no holding confirmed) [\[R 2017-12 L\]](#)

artillery brigade, MRL battalion (Uragan 'being succeeded' by Uragan-M1; no holding confirmed) [\[R 2019-10 L\]](#)

INDICATES Planned for divisional artillery regiments and army artillery brigades as the Uragan successor. Few were built and front-line use is unconfirmed, so a sighting is notable and should be checked against the 6x6 9K57-1.

RECOGNISE Cab Forward-control cab on the MZKT-7930 8x8 chassis, seating the driver and two passengers.

Body Launcher on a turntable at the rear carrying two launcher pods instead of a fixed tube rack.

Axles Four axles (8x8), all driven; same chassis family as the 9T249 transporter-loader.

Details 300 mm pods hold 6 tubes each in two layers of three; 220 mm pods hold 15 tubes each. Pods can be mixed, one of each calibre.

NOT TO BE CONFUSED WITH 9K57-1 Uragan-1 p. 5 6x6 BAZ-69092 chassis with armoured cab, three axles not four; Russian press also calls it Uragan-1M.

9P140 Uragan p. 5 Fixed rack of 16 tubes on a ZIL-135 8x8; no detachable pods.

9A52-2 Smerch p. 6 Fixed 12-tube block in 3 rows of 4 on a MAZ-543M 8x8; Uragan-1M has two separate 6-tube pods.

9A52-4 Tornado p. 6 KamAZ-63501 8x8 truck with a replaceable 6-rocket module (WEG 2015); Uragan-1M carries two pods on the larger MZKT-7930.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Heavy 8x8 truck; fords 1.4 m, climbs 45% gradient, crosses 2.0 m trench.

Weight, combat	40 t
L × W × H	13.0 × 3.0 × 3.29 m travelling
Road speed	70 km/h maximum, maximum, road
Road range	1,000 km maximum, road

Range, 9M27F HE-frag 220 mm	35 km maximum effective, 9M27F unguided HE-frag rocket
Minimum range, 9M27F HE-frag 220 mm	10 km minimum effective, 9M27F unguided HE-frag rocket
Range, 9M27K cluster 220 mm	35 km maximum effective, 9M27K rocket, 30 9N210 fragmentation submunitions
Range, 9M55F HE-frag 300 mm	70 km maximum, 9M55F HE-frag rocket with trajectory correction
Minimum range, 9M55F HE-frag 300 mm	20 km 9M55F HE-frag rocket

SEEN WITH 9T249

SOURCES AR22 CNA D24A D24B IISS25 IZ17 ORYX RAND RWW U244 W1A W1C W1E WEG45 WEG2015 WP-EN48 WP-RU61 ZR

BM-30 Smerch 9A52-2 «Смерч»

NATO: 280mm M1983

WH-AT

Heavy 300 mm 12-tube rocket launcher for deep area fire; Tornado-S adds guided rockets In service 1987

OPERATED BY rocket artillery brigade, three groups of 4 x 9A52-2 in ATP fig. 5-16; whether each group is a battalion or a battery is not readable from the text, 12 x 9A52-2 shown in ATP fig. 5-16 (3 groups of 4) **D 2024 M**

rocket artillery brigade, 7 delivered April 2017, 5 more announced

O 2017-04 M seen with rocket artillery brigade, Tver Oblast, Western MD (named as the 79th Guards Rocket Artillery Brigade by Wikipedia only)

rocket artillery brigade **O 2026-09 M** seen with 79th Guards Rocket Artillery Brigade (Zapad group), Tornado-S, 2026-09, 232nd Guards Rocket Artillery Brigade (Tsentri group), Tornado-S, 2026-08 and 2026-09

INDICATES *Smerch or Tornado-S points to an army-level rocket artillery brigade, not a manoeuvre brigade, whose MRL battalion holds BM-21. Expect deep fires out to 70-120 km and ammunition resupply by MAZ-543 transloaders.*

RECOGNISE **Cab** MAZ-543M 8x8 truck; driver cab front left, commander behind him; launch and fire-control cabin to the rear of the driver cab, one door each side, two small front windows.

Body Single package of 12 long 300 mm tubes over the rear deck, 3 rows of 4 (two banks of four with four more on top); traverses 30 deg each side, elevates to 55 deg.

Axles Four axles, 8x8; hydraulic jacks between the wheels of the 3rd and 4th axles are lowered to steady the rear of the launcher when firing.

Details Tornado-S (9A54): satellite navigation antenna on the roof of the fire control cab; otherwise laid out like the 9A52-2.

NOT TO BE CONFUSED WITH **BM-27 Uragan (9P140)** p. 5 16 tubes of 220 mm on a ZIL-135 chassis; Smerch has 12 larger tubes on the MAZ-543.

9A53 Uragan-1M p. 5 Two quick-change pods of six tubes on an MZKT-7930 8x8; Smerch has one fixed 12-tube package.

9A52-4 Tornado KamAZ-63501 8x8 truck with a single six-tube pod; Smerch has one fixed 12-tube package on the MAZ-543M.

9T234-2 and 9T255 transloaders Same MAZ-543A family chassis; carry 12 rockets and a crane (850 kg on the 9T255) with no launch tube package.

Vilkha (Ukraine) Ukrainian development that uses the same Smerch launcher vehicle; not separable by shape alone.

SEEN WITH MAZ-543M 8x8 chassis (1 per launcher); 9T234-2 transloader (TZM) (INA; carries 12 rockets); 9T255 transloader (Tornado-S) (INA; carries 12 rockets); 1K123 Vivary automated fire control system (up to 3 battalion and 18 battery vehicles per brigade system); 9S936 command and staff vehicle (Tornado-S); 1V168 ASUNO-S fire control; 1T12-2M survey vehicle; 1B44 meteorological complex; Zoopark counter-battery radar (INA; reported cueing Smerch fire 2024-03)

VARIANTS **9A54 Tornado-S** Satellite antenna on the fire control cab roof; same MAZ-543M chassis and 12-tube package. GLONASS, automated laying and fire control; fires 9M544 and 9M549 guided rockets to 120 km; crew 3. **9A52-4 Tornado** KamAZ-63501 8x8 truck with one six-tube 300 mm pod. Half the load; pod replaced after firing; lighter and more mobile. ISS25 lists none in Russian service.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Heavy 43.7 t 8x8; fords 1.1 m unprepared; not amphibious; rear jacks must be lowered to fire.

Weight, combat, 9A52-2 **43.7 t** loaded
Weight, empty **33.7 t** 9A52-2 (WEG) and 9A54 (MISSTS)
L × W × H **12.1 × 3.05 × 3.05 m** travelling
Road speed **60 km/h** maximum
850 km 9A52-2; also 850 km on the WEG
Tornado-S page

Fording **1.1 m** unprepared
Time into action **3 min** from march to ready to fire
Time out of action **3 min**
Ground pressure **INA**

Range, 9M55K cluster	70 km maximum, unguided, range-corrected rocket with fragmentation submunitions
Minimum range, 9M55K cluster	20 km unguided, range-corrected rocket with fragmentation submunitions
Range, 9M55F HE-frag	70 km maximum, unguided, separable HE-frag warhead
Minimum range, 9M55F HE-frag	20 km unguided, separable HE-frag warhead
Range, 9M528 HE-frag	90 km maximum, unguided extended-range rocket, separable HE-frag warhead, claim
Range, 9M544 and 9M549 guided	120 km maximum, Tornado-S guided cluster rockets, inertial and GLONASS, claim
Minimum range, 9M544 and 9M549 guided	20-40 km Tornado-S guided rockets; crew statement, claim
Employment range, Smerch	35-70 km practical, unguided rockets, Ukraine 2022, used at or near maximum range, observed
Employment range, Tornado-S	about 100 km practical, guided rockets against a command post (2026-08) and a fortified strongpoint (2026-09); MoD figures via state media, claim
Crew, 9A52-2	4
Crew, 9A54 Tornado-S	3
Tubes	12 × 300 mm 9A52-2 and 9A54
CEP, 9M55K cluster	150 m inertial range and attitude correction, no terminal guidance; also given for 9M528
CEP, 9M544 and 9M549 guided	5-10 m inertial and GLONASS; official claim, claim

TOS-1A Solntsepek TOC-1A «Солнцёк»

TR

Tracked 220 mm thermobaric and incendiary rocket launcher of the RKhBZ troops In service 1988

OPERATED BY RKhBZ brigade, flamethrower battalion of 3 batteries, each 6 BM-1 and 6 TZM-T; RWW places these battalions in NBC defence brigades and regiments and does not state the echelon, 18 per battalion (6 per battery) [D 2016 M]

RKhBZ brigade, TOS battalion of 3 companies of 6 launchers (ATP para 5-146), 18 per battalion [D 2024 M]

INDICATES TOS launchers belong to army or district RKhBZ units, not manoeuvre brigades. Seeing one suggests an attached flamethrower company or battalion supporting an attack or breach, since it must close to within about 6 km of the target.

RECOGNISE Hull T-72 tank hull with no gun turret; a large launcher box fills the hull roof. Paired with look-alike T-72-based TZM-T reload vehicles.

Armament 24 tubes of 220 mm in a heavy armoured box. Launchers shipped from Nov 2025 add a factory armour shell over the tubes that lifts open when the tubes elevate.

Turret No turret: the launcher box sits on a rotating platform and elevates to fire. Improvised anti-drone cover was fitted in the field before factory kits.

Suspension Standard T-72 tracked running gear with large-diameter road wheels. Road wheel and return roller counts not confirmed in cited sources.

NOT TO BE CONFUSED WITH TOS-1 Buratino 30 tubes, without the heavy armoured housing added on TOS-1A, which has 24 tubes.

TZM-T reload vehicle Same T-72 hull but carries two rocket racks with a crane between them instead of a trainable launcher.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Tracked T-72 chassis, not amphibious; fords 1.2 m; must close to within 6 km of the target to fire the longest-range thermobaric rocket.

Weight, combat	44.3 t
L × W × H	6.86 × 3.46 × 2.60 m
Road speed, maximum	60 km/h
Off-road speed, maximum	45 km/h

Range, thermobaric rocket	6,000 m maximum, MO.1.01.04M thermobaric rocket, 3.7 m, 217 kg
Range, incendiary rocket	6,000 m maximum effective, MO.1.01.04M.OP incendiary rocket, 3.7 m, 204 kg
Range, original short rocket	2,700 m maximum, MO.1.01.04 thermobaric rocket, 3.3 m, 173 kg, fired from TOS-1A
Range, minimum Launch tubes	600 m MO.1.01.04M rocket 24 × 220 mm

SEEN WITH TZM-T transporter-loader (T-72 chassis, crane, rocket racks) [2 per launcher (ATP, MISS); RWW gives 1 per launcher]

SOURCES AR20 AR21 ATP CNA EMP GS17 MIL28 MIL29 MIL30 MISS14 RWW TASS13 TASS14 TASS15 TASS16 WEG42 WMC17 WP-EN42 WP-EN44 WP-EN45

2S5 Giatsint-S 2C5 «Гиацинм-С»

NATO: M1981

TR

Army-level 152 mm self-propelled gun for long-range counter-battery and deep fire In service 1975

OPERATED BY artillery brigade, gun battalion (152 mm), 18 per battalion (24 in forward armies); 2S5 or 2A36 or M-46 [D 1991 M] unit not identified, Airborne Forces artillery; unit not named, 18 (IISS 2024 figures) [R 2024 M]

INDICATES In Soviet doctrine the 2S5 marked the gun battalion of an army artillery brigade. Russian MoD reporting (2026) also places it with motor rifle brigade artillery, so alone it no longer points to army-level artillery.

RECOGNISE Hull Low tracked hull, engine and driver at front; front dozer blade; large spade or baseplate at rear, lowered to the ground before firing.

Armament Very long 152 mm barrel with five-part multi-baffle muzzle brake, held by a travel lock; with gun forward, length 8.95 m against a hull of 8.33 m.

Turret No turret: gun on an open mount at the rear of the hull roof, crew exposed; 15 deg traverse each side. Commander's cupola carries a 7.62 mm PKT and a searchlight.

Suspension Six rubber-tyred double road wheels and four return rollers per side; drive sprocket front, idler rear; torsion bars, track shoes 484 mm wide.

NOT TO BE CONFUSED WITH 2S3 Akatsiya Large enclosed turret at hull rear, 360 deg traverse, cupola offset left; double-baffle muzzle brake and fume extractor. 2S5: open mount, no turret, multi-baffle brake.

2S4 Tyulpan p. 9 Same chassis family, but a 240 mm mortar travels folded with muzzle forward, not reaching past the first road wheel; no long barrel over the front.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Tracked; fords about 1 m; rear spade must be lowered to fire; gun traverse only 15 deg each side, so the hull is laid on the target line.

Weight, combat	28.2 t
L × W × H	8.33 × 3.25 × 2.76 m
Road speed	63 km/h maximum, road
Range	500 km maximum, road

Range, HE-frag 30F29	28.4 km maximum, HE-frag 30F29, full charge
Range, HE-frag 30F29 reduced charge	21.5 km maximum, HE-frag 30F29 with reduced charge 4Zh48
Range, HE-frag 30F59	30.5 km maximum, HE-frag 30F59, improved aerodynamic shape, full charge
Range, rocket-assisted HE-frag 30F30	33 km maximum, rocket-assisted HE-frag
Range, Krasnopol guided 30F39	20 km maximum, laser-guided HE-frag Krasnopol
Minimum range, HE-frag 30F29	8,600 m HE-frag, indirect fire

SEEN WITH 2A37 152 mm gun [1]

SOURCES BMPD2 FS2 FAS3 GS4 IISS25 IZ26 IZ233 JANESRG MF RIA22 RUSIPL RWW WEG3 WP-EN4 WP-RU7

2S19 Msta-S

2C19 «Mcma-C»

NATO: M1990 Farm

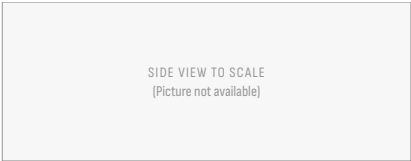
TR

152 mm tracked self-propelled howitzer of brigade, division and army artillery In service 1989

OPERATED BY motor rifle brigade, two self-propelled artillery battalions, each of three batteries of 6 howitzers; ATP notes 2S3 may be held instead, 18 per battalion (6 per battery), 36 per brigade; ATP: some battalions up to 24 systems [\[D 2024 M\]](#)
tank brigade, two howitzer (self-propelled artillery) battalions, three batteries of 6, 18 per battalion, 36 per brigade [\[D 2016 M\]](#)
INDICATES A battery of 6 or battalion of 18 Msta-S points to a self-propelled artillery battalion of a motor rifle or tank brigade, a division artillery regiment or an army artillery brigade. Wartime batteries may be smaller.

RECOGNISE Hull Tank-type tracked hull with T-72 hull geometry and T-80 running gear, built-in dozer blade, travel lock for the barrel. About 11.9 m long overall with the gun forward.
Armament Very long 152 mm L/47 barrel with double-baffle muzzle brake and a bore evacuator. 12.7 mm NSVT machine gun on the commander's cupola.
Turret Large turret over the hull centre; vehicle about 3 m high. Six 81 mm smoke launchers. 2S19M2: two stowage boxes on the right turret side.
Suspension T-80 type running gear: six road wheels and five return rollers per side, 580 mm tracks.
NOT TO BE CONFUSED WITH 2S3 Akatsiya (152 mm SP howitzer) Engine at front and turret toward the rear; much shorter L/27 barrel; four return rollers; about 28 t against 42 t.
2S5 Giatsint-S (152 mm SP gun) p. 7 152 mm 2A37 L/54 gun; most of the crew, all but the gunner, work outside the vehicle when firing; about 28 t.

SOURCES AG4 ARMYT ATP CNA GS9 IIS25 MIL17 ORYX RWW TASS4 TASS5 WEG20 WP-EN4 WP-EN7 WP-EN20 WP-EN21 WP-RU25



SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Heavy tracked vehicle, not amphibious. Fords 1.2 m unprepared; WEG gives 5 m with preparation. About 42 t, needs heavy bridging.	
Weight, combat	42 t 2S19 and 2S19M1
Weight, combat, 2S19M2	43.24 t 2S19M2
L × W × H	11.917 × 3.584 m overall; 3.38 m without skirts × 2.985 m
Road speed, maximum	60 km/h
Range, HE-frag	24.7 km maximum, 30F45 (OF-45) HE-frag, full charge
Range, HE-frag base bleed	29 km maximum, 30F61 HE-frag base bleed (round 3V0F91)
Range, rocket-assisted	36 km HE rocket-assisted projectile (HERA/RAP), type not named
Range, guided	20 km maximum, Krasnopol laser-guided projectile; laser designation by an observer required
Range, jammer round	22 km 3R830 radio jamming projectile; jamming radius 700 m, claim

2S35 Koalitsiya-SV

2C35 «Коалиция-СВ»

TR

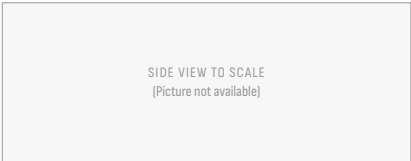
152 mm self-propelled howitzers: tracked autoloaded 2S35 and wheeled 8x8 2S43 In service 2020

OPERATED BY self-propelled artillery battalion, self-propelled howitzer battery, 6 per battery, 18 per battalion [\[D 2016 M\]](#)
self-propelled artillery regiment, 8 2S35 on troop trials [\[R 2020-05 M\]](#)
artillery brigade, self-propelled artillery battalions, INA; planned replacement of the brigade's 2S19 by 2S35, reported 2017, not carried out by late 2024 [\[R 2019 L\]](#)

INDICATES 2S35 or 2S43 suggests new-build guns above manoeuvre-brigade level: 2S35 was planned for army artillery brigades and trialled in a division SP artillery regiment; 2S43 is reported with the 273rd Artillery Brigade, 34th Artillery Division. Rare (IIS: 8, 20).
RECOGNISE Hull 2S35: T-90-type tracked hull with reshaped front; driver, commander and gunner sit side by side in the front, three hatches across the hull front; engine at rear.
Armament 2S35: very long 152 mm 2A88 barrel (52 calibres) with muzzle brake, held by a travel lock on the glacis; Kord 12.7 mm remote weapon station on turret roof.
Turret 2S35: large unmanned turret mid-hull; 902 smoke dischargers on both sides and roof; laser warning sensors at front and rear of roof. From 2021 two boxes on turret front.
Suspension 2S35: six road wheels each side, idler front, drive sprocket rear, torsion bars; upper track covered by a skirt. Return roller count INA.

NOT TO BE CONFUSED WITH 2S19 Msta-S / 2S19M2 Msta-SM2 p. 8 2S19 has a crew of five with gun crew in the turret and a shorter 47-calibre barrel; 2S35 has three crew hatches across the hull front.
2S44 Giatsint-K Same BAZ 8x8 chassis as 2S43 but carries the gun of the 2A36 Giatsint-B towed gun.

SOURCES AG2 AR1 ATP BMPD1 CNA IIS25 IZ23 IZMAL KZ1 KZ2 ORYX REDSTAR RG24 RGMAL RGMAL2 RGPAP RWW SHEP WEG1 WP-EN7 WP-EN8 WP-RU4 WP-RU5 WSN2



SIDE VIEW TO SCALE
(Picture not available)

MOBILITY 2S35 tracked, 48–55 t, not amphibious. 2S43 wheeled 32 t, for road and off-road use, fits I1-76; fords 1.4 m.	
Weight, 2S35	48–55 t combat
Weight, 2S43	32 t combat
L × W × H	9.63 × 3.78 × 2.2 m
Road speed, 2S35	65 km/h maximum
Range, 2S35 standard HE-frag	40 km maximum, standard (unassisted) rounds
Range, 2S35 guided or extended-range shell	80 km maximum, precision-guided or rocket-assisted projectile; munition not named, claim
Range, 2S43 HE-frag	24.5 km maximum, 2A64 gun; munition not stated by WEG or WPMALRU (Rostec: HE-frag more than 24 km)
Rate of fire, 2S35	16 rds/min maximum, claim
Rate of fire, 2S43	7 rds/min maximum, claim
Crew, 2S35	3
SEEN WITH 2F66-1 transport-loader vehicle (KamAZ-6560), 92 rounds, loads 2S35 in 15 min; T-90-based tracked chassis (2S35) (1)	

2S7 Pion / 2S7M Malka 2C7 «Пион»

NATO: M-1975

TR

203 mm heavy self-propelled gun for long-range fire on high-value targets

In service 1975

OPERATED BY high-power artillery brigade, 2S7 battalions (45th bde: two battalions of eight, 2017), ATP fig. 5-18: 24 x 2S7 per brigade, 8 x 2S7 battalion block (OCR uncertain); 16 reported in 45th bde, 2017

[D 2024-02 H](#)

high-power artillery brigade, 16 x 2S7 Pion (with 8 x 2S4 Tyulpan)

[O 2017-12 M](#)

artillery brigade, 2S7 battalion (IZ17, 2017); 2S7M Malka (mil.ru via

RDP18), 1 battalion; 12 x 2S7 (RUWIKI, 2019) [O 2018-04 M](#)

artillery brigade, 12 x 2S7 each (RUWIKI, 2019) [O 2019 L](#)

INDICATES A 203 mm tracked gun points to army-level artillery: a high-power artillery brigade or an army artillery brigade's heavy battalion, not a manoeuvre brigade's guns. Used on high-value targets, in urban fighting and, since 2022, for counter-battery fire.

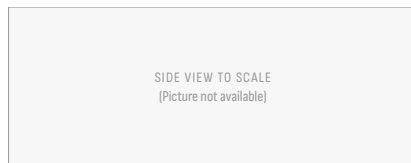
RECOGNISE **Hull** Purpose-built tracked chassis using T-72/T-80 automotive parts. Four compartments: driver, power pack, crew, rear. Front dozer blade.

Armament Very long 203 mm 2A44 gun (about 12 m) on an open mount outside the hull; on the march it lies forward, overhanging the nose. No muzzle brake, no fume extractor.

Turret No turret. Gun on open external mount at hull rear. Large recoil spade at rear lowered to the ground when firing.

Suspension Seven twin rubber-tyred road wheels and six return rollers per side; drive sprocket front, idler rear, idlers lowered to the ground when firing. Torsion bar suspension.

NOT TO BE CONFUSED WITH 2S4 Tyulpan [p. 9](#) Also given the NATO name M-1975. A 240 mm self-propelled mortar, not a gun: no 12 m barrel overhanging the hull.



SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Not amphibious. Needs firm ground to emplace; recoil spade and dozer blade used to dig in.

Weight, combat 46,500 kg
L × W × H 10.5 × 3.38 × 3.0 m
Road speed 51 km/h maximum
Road range 500 km cruising

Range, HE-frag 37.5 km maximum, Conventional HE-frag, 110 kg projectile

Range, HE rocket-assisted 47.5 km maximum, RAP projectile, 103 kg
Rate of fire 1.5 rds/min (2S7); 2.5 rds/min (2S7M) maximum

Ammunition carried 4 (2S7); 8 (2S7M) On the vehicle
Crew 7 (2S7); 6 (2S7M) on the vehicle; 2S7 crew is 14 with the auxiliary vehicle crew (WIKI)

Elevation 0 to +60 deg; traverse 30 deg

SEEN WITH Ammunition transport vehicle with the rest of the crew; carries the main load (40 rounds, RUWIKI; WIKI: 4); 1V12M command vehicle (2S7M)

SOURCES ATP CNA CNA21 DEF24 FAS3 IISS25 IZ17 IZ23B IZ23C LENTA16 LENTA23 MILN RDP1 TASS1 TWZ1 WEG4 WP-EN10 WP-RU8 WWW1

2S4 Tyulpan 2C4 «Тюльпан»

NATO: M-1975

TR

240 mm self-propelled breech-loading heavy mortar for destroying fortified targets

In service 1971

OPERATED BY high-power artillery brigade, 2S4 battalion (ATP figure 5-18 shows two 2S4 groups; subunit name not given), ATP fig. 5-18: groups of 6 and 4 2S4 (OCR uncertain); 8 reported in 45th brigade (2017) [D 2024 H](#)

high-power artillery brigade, heavy mortar battalion (240-mm); brigade allocated to a front (Soviet RVGK) [D 1991 M](#)

INDICATES Held by high-power artillery brigades and by heavy battalions of army-level artillery brigades. A 2S4 sighting points to army-level heavy artillery tasked against fortified positions or built-up areas.

RECOGNISE **Hull** Turretless tracked hull; engine front right, driver front left, commander behind driver; ammunition drums in mid-hull; open, externally mounted mortar at rear.

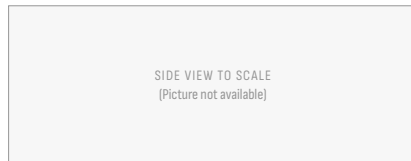
Armament Very large 240 mm breech-loading mortar, barrel about 5 m, carried with its baseplate on the hull roof; in action it swings back onto a baseplate on the ground.

Turret No turret. Commander's rotating cupola behind the driver with an external 7.62 mm PKT and a searchlight.

Suspension Six rubber-tyred road wheels and four return rollers per side, drive sprocket front, idler rear, torsion bars. Road wheel size and spacing INA.

NOT TO BE CONFUSED WITH 2S3 Akatsiya Fully enclosed turret with 152 mm gun at the rear on a similar six-road-wheel chassis; 2S4 has no turret and a roof-carried mortar.

2S5 Giatsint-S [p. 7](#) Same chassis family, also turretless, but a long 152 mm gun overhangs the front; the 2S4 mortar muzzle is only level with the first road wheel, baseplate folded above it.



SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Tracked; heavy. Fords about 1 m unprepared. Mortar must be lowered onto a ground baseplate to fire.

Weight, combat 27.5 t
L × W × H 7.94 × 3.25 × 3.225 m travelling
Road speed 60 km/h maximum
Road range 500 km

Range, 53-F-864 HE mine 9.65 km maximum, HE mine 130.7 kg, full charge

Minimum range, 53-F-864 HE mine 0.8 km HE mine, smallest charge

Range, 3F2 rocket-assisted HE mine 18 km maximum, 228 kg rocket-assisted HE mine; WEG and GS do not name the round, 3F2 per RUWIKI

Minimum range, 3F2 rocket-assisted HE mine 0.8 km rocket-assisted HE mine

Range, 3F5 SmeI'chak laser-guided mine 9.2 km maximum, laser-guided mine, target designated by 1D15 from 0.2-5 km

SEEN WITH 6x6 truck or MT-LB carrying the rest of the crew

SOURCES AG3 ATP CNA CULT DB17 FM32 GS3 IISS25 IZ17 IZ232 JANESRG MK26 ORYX PRIME26 RG17 RG22 WEG2 WP-EN9 WP-RU6 VZ22

2A36 Giatsint-B 152 mm towed gun 2A36 «Гуацунм-Б»

NATO: M1976

TW

152 mm towed gun for long-range and counter-battery fire at army level

OPERATED BY artillery brigade, gun battalion (152 mm) of the army artillery brigade; 2A36 or 2S5 or M-46, 36 per brigade (two 18-tube gun battalions) or 48 (four 24-tube battalions) [D 1991 M]

INDICATES ATP: towed artillery remains in some units. The 2A36 went to the gun battalions of artillery brigades and divisions at army and front level (FM 100-2-3, Victory Museum), so it indicates an artillery formation rather than a manoeuvre unit.

RECOGNISE Barrel Long 152 mm barrel, 8197 mm with muzzle brake; multi-slot slit-type muzzle brake; recoil brake and recuperator sit above the barrel.

Shield Shield on the upper carriage protects the crew against bullets and fragments.

Carriage Split box-section welded trails; gun rests on a base plate for firing, wheels raised by a hydraulic jack.

NOT TO BE CONFUSED WITH 2A65 Msta-B p. 10 2A65 has a two-wheeled carriage (2A36 has four wheels), a shield sloping to the rear over the wheels and a caster wheel on each trail.

SOURCES ATP FM3 GS1 IISS25 ORYX RUSARMY RUSIPL RWW VMUS WP-EN1 WP-EN2 WP-EN3 WP-EN4 WP-RU2 WSN1 not in WEG

In service 1976

MOBILITY Towed by truck or tracked tractor; fired from a base plate with wheels raised, so needs a prepared firing position. Ground pressure and fording not found.

Weight, firing position 9,760 kg
L × W × H 12.92 × 2.788 × 2.76 m travel position
Road speed, towed 80 km/h maximum, on road
Off-road speed, towed 35 to 45 km/h

Range, HE-frag 3VOF39 28.5 km maximum, HE-frag 3OF29, full charge
Range, rocket-assisted 33.5 km maximum, active-rocket
3VOF41 projectile 3OF30
Minimum range, HE-frag 8.6 km ammunition not stated; listed
3VOF39 with HE-frag maximum
Range, direct fire 1,530 m ammunition and target not stated in source

SEEN WITH KrAZ-260 6x6 truck (or KrAZ-255B, Ural-4320); AT-T, ATS-59, AT-S tracked artillery tractors

2A65 Msta-B 2A65 «Мсма-Б»

NATO: M1987

TW

Towed 152 mm howitzer for army-level artillery brigades and regiments

OPERATED BY artillery brigade, artillery battalions (several per brigade; 2S19 or 2A65) [D 2019 M]

INDICATES A towed 152 mm howitzer points to an army-level artillery brigade or a division artillery regiment. CNA says replaced 2S19s and 2A65s move to division regiments and brigade battalions.

RECOGNISE Gun Conventional towed howitzer, 152.4 mm, 47 calibres, monoblock barrel with a three-chamber muzzle brake and no bore evacuator.

Carriage Split trail with box-section trails, each with a caster wheel that swings up over the trail when firing. Hydraulic firing jack under the front.

Shield Gun shield on the upper carriage.

NOT TO BE CONFUSED WITH 2S19 Msta-S p. 8 Self-propelled turret version of the same gun; the 2S19 barrel has a bore evacuator, the 2A65 does not.

2A36 Giatsint-B p. 10 Four-wheeled carriage, two wheels each side; the 2A65 has two-wheeled running gear.

SOURCES ATP CNA FM3 GS2 IISS25 JANESRG ORYX RWW WP-EN2 WP-EN5 WP-EN6 WP-RU3 not in WEG

In service 1986

MOBILITY Towed. Emplaced with hydraulic jack and spades. Towing speed 80 km/h on highway, 60 km/h on dirt roads, 20 km/h cross-country.

Weight, travel position 7,000 kg claim
L × W × H 12.7 × 2.5 × 2.95 m claim
Towing speed, road 80 km/h 20 km/h cross-country, 60 km/h dirt road, claim
Time into action 2 to 2.5 min claim

Range, OF45 HE-frag 24.7 km maximum, OF45 (3OF45) HE-frag projectile, claim
Minimum range, OF45 HE-frag 6.5 km OF45 HE-frag projectile, claim
Range, OF61 HE-frag base bleed 29 km maximum, OF61 (3OF61) HE-frag with base-bleed gas generator, round 3VOF91, claim
Rate of fire up to 7 rounds/min maximum, claim

SEEN WITH Ural-4320 6x6; KrAZ-260 6x6 truck

Iskander-M 9K720 «Искангеп-М»

NATO: SS-26 Stone

WH-AT

Army-level operational-tactical ballistic and cruise missile system for deep strike In service 2007

OPERATED BY missile brigade, 3 missile battalions of 2 batteries each; 2 launchers per battery, 12 launchers and 12 transloaders per brigade; 4 of each per battalion [D 2021 H](#)

combined arms army, army missile brigade, 1 missile brigade (12 launchers) per army [D 2024 H](#)

tank army, army missile brigade, 1 missile brigade (12 launchers) [O 2021 M](#)

army corps, corps missile brigade, 1 missile brigade (12 launchers) [O 2021 M](#)

combined arms army, army missile brigade, 12 launchers per brigade [O 2023 M](#)

missile battalion, 4 launchers [O 2008 M](#)

INDICATES Iskander vehicles indicate an army-level missile brigade of 12 launchers. The army commander holds it as a reserve for priority strikes; it is not part of brigade artillery groups, so it does not by itself mark a manoeuvre brigade.

RECOGNISE Launcher Iskander-M: two ballistic missiles lie in the rear body on separate lift arms, raised to 85 deg to fire. Iskander-K: 2 x 9M728 or 4 x 9M729 in canisters.

Radar No radar is described on the launcher; it has satellite navigation and its own fire-control post. Target data comes from the 9S920 and 9S552 vehicles.

Chassis MZKT-7930 8x8 heavy truck, about 13 m long and 3 m wide, first two axles steer. Four hydraulic jacks lower to level the vehicle before launch.

Complex Launchers move with 9T250 transloaders on the same 8x8 chassis, which carry a crane. Command, data preparation, maintenance and living vehicles are KamAZ trucks.

NOT TO BE CONFUSED WITH 9T250 transporter-loader (same complex) Same MZKT-7930 chassis and two missiles, but has a loading crane and no launch equipment.

K-300P Bastion-P coastal launcher (MZKT-7930) Coastal troops system: two Oniks anti-ship missiles fired vertically; battery of four launchers with KamAZ command vehicles.

Bal coastal missile launcher (MZKT-7930) Carries eight Kh-35 anti-ship missiles, not two large missiles.

9A53 Uragan-1M MRL (MZKT-7930) p. 5 Rear carries two rocket pods of six tubes each instead of two missiles.

SEEN WITH 9P78-1 self-propelled launcher, MZKT-7930 8x8 (2 per battery, 4 per battalion, 12 per brigade); 9T250 transporter-loader, MZKT-7930 8x8 with crane, 2 missiles (2 per battery, 4 per battalion, 12 per brigade); 9S552 command and staff vehicle, KamAZ truck (1 per battery; 11 per brigade); 9S920 information preparation point (mission planning), KamAZ truck (1 per brigade); Maintenance vehicle (MRT0), KamAZ truck (1 per brigade in the 51-vehicle brigade set); Life support vehicle, KamAZ truck, living space for up to 8 (14 per brigade); 9M723 ballistic missile (2 per launcher, 2 per transloader; 48 per brigade); 9M728 (R-500) and 9M729 cruise missiles (2 x 9M728 or 4 x 9M729 per launcher); MZKT-7930 8x8 chassis

VARIANTS 9K720 Iskander-M with 9M723 Baseline: two ballistic missiles about 7.3 m long and 0.92 m wide on lift arms. Quasi-ballistic missile, up to about 500 km; reported to release decoys during flight. Iskander-K with 9M728 (R-500) Same 9P78-1 launcher, loaded with two cruise missile canisters instead of two ballistic missiles. Low-flying subsonic cruise missile; official range up to 500 km. Used against Chernihiv in Aug 2023. Iskander launcher with 9M729 Special launcher configuration with four cruise missile canisters; the missile is 530 mm longer than 9M728. Range disputed: US assessed over 500 km in breach of the INF Treaty; Russia said it was within limits.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Heavy 8x8 truck: 70 km/h on roads, about 40 km/h on dirt roads. Must halt and level on jacks to fire. Fords 1.4 m. A battery has been airlifted by An-124.

Weight, launcher loaded	42.85 t
Weight, 9T250 transloader loaded	40 t
L × W × H	13.07 × 3.07 × 3.55 m
Road speed	70 km/h
Road range	1,000 km
Fording	1.4 m
Time into action	16 min to first launch from the march; 4 min from highest readiness claim

Military load class

Levelling and erecting

41 Derived: 42.85 t wheeled x 0.94, rounded up, estimated
30 to 80 s to level on jacks; about 20 s to raise missiles

Range, 9M723 ballistic missile
Minimum range, 9M723

up to 500 km maximum, 9M723, domestic Iskander-M
50 km From the MIS Iskander-E (export) data table, max 280 km; domestic 9M723 minimum not confirmed
500 km official maximum, 9M728 cruise missile, Iskander-K, claim
10 to 30 m reported, With electro-optical terminal seeker

Warhead mass, 9M723
Warhead types, 9M723

480 to 700 kg Depends on warhead type
Cluster, unitary HE, penetrator, fuel-air and others; nuclear capable
3,800 to 4,020 kg Depends on payload
7.3 m; 0.92 m

Launch weight, 9M723
Missile length and diameter, 9M723
Missiles per launcher

2 ballistic (9M723) or 2 cruise (9M728); 4 x 9M729 in special configuration claim

Interval between launches

about 1 min Two missiles from one launcher

Crew
Brigade vehicle set

Launcher 3; transloader 2
51 vehicles 12 launchers, 12 transloaders, 11 command, 14 life support, 1 maintenance, 1 information preparation
INA maximum, 9M729 cruise missile

AIR DEF D

S-400 Triumph 40P6 «Триумф»

NATO: SA-21 Growler

WH-AT

Long-range SAM system against aircraft, cruise missiles and ballistic missiles In service 2007

OPERATED BY military district, SAM regiments of the district's Air Force and Air Defence Army, IISS late 2024, SAM regiments listed with S-400 in each district's Air Force and Air Defence Army: 45th (Leningrad MD) 5, plus 4 listed with S-300PS, S-300PM1 and S-400; 14th (Central MD) 4; 4th (Southern MD) 3; 11th (Eastern MD) 4 listed with S-300PS and S-400, plus 1 with S-300V and S-400; 6th (Moscow MD) none (Moscow area: 5 regiments under Aerospace Defence Command). Launchers per district **INA** D 2024 M

air defence missile regiment, about 12 launchers reported at the unit site; 4 launchers and 96L6 and 92N6 radars reported destroyed O 2025-11 M

air defence missile regiment, air defence missile battalion (division), usually 2 battalions of 8 launchers each (16 per regiment); 2012 press report relaying MoD press service R 2012 M

aerospace forces air defence regiment, IISS: 13 Aerospace Forces regts with S-400 and Pantsir-S1, plus 2 mixed with S-300 R 2024 M

air defence missile regiment, Navy: regiments under fleet air defence divisions, IISS: 64 S-400 in Navy; 3 SAM regts: 2 Baltic Fleet, 1 Pacific Fleet (naval aviation) R 2024 M

air defence missile battalion, IISS: 2 SAM bn in Belarus; 1 SAM bty in Syria R 2024 M

INDICATES S-400 belongs to Aerospace Forces or Navy SAM regiments, not ground-force brigades. A regiment usually has two battalions of eight launchers; launchers imply a 92N6 engagement radar nearby, and often Pantsir-S1 in the same regiment.

RECOGNISE **Launcher** Four large missile canisters on an erector, raised vertical to fire. One canister can hold four small 9M96 missiles instead of one 48N6.

Radar 92N6 Grave Stone engagement radar: phased array on MZKT-7930 8x8. 91N6 acquisition radar: back-to-back phased array, MZKT-7930 tractor. 96L6 may sit on a 40V6M mast.

Chassis Launchers self-propelled on MAZ-543M 8x8 (5P85SM2-01), or on a semitrailer towed by a BAZ-6402-series 6x6 tractor (5P85T2; export 5P85TE2 on BAZ-64022).

Complex 92N6 and 96L6 radars on MZKT-7930 8x8; 55K6 command post on Ural 8x8 truck. A battalion site holds about eight launchers around one 92N6.

NOT TO BE CONFUSED WITH S-300PM1/PM2 (SA-20 Gargoyl) Same 5P85 launcher family, 96L6 radar and a look-alike command post; S-400 has the redesigned 92N6 Grave Stone, not the 30N6 Flap Lid/ Tomb Stone.

64N6E Big Bird (S-300PMU family) Same two-faced array layout; the S-400 91N6E is readily told apart by its new-build MZKT-7930 tractor (APA).

SEEN WITH 55K6E command post (30K6E control system) (1 per 30K6E system of up to 6 fire units (export composition)); 91N6E Big Bird acquisition radar (1 per 30K6E system (export composition)); 92N6E Grave Stone engagement radar (domestic 92N6A) (1 per 98Zh6E fire unit); 96L6E all-altitude radar (domestic 96L6-TsP); 5P85SM2-01 / 5P85T2 launcher (APA export list: 5P85SE2 / 5P85TE2) (8 per battalion (Russian regiments, 2012); up to 12 per fire unit (export composition)); BAZ-64022 6x6 tractor (1 per 5P85TE2 semitrailer launcher); 40V6M/MD mast (INA (optional)); 5T58-2 missile transporter and 22T6-2 loading crane; 96K6 Pantsir-S1 (INA; listed in the same regiments)

VARIANTS 5P85SM2-01 Self-propelled launcher on MAZ-543M 8x8; no separate tractor. INA. 5P85T2 Launcher on a semitrailer towed by a BAZ-6402-series 6x6 tractor. INA. 5P90S Self-propelled TEL on BAZ-6909-022; new gantry, folding mast with directional antenna, satnav antennas on a longer baseline. Intended to carry a heavier missile payload than the MAZ-79100-series TELs (APA). 51P6 Launcher on MZKT-7930 chassis (RU Wikipedia only); Oryx lists 51P6 TELs among S-400 losses. INA.

SIDE VIEW TO SCALE

(Picture not available)

MOBILITY Wheeled 8x8 vehicles and tractor-semitrailer launchers; WEG estimates 30 km/h on dirt roads for the semitrailer launcher.

Road speed, 5P85TE2 launcher

Cruising range, 5P85TE2 launcher

Time into action

Weight, launcher

Fording

Time out of action

Ground pressure

Length, launcher

Width, launcher

60 km/h

road; dirt road 30 km/h, estimated

800 km

estimated

5-10 min

from travelling position; 3 min to ready from deployed position

INA

INA

INA

INA

INA

INA

Range, 40N6

Altitude, 40N6

Range, 48N6DM (48N6E3)

Altitude, S-400 (missile not stated)

Range, 48N6 (ballistic targets)

Range, 9M96E2

Altitude, 9M96E2

Range, 9M96E

Altitude, 9M96E

Detection range, 91N6E

Detection range, 96L6E

Fire unit distance from command post

Warhead, 48N6

Missiles per launcher, 48N6 family

400 km

maximum, aerodynamic targets (STRASAM); APA brochure figure is for export 40N6E, 215 nmi, claim

30-35 km

maximum, 40N6; the 27 km often quoted is reported for the 48N6DM missile specifically, claim

250 km

maximum, aerodynamic targets; RUSIAD names export 48N6E3 (Army Technology lineage); TASS 2018 state_claim: 48N6DM up to 250 m, claim

0.1-27 km

reported, reported engagement altitude band

60 km

ballistic missile targets, 48N6 series

120 km

export 9M96E2; aerodynamic targets; APA gives 64.8 nmi, claim

0.005-30 km

range, export 9M96E2; APA gives 15 ft to 100 kft, claim

40 km

export 9M96E; aerodynamic targets; APA gives 21.6 nmi, claim

0.005-20 km

range, export 9M96E; APA gives 15 ft to 66 kft, claim

580-600 km

detection/tracking, target RCS >10 m2 (strategic bomber)

300 km

max, export 96L6E in S-300PMU family, target not stated; JREW-p1877, PDF p.1877, claim

up to 100 km

maximum, 98Zh6E fire unit from 55K6E, export composition, claim

143 kg HE-frag

4 or up to 16 9M96E/E2, or mixes such as 3 x 48N6 + 4 x 9M96

SOURCES

APA1 ATP CSIS6 ISS25 JREW02 LENT2 MIL25 MIL26 ORYX RUSIAD STRASAM TASS11 WEG34 WP-EN37 WP-RU51 WP-RU52

S-300V4 C-300BA

NATO: SA-23 **TR**

Tracked long-range SAM and theatre anti-ballistic-missile system of Army air defence In service 1988

OPERATED BY air defence missile brigade, air defence missile battalion (up to 4 per brigade, under Polyana-D4M1 brigade CP); battalion CP 9S457M controls up to 4 batteries; WEG: an upper-level 9S52 Polyana-D4 command post may integrate S-300V with Buk into one brigade, S-300V battery: 1 9S32, 4 9A83, 2 9A82, 1 9A84, 2 9A85; S-300V4 battery counts INA [\[D 2017 M\]](#)

air defence missile brigade, INA. IISS lists S-300V and S-300V4 in Army holdings without a count and counts 16 SAM brigades, but does not say which hold them; Izvestia reports S-300V4 brigades under the OSK [\[R 2024 M\]](#)

INDICATES S-300V4 vehicles point to an Army air defence missile brigade: a front-level asset directly under a district (OSK) or front commander, covering main troop groupings. Older S-300V also serves in Aerospace Forces SAM regiments and at the Armenia base (IISS).

RECOGNISE Launcher 9A83 TELAR: four large tubes carried flat, raised vertical to fire; illuminator radar at the front on a folding mast. 9A82: two tubes, radar fixed over the cab.

Launcher Loader-launchers (9A84, 9A85) look like TELARs but carry a hydraulic crane in place of the radar. In S-300V4 heavy missiles ride only on 9A84-2 loader-launchers.

Radar 9S32 guidance station: large phased array raised hydraulically on its own tracked vehicle. S-300VM/V4 9S32M uses a larger High Screen-type space-fed array.

NOT TO BE CONFUSED WITH S-300PS/PM, S-400 [SA-10, SA-20, SA-21] p. 12 Launchers are wheeled: S-300P on 8x8 trucks, S-400 on trailers behind 6x6 tractors. The S-300V family is on tracked vehicles.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Heavy tracked vehicles, not amphibious; ford 1.2 m. Deploy from march and fire from unprepared sites in about 5 min; ready to move about 5 min after shutdown.

Weight, 9A83 TELAR	50 t combat weight
L × W × H	9.30 × 4.50 × 3.40 m WEG 'hull length'
Road speed	53 km/h maximum
Road range	250 km cruising range

Range, 9M83	75 km maximum, aircraft targets; S-300V
Minimum range, 9M83	6 km aircraft targets; S-300V
Altitude, 9M83	250 m-25 km maximum and minimum, aircraft targets; S-300V
Range, 9M82	100 km maximum, aircraft and jammer targets; S-300V
Minimum range, 9M82	13 km aircraft targets; S-300V, claim
Altitude, 9M82	1-30 km maximum and minimum, aircraft targets; S-300V

SEEN WITH 9S457M command post (1 per battalion); 9S15M/9S15MV Obzor-3 all-round surveillance radar (Bill Board) (1 per battalion)

SOURCES APA10 AR16 ATP BMP07 CSIS5 IISS25 IZV3 JREW02 MISS9 RUSIAD RWW TASS10 VKS WEG4 WEG33 WEG34 WP-EN34 WP-RU48 WP-RU49 WSP0T11 WSP0T12

S-350 Vityaz C-350 «Бумраг3б»

NATO: SA-28 **WH-AT**

Wheeled medium-range SAM against aircraft, cruise missiles, UAVs and ballistic missiles In service 2019

OPERATED BY air defence missile battalion, Battalion set: 1 command post 50K6A, 2-3 multifunction radars 50N6A, 12 launchers 50P6A, 12 launchers (12 missiles each) per battalion, per Russian MoD statement [\[R 2020 M\]](#)

unit not identified, training centre of the Anti-aircraft Missile Forces, Zhukov Military Academy of Aerospace Defence, first S-350 set (crews and instructors training) [\[R 2020-02 M\]](#)

INDICATES An S-350 battalion covers important sites or groups of forces against cruise missiles and aircraft. Russian statements tie deliveries to VKS air defence regiments and brigades and to Navy units; units seen near the front in 2022-25 are unidentified.

RECOGNISE Launcher 50P6 launcher carries 12 missiles in vertical-launch containers; the container pack is raised upright to fire ('cold' vertical launch).

Launcher From photograph: forward-control boxy cab; container pack in two horizontal tiers behind a machinery box. Four axles (8x8 in text sources): two front, gap, two rear.

Radar 50N6A multifunction radar: rotating phased-array antenna, 360° cover, circular and sector modes; unmanned, remote-controlled from the command post.

Chassis Launcher on 8x8 BAZ-6909 (WEG, Almaz-Antey 2023). WEG puts the radar on 6x6 BAZ-69092; Aif puts all modules, launcher included, on BAZ-69092-012.

NOT TO BE CONFUSED WITH S-300PS/PM and S-400 launcher p. 12 Four missile canisters per launcher, against twelve containers on the 50P6.

50P6T launcher-loader Described as very similar to the 50P6 launcher; no distinguishing mark found in text sources.

SOURCES AIF5 AINF2 AR17 ATO GS13 GS14 IISS25 IZ21 IZ202 RG2 RUSIAD TW27 WEG35 WMC14 WMC15 WP-EN35 WP-EN36 WP-RU50

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Not amphibious; fords about 1.2 m (WEG). Wheeled 8x8 launcher; into and out of action about 5 minutes. Cannot fire on the move.

Weight, launcher vehicle empty	22 t BAZ-6909 launcher vehicle
L × W × H	11.3 × 2.75 × 2.85 m BAZ-6909 launcher vehicle
Road speed	70 km/h maximum
Road range	1,000 km maximum

Range, 9M96E2	120 km maximum, aerodynamic target; export missile, claim
Minimum range, 9M96E2	1 km export missile, claim
Altitude, 9M96E2	5 m to 30 km maximum, aerodynamic target; export missile, claim
Range, 9M96E	40 km maximum, aerodynamic target; export missile, claim
Minimum range, 9M96E	1 km export missile, claim
Altitude, 9M96E	5 m to 20 km maximum, aerodynamic target; export missile, claim

SEEN WITH 50K6A command post (1 per battalion)

Buk-M1 / Buk-M2 / Buk-M3 9K37M1; 9K317; 9K317M «Buk-

M1»; «Buk-M2»; «Buk-M3»

NATO: SA-11 Gadfly, SA-17 Grizzly, SA-27

TR

Tracked medium-range SAM complex; army-level air defence of ground forces In service 1983

OPERATED BY air defence missile brigade, 4 air defence missile battalions, each 3 batteries; brigade command post Polyana-D4, Buk-M1: 6 TELARs, 3 loader-launchers, 1 CP, 1 search radar per battalion; 24 TELARs per 4-battalion brigade (JIT: 53rd Bde had 3 operational battalions in 2014) [D 2016 H]

air defence missile brigade, battalion TELAR, INA per battalion; brigade had 3 operational battalions in 2014 (JIT) [O 2014-07 M]

air defence missile brigade, air defence missile battalion (brigade set of Buk-M3), Buk-M3: up to 6 TELARs or 9S36M radars and up to 12 launchers per battalion [R 2019 M]

aerospace forces air defence regiment, 4 Aerospace Forces regiments with Buk-M1-2, Buk-M2 and S-300V; about 80 Buk-M1-2/Buk-M2 in Aerospace Forces holdings [R 2024 M]

air defence missile brigade, air defence missile battalion [R 2018 L]

INDICATES Buk vehicles usually indicate an army-level air defence missile brigade, nominally of four battalions (the 53rd had three in 2014). The army may detach Buk subunits to its divisions, so a battery alone does not fix the brigade's position.

RECOGNISE **Launcher** M1 and M2 TELAR: four bare missiles on a trainable launcher above the hull, radar at its front. M3 TELAR 9A317M: six missiles in cylindrical containers.

Launcher Loader-launcher has no radar. M1/M2: 8 missiles (4 on rails, 4 on cradles) and a crane. M3 9A316M: 12 containers on two arms. TELAR: 4 missiles (M3: 6).

Radar M1 TELAR carries the 9S35 radar, not a phased array. M2 and M3 TELARs carry a flat phased-array panel. Tell M2 from M3 by missiles: 4 bare vs 6 in tubes.

Radar 9S18M1 search radar: flat array on a tracked hull, 8 m high when raised, turns 360 deg. M2/M3 9S36: array on a telescopic mast up to 21–22 m, no missiles.

Chassis All combat vehicles on similar heavy tracked chassis, about 8–9.6 m long, 3.3 m wide, 30–38 t. M1/M2: GM-567M, 569, 577, 579. M3: GM-5969.

NOT TO BE CONFUSED WITH **Tor-M1/M2 (SA-15)** p. 15 One self-contained vehicle with its own search radar; missiles in 4-round container modules, not on an arm. No separate radar vehicle; battery has two transloader trucks.

2K12 Kub (SA-6) 2P25 launcher carries three missiles on a lighter 19.5 t chassis and has no radar; radar is on a separate 1S91 vehicle.

S-300V 9A83/9A82 (SA-12) p. 13 Launcher carries four (9A83) or two (9A82) large missiles fired vertically, on an MT-T-based chassis. Buk fires from an inclined arm.

SEEN WITH 9S52 Polyana-D4 brigade command post (1 per brigade); 9S470M1 command post (Buk-M1) (1 per battalion); 9S18M1 Kupol-M1 target acquisition radar (Buk-M1) (1 per battalion); 9A310M1 TELAR (Buk-M1) (2 per battery, 6 per battalion); 9A39M1 loader-launcher (Buk-M1) (1 per battery, 3 per battalion); 9S510 command post (Buk-M2) (1 per battalion; controls up to 6 fire sections); 9S18M1-3 target acquisition radar (Buk-M2) (1 per battalion); 9A317 TELAR (Buk-M2) (Up to 4 per battalion, each with 1 loader-launcher); 9S36 mast-mounted illumination and guidance radar (Buk-M2) (Up to 2 per battalion, each with 2 loader-launchers)

VARIANTS 9K37M1 Buk-M1 TELAR 9A310M1 with 9S35 radar and four bare 9M38M1 missiles; loader-launcher 9A39M1 with 8 missiles. 9K37M1-2 Buk-M1-2 Looks like Buk-M1: TELAR 9A310M1-2 with 9S35M1-2 radar and bare missiles. 9K317 Buk-M2 TELAR 9A317 with flat phased-array radar, four bare missiles; adds 9S36 radar on a 21–22 m mast. 9K317M Buk-M3 Missiles in containers: 6 on TELAR 9A317M, 12 on launcher 9A316M. All vehicles on GM-5969 chassis.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Heavy tracked, not amphibious; fords 1 m. TELAR section ready 5 min from march; 9S36 mast-radar section needs 10–15 min.

Weight, combat 34 t Buk-M1; WEG does not say which vehicle (MISM1 gives 35 t for the 9S18M1 radar)
L × W × H 9.3 × 3.25 × 3.8 m Buk-M1; WEG does not say which vehicle (MISM1 gives 9.59 m for the 9S18M1 radar)
Road speed 65 km/h maximum
Road range 500 km maximum
Fording 1 m
Time into action 5 min TELAR/battery from march
Time out of action 5 min
Ground pressure 0.8 kg/cm2
Survivability move 20 s reposiion 100–200 m with equipment running

Range, 9M38M1 (Buk-M1) 3.25–34 km maximum, air target
Altitude, 9M38M1 (Buk-M1) 30 m–22 km intercept altitude, air target
Range, 9M317 (Buk-M2) 3–50 km engagement zone, aircraft
Range, 9M317 vs ballistic and cruise missiles 20 km maximum, ALCM-type cruise missile at 30 m (MISM2; ROE gives 100 m, unchecked); Lance-type tactical ballistic missile 15–20 km (MISM2), claim
Altitude, 9M317 (Buk-M2) 15 m–25 km intercept altitude, air target
Range, 9M317M (Buk-M3) 2.5–70 km maximum effective, air target
Altitude, 9M317M (Buk-M3) 35 km maximum, air target
Detection range, 9S18M1 search radar (M1) 100–150 km target size and altitude not stated
Detection range, 9S18M1-3 / 9S18M3 search radar (M2/M3) 160 km RCS 1–2 m2; Buk-M2 9S18M1-3
Detection range, 9S35 TELAR radar (M1) 80 km RCS 2 m2; 100 km vs RCS 3 m2
Detection range, 9A317 / 9A317M TELAR radar (M2 / M3) 120 km RCS 1–2 m2 at 3 km altitude and 18–20 km vs same target at 10–15 m (M2); M3: RCS 1–2 m2, altitude not stated
Detection range, 9S36 mast radar (M2) 120 km RCS 1–2 m2 at 3 km altitude; 30–35 km vs same target at 10–15 m
Detection range, 9S36M mast radar (M3) 70 km cruise missile flying at 5 m
Crew, TELAR 4

Tor 9K330 «Top»

NATO: SA-15 Gauntlet

TR

Short-range tracked SAM with own radars; defends troops against aircraft, PGMs and UAVs In service 1986

OPERATED BY motor rifle division, air defence missile regiment, 16 per regiment (4 batteries of 4, bmpd); ATP: two battalions; ru.wikipedia: Tor-M2 'divisional' set of 3 batteries, 12 vehicles [\[D 2024 H\]](#)

motor rifle brigade, air defence missile battalion, three batteries, 12 launcher vehicles per battalion (4 per battery) [\[D 2024 M\]](#)

tank brigade, air defence missile battalion, three batteries, 12 launcher vehicles per battalion (4 per battery) [\[D 2024 M\]](#)

motor rifle division, air defence missile regiment, one regimental set of Tor-M2 (16 combat vehicles) [\[O 2017-11 M\]](#) seen with 933rd Air Defence Missile Regiment, 150th Motor Rifle Division (Millerovo)

tank division, air defence missile regiment, one regimental set of Tor-M2 (16 combat vehicles) [\[O 2017-03 M\]](#) seen with 538th Guards Air Defence Missile Regiment, 4th Guards Tank Division

motor rifle brigade, air defence missile battalion (Tor-M2DT); brigade link by location and retraining report only [\[R 2019-11 L\]](#) seen with Tor-M2DT battalion of Northern Fleet ground and coastal troops, on duty in Pechenga district where a motor rifle brigade is based (number not given)

INDICATES Four Tor vehicles together suggest a battery of a brigade air defence missile battalion (12 vehicles) or of a division's air defence missile regiment (16). Doctrine sites Tor-M1 5-10 km behind the forward edge.

RECOGNISE Launcher Missiles stand vertically inside the central unmanned turret. Tor/Tor-M1: two boxes of four (8). Tor-M2: 16 missiles.

Radar Rotating search antenna at turret rear, lowered behind turret when stowed. Phased-array tracking radar on turret front folds down; dome antenna and TV sight nearby.

Chassis Tracked GM-355 chassis (Tor-M2: GM-5955): three crew in front, turret amidships, engine at rear. Not amphibious. Road wheel count INA.

Complex Battery: command post, four combat vehicles, two transloaders on Ural-4320 trucks. Tor-M2DT rides a two-section articulated tracked DT-30 Vityaz carrier.

NOT TO BE CONFUSED WITH 2K22 Tunguska (SA-19 Grison) p. 17 Tunguska carries two 30 mm cannon on its turret; Tor has no guns and its missiles sit inside the turret.

96K6 Pantsir-S1 (SA-22 Greyhound) p. 16 Pantsir-S1 combines guns with 12 missiles and is usually on a wheeled chassis; Tor (tracked) has no guns and its missiles sit inside the turret.

9K33 Osa (SA-8 Gecko) p. 17 Osa guidance group has two small dishes flanking a central tracker; Tor has one flat phased-array tracking antenna.

SEEN WITH 9S737M Ranzhir-M battery command post (9S737MK with Tor-M2E) (1 per battery); 9T244 transloader (2 per battery); 9T245 transport vehicle; 9V887M2U / 9V887-1M2U maintenance vehicles (9V887M2K / -1M2K in the Tor-M2E export set); GM-355 / GM-5955 tracked chassis (1 per combat vehicle); DT-30 Vityaz articulated tracked carrier (Tor-M2DT; DT-30PM per Army Recognition) (1 per combat vehicle)

VARIANTS 9K331 Tor-M1 Missiles in two four-round containers (9Ya281) in the turret; some vehicles on GM-5955 chassis (9A331-1). Second target channel: 2 targets at once; crew 3. 9K331MU Tor-M1-2U INA. Interim upgrade from 2012: 4 targets at once, ceiling 10 km. 9K332 / 9K331M Tor-M2 New surveillance antenna; GM-5955 chassis; 2025 build adds armour and EW fit. 16 9M338K missiles, 4 targets at once, fires on the move; claimed 32 km detection and 16 km range. 9K331MDT Tor-M2DT Two-section articulated tracked DT-30 carrier; weapon module on rear section. Arctic mobility; seen in Ukraine from December 2022.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Tracked, not amphibious; fords 1.0 m. Tor-M1 halts to fire; Tor-M2 claimed to fire on the move. Tor-M2DT for snow and bog.

Weight	32 t combat; WEG figure for the 9K330 page, variant not stated
L × W × H	8.0 × 3.2 × 4.0 m
Road speed	65 km/h maximum
Road range	580 km cruising range
Fording	1.0 m
Swimming	not amphibious
Time into action	less than 3 min
Time out of action	less than 5 min
Bridge load class	36 derived from 32 t tracked x 1.10, estimated

Range, 9M330/9M331	12 km maximum, aircraft and PGM targets, Tor/Tor-M1
Minimum range, 9M330/9M331	1.5 km Tor/Tor-M1
Altitude, 9M330/9M331	10 m to 6 km maximum, Tor/Tor-M1
Range, 9M338K	16 km maximum, Tor-M2 domestic, air targets, claim
Altitude, 9M338K	10 m to 10 km maximum, Tor-M2 domestic, claim
Range, 9M331 (Tor-M2E)	1 to 12 km maximum, Tor-M2E export; target 300 m/s, claim
Altitude, 9M331 (Tor-M2E)	10 m to 10 km maximum, Tor-M2E export; target 300 m/s (10 m to 6 km at 600 m/s), claim
Detection range, surveillance radar (Tor/Tor-M1)	25-27 km maximum, aircraft; WEG does not state the variant
Detection range, surveillance radar (Tor/Tor-M1), helicopter	12 km maximum, helicopters
Detection range, surveillance radar (Tor/Tor-M1), UAV	9-15 km maximum, UAVs
Detection range, surveillance radar (Tor-M2)	32 km air targets, size not stated, claim
Detection range, tracking radar (Tor/Tor-M1)	15 km maximum, 60-degree sector; target and variant not stated
Detection range, TV sight	20 km maximum, target not stated
Deployment distance, Tor-M1 in defence	5-10 km behind forward edge practical, first defensive position; 5-10 km between vehicles

Pantsir-S1 96K6 «Панцирь-С1»

NATO: SA-22 Greyhound

WH-AT

Short-range gun and missile air defence of S-400/S-300 battalions and key sites In service 2012

OPERATED BY aerospace forces air defence regiment, Pantsir vehicles deployed with each S-400 battalion (Armada); Lenta: batteries attached to S-400 regiments, 3 per S-400 battalion (Armada, 2024)

D 2024 H

aerospace forces air defence regiment, Pantsir-S battery, 6 Pantsir-S delivered to Eastern MD air defence units on Kamchatka, Dec 2015; the regiment was reported due one battery O 2015-12 M

aerospace forces air defence regiment, Pantsir-S1 gun-missile battalion; each battery covers one S-400 battalion O 2020-02 M

air defence missile regiment, 30 held by the Navy (IIS p. 187)

R 2024 M

aerospace forces air defence regiment, INA per regiment; IIS lists 9 SAM regts with Pantsir-S1 in the 45th, 4 in the 14th, 3 in the 4th and 4 in the 11th AF&AD Army R 2024 M

air defence missile regiment, INA (1 SAM regt with S-400 and Pantsir-S1) R 2024 M

INDICATES Usually marks a nearby S-400 (or S-300) battalion or a defended strategic site: VKS regiments deploy Pantsir with their long-range SAM battalions (three per S-400 battalion, Armada). IIS lists none in Ground Forces holdings.

RECOGNISE Launcher Turret with two blocks of six missile tubes, one each side, and a twin-barrel 30 mm 2A38M gun beside each block; 12 missiles ready.

Radar Flat rectangular search array, about 1.8 x 0.9 m, rotating on the turret roof; folds forward flat on the roof when stowed.

Radar Tracking radar on the turret front face; small electro-optical post on a pedestal on the turret top, ahead of the search radar.

Chassis Russian systems on an 8x8 KAMAZ-6560 truck; four hydraulic jacks level the vehicle to fire. Some later ones on MZKT-7930 with armoured two-piece-windshield cab.

Complex Battery of up to six combat vehicles linked by data link, with one transloader truck per two combat vehicles.

NOT TO BE CONFUSED WITH 2K22M1 Tunguska-M1 (SA-19 Grison) p. 17 Tracked chassis and 8 missiles; Pantsir-S1 in Russian service is on an 8x8 truck with 12 missiles.

Pantsir-SM-SV Same Pantsir turret family but on a tracked GM-5970-series chassis.

SEEN WITH S-400 (SA-21 Growler) battalion (one Pantsir battery per S-400 battalion (Red Star); 3 Pantsir per battalion (Armada)); Transloader vehicle (designation not confirmed) (1 per 2 combat vehicles); Maintenance, repair, alignment and spares vehicles; KAMAZ-6560 8x8 truck (1 per combat vehicle); 1L122 Garmon radar (INA; reported ringing Pantsir positions in Ukraine)

VARIANTS Pantsir-S2 INA. Modernised Pantsir for Russian forces, in service 2015; IIS lists VKS holdings as Pantsir-S1/S2. Details INA. **Pantsir-SM** New multifunction radar with active phased array; reported on a new 8x8 KAMAZ with armoured cab. KBP claim: engagement range 40 km and detection 75 km, against 20 km and 40 km for Pantsir-S1. **Pantsir-SM-SV** Tracked GM-5970-series chassis instead of a truck. For Ground and Airborne Forces; 1RS2-3 radar; 57E6M and TKB-1055 missiles; imagery with army units reported March 2025 (likely taken 2024). **Pantsir-S1 (MZKT-7930)** MZKT-7930 8x8 chassis with armoured cab and two-piece windshield; one destroyed near Mariupol Feb 2026. Same 72V6 combat module as the KAMAZ version; chassis ordered from Belarus in 2024. **Pantsir-S1M** INA. Export version of Pantsir-SM; new missile with 30 km range (claim). Oryx records one destroyed in Ukraine.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY 8x8 truck, fast on roads. KAMAZ chassis has limited off-road mobility and tends to roll over (high centre of gravity, narrow track). Jacked level to fire guns accurately.

Weight	14.85 t WEG 'curb weight'; basis unclear also up to 30 t depending on chassis WP-RU29
L × W × H	10.3 × 2.5 × 3.27 m
Road speed	90 km/h maximum
Road range	1,000 km maximum also 500 km WP-RU29
Fording	1.8 m
Time into action	3 min WEG 'time in action'; WEG also gives deployment time 5 min; which is time into action is not stated also about 15 min WP-RU29
Time out of action	INA
Ground pressure	INA

Range, 57E6 missile	20 km maximum, aircraft-type targets; KBP figures for export 57E6-E, claim also 12 km CSIS4
Minimum range, 57E6 missile	1.2 km KBP figures for export 57E6-E, claim also 1.5 km CSIS4
Altitude, 57E6 missile	5 m to 15 km maximum, KBP figures for export 57E6-E; 5 m only in optical mode over water, claim also 10 km CSIS4
Range, 30 mm 2A38M gun	4 km maximum, air and ground targets, claim
Minimum range, 30 mm 2A38M gun	200 m air targets, claim
Altitude, 30 mm 2A38M gun	0 to 3 km maximum, air targets, claim
Detection range, 1RS1 search radar	36 km maximum, 2 m2 target; S-band; up to 20 targets tracked, claim also 47 km APAB
Detection range, 1RS2-1 tracking radar	28 km maximum, target size not stated; WEG and Armada, claim also 24 km MISSB APAB
Detection range, optical post	17-26 km auto-tracking, F-16 autotracking, 10 km visibility; 13-15 km for HARM missile, claim
Missile speed, 57E6	1,300 m/s maximum, claim
Target speed	1,000 m/s maximum, claim
Range, Pantsir-SM missile	40 km claimed, Pantsir-SM with new missile; KBP chief designer, 2016, claim
Detection range, Pantsir-SM radar	75 km claimed, target size not stated, claim
Range, Pantsir-S1M missile	30 km claimed, export Pantsir-S1M; new missile, designation not given by source; High Precision Systems, 2019, claim

SOURCES APAB AR8 AR9 ARMADA7 CSIS4 IIS25 LENTA MIL21 MISSB ORYX RSTAR RUSIAD RWW TASS7 TASS8 WEG23 WP-EN22 WP-RU29

Tunguska 2K22 «Тунгуска»

NATO: SA-19 Grison

TR

Tracked short-range gun and missile air defence vehicle, 30 mm guns and SAMs In service 1982

OPERATED BY motor rifle brigade, air defence missile-artillery battalion, Tunguska battery, 6 launcher vehicles per battery

[D 2016 M](#)

tank brigade, anti-aircraft missile-artillery battalion, SA-19 battery; ATP 4-161 gives this battalion to tank brigades, but the battery figure is for brigades generally, 6 per battery (generic brigade figure: RWW 'maneuver brigade', ATP fig. 4-19) [D 2024 L](#)

INDICATES A Tunguska battery in a forward position points to a manoeuvre brigade's air defence missile-artillery battalion covering its first-echelon battalions; doctrine places it in brigade air defence, not in area air defence units.

RECOGNISE Launcher 2S6M and 2S6M1 carry eight missile tubes, four each side of the turret; original 2S6 had four, two each side (APA and FM 100-2-3: eight).

Radar Search radar at turret rear, folded over the rear when travelling; tracking radar at turret front; search antenna a cylindrical parabolic section, tracker a dish.

Chassis Tracked, six road wheels per side, three return rollers, rear sprocket; driver front left, engine at rear. GM-352 chassis; GM-5975 on 2S6M1.

NOT TO BE CONFUSED WITH **Pantsir-S1** p. 16 Usually an 8x8 truck with 12 missile tubes in two blocks of six, one each side of the turret; the tracked GM-352M1E Pantsir is a rarer offer with 8 missiles. ZSU-23-4 Shilka Gun-only predecessor, four 23 mm barrels, no missile tubes; Tunguska has two twin 30 mm guns with missile tubes outboard and a second radar at the turret front.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Fires guns on the move but must be stationary to fire missiles. Not amphibious (WEG); fords 1.0 m (WEG 2S6M asset, RUWIKI).

Weight, combat

34 t

L × W × H

7.93 × 3.24 × 4.02 m radar up; 3.36 m radar down m

Speed, maximum road

65 km/h

Range, road

500 km cruising range

Range, 9M311-M1 missile

10 km maximum, Tunguska-M1, air target

Minimum range, 9M311-M1 missile

2.5 km Tunguska-M1, air target; APA gives

2.5 km for 2K22 and 2K22M too

Altitude, 9M311-M1 missile

3.5 km maximum, air target; ceiling for 9M311 family

Range, 30 mm 2A38M guns

4 km maximum, air target

Minimum range, 30 mm 2A38M guns

200 m air target

Altitude, 30 mm 2A38M guns

3 km maximum, air target

SEEN WITH 2F77 / 2F77M transporter-loader (6 per battery

(WEG; RUWIKI: one per 2S6); APA 2 to 3, WIKI up to 3)

SOURCES APA8 ATP FM32 IISS25 JREW02 MISS8 ROE2 RWW WEG43 WP-EN46 WP-RU59

D AIR DEF

Strela-10 and Osa short-range SAM 9K35, 9K33 «Смерель-10», «Оса»

NATO: SA-13 Gopher

TR

Short-range SAM: tracked Strela-10 (optical/IR) and wheeled Osa (radar command guidance) In service 1971

OPERATED BY motor rifle brigade, air defence missile-artillery battalion, missile battery (SA-13), 6 launchers per battery [D 2024 M](#)

INDICATES SA-13 in a brigade fits the missile battery of its air defence missile-artillery battalion, six launchers in doctrine; two platoons cover the artillery group. Osa-AKM fits a division air defence regiment, being withdrawn in 2019; it covers troops on the march.

RECOGNISE Launcher Strela-10: turret with four missile canisters in two pairs, range radar between them. Osa-AK/AKM: six ribbed missile containers; first Osa had four exposed missiles.

Radar Strela-10 9A35 adds Flat Box-B passive antennas on hull and turret; 9A34 has none. Osa: surveillance array folds back 90 degrees at rear, tracking group at front.

Chassis Strela-10: modified MT-LB, tracked, six equally spaced road wheels, no return rollers. Osa: BAZ-5937, 6x6 wheeled, amphibious; 4.20 m high with mast stowed.

NOT TO BE CONFUSED WITH 9K31 Strela-1 (SA-9 Gaskin) Four-wheeled BRDM-2 chassis; Strela-10 is on a tracked MT-LB.

SOURCES APA11 APA12 ATP CNA IISS25 JANESRG JREW02 MISS12 RWW WEG38 WP-EN40 WP-EN41

MOBILITY Strela-10 is tracked and floats only with fitted buoys (M2). Osa is a wheeled amphibious 6x6 (WH-AT-A class); it fires from short stops, not on the move.

Weight, combat

12.1 t

Weight, combat

18.68 t Osa-AKM, APA8 comparative table

L × W × H

6.45 × 2.85 × 2.22 m Strela-10

Range, road

60 km/h maximum, Strela-10

Range, 9M37M/9M333

0.8 to 5 km engagement envelope, Strela-10 engagement zone, all variants; target closing at up to 415 m/s or receding at 310 m/s

Altitude, 9M37M/9M333

25 m to 3.5 km engagement envelope, Strela-10; 9M333 minimum 10 m

Range, 9M33M3

15 km operational range, Osa-AKM missile

Altitude, 9M33M3

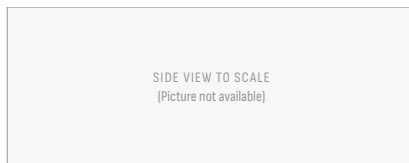
4 km maximum flight altitude, Osa-AKM missile; WEG figure

SEEN WITH 9A35 launcher vehicle (with Flat Box-B) (1 per battery); 9A34 launcher vehicle (3 per battery)

Nebo-M 55Ж6М «He60-M»

NATO: Tall Rack **WH-AT**

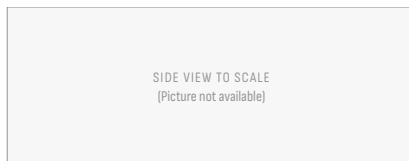
Mobile multiband air surveillance radar complex; paired low-altitude radar Podlet-K1 In service 2012

OPERATED BY unit not identified, 4 Nebo-M and Nebo-UM combined [R 2015 L]unit not identified, air defence and radio-technical crews with a division-level command post (MoD release; unit not named) [R 2017-09 L]**INDICATES** Nebo-M or Podlet-K1 indicates an Aerospace Forces radio-technical (radar) unit building the air picture, often near an S-300 or S-400 position that the radars cue. One reported Podlet-K1 site (Kherson, Jul 2022) lay over 100 km behind the front.**RECOGNISE Antenna** RLM-M: very large VHF array of short Yagi elements, 24 x 7. RLM-D: flat L-band panel, 38 x 48 elements; outer segments fold and stow flat under clamshell doors.**Antenna** Podlet-K1: one large upper phased array, two smaller back-to-back IFF arrays below it, a narrow anti-jam antenna beside it. Antenna lies lowered on the truck in travel.**Mast** Nebo-M antennas raised and folded hydraulically on a mast with a compact equipment cabin at its base; small navigation antenna package at the top of each array.**Chassis** Nebo-M: each module on a BAZ-6909-015 8x8 truck, the chassis of the S-400 TEL. Podlet-K1: KamAZ-6350 8x8 truck.**NOT TO BE CONFUSED WITH** 55Zh6U Nebo-U Transportable, not self-propelled: six transport units and hours to deploy. RLM-M is one 8x8 truck that deploys in about 15 min.**11L19 Nebo-SVU** Box-shaped T/R module enclosures on the mast and rear backlobe elements; both absent on RLM-M.**59N6E Protivnik-GE** Semi-trailer mounted L-band radar; RLM-D uses a smaller antenna on a self-propelled 8x8.**MOBILITY** Self-propelled 8x8 trucks; Nebo-M can also move by rail. Each Nebo-M module has its own 100 kW generator. Hydraulic levelling and antenna deployment.Road speed, KamAZ-6350 **95 km/h** maximum, base truck, Podlet-K1 carrierHeight, Podlet-K1 antenna deployed **12 m** Podlet-K1, radar fully deployedTime into action, Nebo-M **15 min** Nebo-M, manufacturer cited stow and deploy time, claimTime into action, Podlet-K1 **20 min** Podlet-K1, deployment or stowage, claimDetection range, Nebo-M **600 km** maximum, complex; target type, size and altitude not stated, claimAltitude coverage, Nebo-M **600 km** maximum, complex, ballistic targets implied; not stated, claimDetection range, Podlet-K1 **200 km** maximum, main mode, 10-200 km; target size not stated, claimMinimum range, Podlet-K1 **10 km** air targets, lower limit of stated 10-200 km detection zone, claim**SOURCES** AIF ANNA APA6 APA7 ARMADA6 ESP GS10 JREW02 RUSIAD TRUCK TWZ5 TWZ6 U242 VPK4 WEG22 WP-RU26 WP-RU27 ZVEZ

96L6 all-altitude acquisition radar 96Л6 «Всевысотный обнаружитель»

NATO: Cheese Board **WH-AT**

3D C-band acquisition and target designation radar for S-300PM/PMU2 and S-400 SAM units In service 1997

OPERATED BY aerospace forces air defence regiment, 3rd SAM battalion, INA (photographed deployed and stowed) [O 2010-02 M]
aerospace forces air defence regiment, S-400 'battery' in Armada's usage; APA export diagram lists it as a battery component, 1 per S-400 battery (Armada, term undefined; APA lists 96L6E as a battery component without a count); S-300PMU2 count INA [R 2014 M]**INDICATES** A 96L6 usually marks a nearby long-range SAM fire unit (S-300PM/PMU2 or S-400) of the Aerospace Forces, which are district-level assets. The maker also offers it to radio-technical units, so on its own it does not confirm a SAM site.**RECOGNISE Antenna** Deployed: one flat rectangular array, slightly taller than wide, near-vertical, tilted back a little, on a rotating pedestal just behind the cab; lattice IFF array below.**Antenna, stowed** Stowed: array folds forward and lies flat and horizontal over the pedestal and cab roof, giving a long, low, box-like vehicle outline.**Mast** Normally no mast. Antenna post can be lifted onto a 40V6M (23.8 m) or 40V6MD lattice mast for low-level cover; the roughly 2 h fitting time is an unverified maker figure.**NOT TO BE CONFUSED WITH** 76N6 Clam Shell Split antenna with a large beak-like shroud, usually raised on a 40V6M mast. A mast-mounted 96L6 shows one flat rectangular array.**64N6E / 91N6E Big Bird** Janus-faced (back-to-back) space-fed array on a semi-trailer towed by an MZKT-7930 or BAZ-6403 tractor. 96L6 is a single self-propelled vehicle with one array.**MOBILITY** Self-propelled 8x8. Single-vehicle version ready 5 min from march. Two-trailer and mast versions are semi-mobile; maker figures of 30 min to about 2 h not verified. Not amphibious.L x W x H **11.7 x 3.07 x 2.92 m** WEG figure for MAZ-543 chassis; radar vehicle is MZKT-7930Road speed **65 km/h** maximum, WEG figure for MAZ-543 chassisRoad range **650 km** cruising, WEG figure for MAZ-543 chassisFording **1.3 m** WEG figure for MAZ-543 chassisDetection range **300 km** instrumented, Range band 5 to 300 km for detected targets; target size and altitude not stated, claimMaximum target altitude **100 km** All-altitude mode, aerodynamic targets, claimBand **C-band** claimFrequency **5.25 to 5.925 GHz**Elevation cover, all-altitude mode **-1.5° to +20°** 360° azimuth**SEEN WITH** MZKT-7930 8x8 carrier (TM966E transport vehicle) (1 per radar (single-vehicle version))**SOURCES** APA1 APA2 APA3 ARMADA1 ATP GS5 JREW02 MIL1 MILRU ORYX PVO RS RT RUS11 RUSIAD RWW U241 WEG5 WMC1 WP-RU9

Zoopark-1 / Zoopark-1M 1A219, 1A219M; 1A260 «3oonapk-1», «3oonapk-1M»

TR

Tracked radar that locates enemy guns, mortars, rockets and missile launchers In service 1970

OPERATED BY motor rifle brigade, fire control and artillery reconnaissance battery, radar platoon (with SNAR-10M1) [D 2016 M] tank brigade, fire control and artillery reconnaissance battery, radar platoon [D 2016 M] motor rifle division [D 2024 L]

INDICATES Doctrinally held in the radar platoon of a manoeuvre brigade's fire control battery and by artillery formations. Seeing one suggests brigade-level or higher artillery is nearby and set up for counter-battery fire.

RECOGNISE **Antenna** One flat rectangular phased-array panel. 1L219/1L219M: raised upright at rear of roof. 1L260: a much taller panel.

Mast No separate mast. Panel is lifted and turned on the roof by a lift and rotation mechanism. For travel it folds down flat along the roof.

Chassis 1L219/1L219M: low MT-LBu tracked carrier, flat sides, raised rear hull. 1L260: larger GM-5955 tracked chassis, 7 road wheels a side, boxy hull with side lockers.

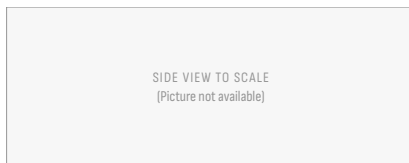
Complex Radar vehicle can work alone. Support: 1I30 (1L219M) or 1I38 (1L260) maintenance vehicle and a generator; same brigade fire control battery as SNAR-10M1 and PRP-4A.

NOT TO BE CONFUSED WITH SNAR-10 / SNAR-10M1 (1RL232) battlefield surveillance radar p. 20 Same MT-LBu hull but has a turret with a PKT machine gun and a rotating parabolic radar, not a large flat panel.

ARK-1 / ARK-1M Rys (1RL239) counter-battery radar p. 19

Predecessor on the same MT-LBu hull and looks alike. No reliable distance cue found; confirm by markings or close imagery.

SOURCES AR24 AR25 AR26 ARMADA9 ATP BMPD10 DOGS FISH GWARA IZ3 MIL34 MIL35 MIL36 ORYX RUMOD RWW SAIDPVO WEG2015 WMC21 WMC22 WP-RU63 not in WEG



MOBILITY Tracked, cross-country capable. Emplaces or displaces in about 5 min. Swimming ability not documented for either chassis.

Weight, 1L261 (1L260) 38.1 t
Road speed, 1L219M 62 km/h maximum
Road speed, 1L260 70 km/h maximum
Road range 500 km Both chassis

Detection range, mortar, 1L219M

Detection range, tube artillery, 1L219M

Detection range, rocket artillery, 1L219M

Detection range, tactical missile, 1L219M

Fire-correction range, 1L219M

20 km detection of launch position, Mortar firing position; calibre not stated
15 km detection of launch position, Field artillery firing position
30 km detection of launch position, MLRS firing position
40 km detection of launch position, Tactical missile launch position
mortar 22, artillery 20, MLRS 35, tactical missile 40 km fire correction range, Adjusting friendly fire, by weapon type

ARK-1M Rys 1PA239 «PblCb»

NATO: Small Fred TR

Artillery weapon-locating and fire-control radar In service 1977

OPERATED BY unit not identified, division's reconnaissance battery (WEG wording, identical to Wikipedia; Soviet-era) [R 2024 L]

INDICATES A single sighting points to an artillery reconnaissance element supporting a formation's guns. Sources give no current Russian subunit or count, so it says little about formation type.

RECOGNISE **Antenna** Parabolic antenna (JREW02); radiator in a compact radio-transparent housing plus a large receiving antenna (TOPWAR). Sizes INA.

Mast No mast described in any read source. Two emitters and a receiving antenna sit outside the hull (MILN); TOPWAR names one emitter housing. Height INA.

Chassis MT-LBu unarmed tracked carrier; electronics mostly inside the armoured hull.

NOT TO BE CONFUSED WITH 1L219 Zoopark-1 p. 19 Also MT-LB based; Zoopark-1 uses a phased array antenna, ARK-1M a parabolic antenna.

MOBILITY Tracked MT-LBu chassis; WEG says it may not be amphibious given radar weight. Forging depth INA.

Weight 15.7 t
L × W × H 7.61 × 2.85 × 2.72 m Chassis overall
Road speed 61 km/h maximum
Road range 500 km cruising

Detection range, ARK-1M vs artillery (howitzers) 8 km unspecified, Artillery (WEG), howitzers (JREW02); target and mode not stated by WEG

Detection range, ARK-1M vs mortars 13 km unspecified, Mortar (WEG); mode not stated

Detection range, ARK-1M vs MRL rockets 25 km unspecified, Rocket (WEG); mode not stated

Detection range, ARK-1M vs battlefield missiles 30 km unspecified, Missile (WEG)

SEEN WITH MT-LBu tracked chassis (1)

VARIANTS ARK-1 INA.

SOURCES ATP JREW02 MIL2 MIL3 ORYX PMUL TOPWAR1 TOPWAR2 WEG6 WEG2015 WP-EN11

1B75 Penicillin

1B75 «Пенициллин»

WH-AT

Passive acoustic-thermal gun locator (1B75); man-portable counter-mortar radar (1L271) In service 2018

OPERATED BY self-propelled artillery battalion, battalion fire control network, conceptual ATP fig. 5-3 (1L271) D 2024 M

tank division, 1L271; holding subunit not stated O 2018-01 M

combined arms army, 1L271 in rocket, missile and tube artillery units R 2017 L

INDICATES A 1B75 points to an artillery regiment or brigade counter-battery element, as deliveries were planned for those units. A 1L271 was issued down to company and battalion level and marks forward mortar or artillery fire control.

RECOGNISE **Antenna** 1L271: transceiver on hand-turned tripod pedestal; reflector described as single-curvature parabolic or as phased array with mirror. 1B75 has no radar antenna.

Mast 1B75: stabilised head with six TV and six thermal cameras on a raising mast. Per GlobalSecurity only, the mast folds rearward onto the van roof in transit.

Chassis 1B75: KamAZ cross-country truck, 8x8 per GlobalSecurity and Wikipedia; van body and four jacks per GlobalSecurity only. 1L271: man-carried or on a UAZ-2966.

Complex 1B75: two KamAZ vehicles plus remote ground acoustic sensors, up to six vehicles. 1L271: 135 kg set of radar, processor, power unit, console and radio.

NOT TO BE CONFUSED WITH **Zoopark-1 / Zoopark-1M** p.19 Radar complex on one armoured amphibious vehicle; 1B75 is a KamAZ truck with a camera mast and no radar.

1B33M AZK-7M Older sound-ranging system, acoustic only; lacks the 1B75 thermal and TV camera mast.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY 1B75 on KamAZ 8x8 truck; mast raised and ground sensors laid out at site, crew relocates daily. 1L271 man-portable (135 kg in loads) or vehicle-carried.

Weight, 1L271	135 kg complete set: radar, processor, power unit, claim
Road speed, 1B75	90 km/h maximum, KamAZ chassis, claim
Time into action, 1L271	5 min maximum, preparation for work, claim
Weight, 1B75	INA

Detection range, 1B75	up to 25 km maximum, radius; artillery, mortar and MLRS firing positions, sound and thermal, claim
Target location time, 1B75	5 s maximum, single firing target, claim
Radio range, 1B75	40 km maximum, to external subscribers, claim
Detection range, 1L271	20 km maximum, tank-type moving ground target, claim
Detection range, 1L271 (mortar positions)	5 km maximum, mortar firing positions, 81-120 mm, claim

SOURCES AIF2 AR10 ATP BMPD5 BMPD6 DEFEX GS11 IZV20 IZV22 ORYX RADTUT RG21 RG152 RG242 RWW SHEP2 TASS9 WP-EN23 WP-EN24 WP-RU30 WP-RU31 not in WEG

RADARS

SNAR-10 / SNAR-10M1 Leopard

CHAP-10M1 «Aeonapp»

NATO: Big Fred

TR

Tracked battlefield surveillance and artillery-adjustment radar In service 1975

OPERATED BY motor rifle brigade, fire control and artillery reconnaissance battery, radar platoon D 2016 M

INDICATES A SNAR-10M1 points to a maneuver brigade's fire control battery radar platoon, so an artillery-adjustment and ground-surveillance element is working for that formation. Ukraine also holds the SNAR-10, so a sighting alone is not proof of a Russian unit.

RECOGNISE **Antenna** Rectangular reflector nearly vehicle width, raised electromechanically for use; folds forward when travelling.

Mast No mast described: radar is built into the vehicle's turret, antenna on the roof (SNAR-10); on the M1 the antenna module sits on the rear roof.

Chassis MT-LB family tracked hull; the Big Fred version is a roof-mounted radar on an MT-LB.

NOT TO BE CONFUSED WITH **MT-LBu command vehicle** MT-LBu has canvas cover extending to rear and extra radios, no folding radar reflector on roof.

MOBILITY Tracked MT-LB derivative; radar deploys from halt, about 5 min into action from prepared position.

Weight, combat	12.7 t SNAR-10M1, developer figure, claim
Road speed	60 km/h maximum, SNAR-10 (WEG); SNAR-10M1 'up to 60 km/h' (developer)
Road range	500 km cruising, SNAR-10
Time into action	5 min claimed, SNAR-10, prepared position; up to 20 min unprepared, claim

Detection range, SNAR-10 radar	16 km maximum, moving vehicles, without MTI; 10 km with MTI, claim
Detection range, SNAR-10 radar	30 km maximum, moving ships, without MTI, claim
Detection range, SNAR-10 radar	10 km maximum, shell impact (100 mm), without MTI, claim

SEEN WITH MT-LB chassis (1 per radar); 1L219M Zoopark-1

SOURCES AR18 ARMYINF3 ATP BMPD8 FOCUS2 JANESRG JREW02 MILUA PERISC RWW SHEPH UKRIN VPK8 VPK9 WEG27 WEG37 WEG2015 WP-RU54

Krasukha-2 / Krasukha-4 1A269; 1PA257 «Kpacuxa-2»; «Kpacuxa-4»

WH-AT

Truck radar jammers: Krasukha-2 vs S-band AEW radar; Krasukha-4 vs X/Ku-band radars In service 2011

OPERATED BY EW brigade, REB-S [counter-air] EW battalion, INA; usually one Krasukha-2 and one Krasukha-4 per brigade (reported 2022) [D 2018 H]

EW brigade, part of 1 Krasukha-4 complex (command-post container found near Makariv); Armada estimates 4 complete Krasukha systems in theatre [R 2022-03 M]

EW battalion, independent REB-S battalion of an Air and Air Defence Army (VKS); FOI infers, does not name Krasukha there [R 2018 L]

unit not identified, 1 Krasukha-4 reported hit [R 2023-11 L]

unit not identified, 1 Krasukha-2 reported [R 2018-08 L]

INDICATES Krasukha jammers belong to EW brigades (REB-S battalions): four under the military districts, the 15th (Tambov) under the General Staff. Manoeuvre brigades have EW companies. A sighting points to an EW brigade element or a protected site such as an air base.

RECOGNISE **Antenna** Krasukha-2: one very large solid circular dish, about twice a wheel diameter across, on a turntable at the rear; rotates 360° and tilts. Stowed look not documented.

Antenna Krasukha-4: box-shaped rotating turret on a van-body roof with three dishes, two large on the sides, one small on the front; all far smaller than the Krasukha-2 dish.

Mast Krasukha-4: telescopic mast with a small antenna head behind the turret; the second vehicle also carries a mast antenna. Krasukha-2: no tall mast in photos.

Stowed Krasukha-4: in travel position the turret is stored inside the shelter (AR); in action two hydraulic stabilisers are lowered on each side of the shelter.

Chassis Krasukha-2: BAZ-6910-022 8x8, angular flat-faced cab, axles 2+2 with a wide gap. Krasukha-4: KamAZ-6350 8x8, standard KamAZ cab-over, spare wheel behind cab.

Complex Krasukha-4: two KamAZ-6350 trucks, jammer plus command-post container vehicle. Krasukha-2: three vehicles (one source); probably procured with Moskva-1 (FOI).

NOT TO BE CONFUSED WITH 1L269 Krasukha-2 vs 1RL257 Krasukha-4 Krasukha-2: one big dish on a BAZ-6910 with angular cab. Krasukha-4: turret with three smaller dishes on a KamAZ-6350 van body.

Pelena-1 (older S-band counter-AEW jammer) Antenna is a tall curved rectangular reflector on a towed wheeled trailer (APA photos), not a solid circular dish on a self-propelled 8x8 truck.

1L267 Moskva-1 ELINT and command complex p. 22 Three KamAZ trucks (FOI: one ELINT, two command; 3G: KamAZ-6350-1335, third with a jamming control station); no Krasukha-4 style dish turret.

GT-01 Murmansk-BN HF jamming complex p. 22 Has KamAZ-6350 support trucks, but its signature is very tall extendable HF antenna masts on KamAZ-53501 trucks, not dishes.

55Zh6M Nebo-M radar modules (BAZ-6909-015) p. 18 Same heavy BAZ 8x8 family, but antennas are a flat rectangular planar array or a Yagi array, not a solid circular dish.

VARIANTS 1L269 Krasukha-2 (Krasukha-20) BAZ-6910-022 8x8 with one large solid circular dish at the rear. S-band jammer aimed mainly at AEW radars such as E-3 and E-8. Analogue equipment per ru.wikipedia. 1RL257 Krasukha-4 (Krasukha-S4) KamAZ-6350 8x8 van with a three-dish turret, plus a second KamAZ-6350 carrying the command post. X/Ku-band jammer vs fighter fire-control and ISR radars, UAVs and LEO satellites. Digital equipment per ru.wikipedia.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Heavy 8x8 trucks. Must halt and emplace to jam: Krasukha-4 needs under 20 min to emplace or displace; Krasukha-2 lowers jacks before operating.

Krasukha-4: weight	39,000 kg also 12,450 kg AR6
Krasukha-2: weight	40,000 kg BAZ-6910-022 chassis full mass; kerb mass 18 t
Krasukha-4: road speed	95 km/h maximum, KamAZ-6350 truck figure (AR)
Krasukha-2: road speed	80 km/h maximum
Krasukha-4: road range	1,000 km KamAZ-6350 truck figure (AR)
Krasukha-2: road range	1,000 km
Krasukha-4: fording	1.75 m KamAZ-6350 truck figure (AR), claim
Krasukha-2: fording	1.4 m BAZ-6910-022 chassis figure (REACH)
Krasukha-4: time into action	less than 20 min

Krasukha-2: jamming range	up to 250 km up to, air target: AEW aircraft radar; target altitude not stated
Krasukha-4: jamming range	150–300 km maximum, air and ground-based radar targets; target size and altitude not stated, claim
Krasukha-2: target band	S-band also 2.3–2.7 GHz CRFS
Krasukha-2: main targets	AEW aircraft radars [E-3 Sentry, E-8 JSTARS] air targets
Krasukha-4: operating band	8–18 GHz (X and Ku band) also 8.5–10.7 and 13.4–17.7 GHz 3G
Krasukha-4: main targets	Fighter and interceptor radars, UAVs, reconnaissance aircraft, LEO satellites air and space targets
Krasukha-4: coverage	360° azimuth; 1° to 60° elevation
Krasukha-4: output power	24 dBW (8 GHz); 23 dBW (13 GHz); 21 dBW (18 GHz)
Krasukha-4: power consumption	30 kW
Krasukha-4: continuous operation	24 h
Krasukha-2: crew	INA also 7 or 3 REACH
Krasukha-4: crew	INA also 3 AR6

SEEN WITH Krasukha-4 command-post vehicle (KamAZ-6350 with container) (1 per Krasukha-4 complex); 1L267 Moskva-1 ELINT and command complex (INA; one Moskva-1 can link up to nine Krasukha-type jammers); KamAZ-6350 8x8 truck (2 per Krasukha-4 complex); BAZ-6910-022 8x8 chassis (1 per Krasukha-2 jammer vehicle); RB-109A Bylina EW brigade command system (brigade-level EW C2, not Krasukha-specific)

SOURCES 3G APAS APAG AR6 ARMADA2 ATP CRFS FOI ICDS MIL12 MIL13 MIL14 REACH RWW TWZ4 WEG17 WMC4 WMC5 WMC6 WMC7 WP-EN19 WP-RU19

Murmansk-BN «Мурманск-БН»

WH-AT

Strategic HF (3-30 MHz) communications intercept and jamming complex, multi-truck In service 2014

OPERATED BY EW brigade, Probably a company of the REB-N (anti-ground) battalion [FOI assessment] [D 2018 H](#)
 unit not identified, Radio-jamming node (y3ex paguonmex); its 1st section named at the 471st (KZ), 2 complexes at the Kamchatka centre (KZ, reported Jan 2016; FOI: delivered 2014) [O 2019 H](#)

EW brigade, At least 1 complex per brigade (Armada); 2 antenna groups observed at 16th EW bde, vehicle count at 19th suggests 2 (PAS imagery, 2019) [O 2019 M](#)

INDICATES Seeing Murmansk-BN indicates an Army EW brigade (IISS lists 5; 15th under General Staff) or a fleet EW centre, with long-range HF intercept and jamming in place; it takes about 72 hours to deploy, so a raised complex marks a deliberate, fixed position.

RECOGNISE **Antenna** Wire HF antennas and mast guy lines run from the masts to the ground; from afar like a giant spider's web. Antenna field about 800 m a side. Stowed: carried on trucks.

Mast Four hydraulic telescopic masts up to 32 m, guyed. Armada: antenna trucks typically about 35 m apart across a 90-degree arc. Mast lowered for travel.

Chassis Heavy multi-axle KamAZ (or Ural) trucks with unified box bodies (kungs); some masts reported on towed trailers.

Complex Seven trucks: four mast vehicles plus generator, stowed-antenna and command post vehicles. Kola site seen with two groups of four antenna vehicles about 350 m apart.

NOT TO BE CONFUSED WITH Krasukha-2 / Krasukha-4 p. 21 Each is housed on a single truck; Murmansk-BN is spread over several trucks with tall guyed masts.

Samarkand Stationary coastal jamming complex, probably a fixed version of Murmansk-BN; not truck-mounted.

SOURCES 3G ARMADA5 B02 CRFS FOI ICDS IISS25 IZ KZ3 MIL18 PAS RG16 RG173 RWW UKRINF2 WEG21

SIDE VIEW TO SCALE
 (Picture not available)

MOBILITY Road-mobile truck column; needs a prepared antenna field about 800 m a side, marking and engineer work; 72 hours to deploy. Trucks up to 35 t need category E drivers.

Weight, vehicle up to 35 t Loaded KamAZ trucks of the complex, claim

Time into action 72 h Normative deployment time, claim

Number of vehicles 7 Complex in travel order, claim

Time out of action INA

Jamming range, HF communications 5,000 km effective (Armada's term), HF (shortwave) radio links, skywave; ship, aircraft and ground emitters, claim

Frequency band 3-30 MHz (HF)

Power up to 400 kW In some operating modes (KZ); generator output (Armada), claim

Mast height 32 m 4 telescopic masts, fully raised, claim

Deployed position area 640,000 m² Antenna field about 800 x 800 m, claim

Crew INA

SEEN WITH Antenna mast vehicle (4 per complex)

EW M

Moskva-1 1A267 «MockBa-1»

WH-AT

Passive ELINT and automated control of counter-air jammers and air defence cueing In service 2014

OPERATED BY EW brigade, REB-S (counter-air EW) battalion, reported [D 2018 H](#)

EW brigade, INA [ARM22 lists one Moskva-1 among five brigade systems] [R 2023-09 L](#)

unit not identified, 1 claimed destroyed [R 2022-09 L](#)

INDICATES Moskva-1 is held by Ground Forces EW brigades, four of which answer to the military districts. Seeing it suggests a brigade counter-air EW element and Krasukha-type jammers at other sites, up to nine of which one complex can task.

RECOGNISE **Antenna** No automatically raised antenna array, unlike Krasukha. The ELINT module (1L265) was withheld from press; its antenna is not described in text (INA).

Chassis KamAZ-6350 8x8 trucks with plain box bodies (kung); a Russian reporter wrote the vehicles are hard to tell from other trucks.

Complex Three KamAZ vehicles per complex: an ELINT module and control vehicles; one account names a power-station truck as the third.

Complex Deployed: the control module stands on hydraulic legs on the ground while its truck drives 30-100 m away.

NOT TO BE CONFUSED WITH 1RL257 Krasukha-4 Krasukha deploys an automatically raised antenna block; Moskva-1 vehicles show plain box bodies.

1L269 Krasukha-20 p. 21 Krasukha-20 is on the larger BAZ-6910 chassis, not KamAZ.

Other KamAZ-6350 box-body vehicles No reliable external mark in text sources; FOI says Krasukha-20 is probably procured together with Moskva-1, so Krasukha jammers nearby are the main clue.

SIDE VIEW TO SCALE
 (Picture not available)

MOBILITY Road and track mobile truck complex; manufacturer gives 45 min to deploy. Control module can be set down on legs apart from its truck.

Weight 12,450 kg chassis only

L × W × H 9.3 × 2.5 × 3.25 m chassis

Road speed 95 km/h maximum

Road range 1,000 km cruising

Detection range, airborne emitters 400 km up to, passive; emitting aircraft and cruise missiles; 360 deg sector, claim

Jamming range, airborne S-band radar 250 km up to, air targets; ARM23 says the 1L267 'can engage targets' per unnamed sources; Moskva-1 itself has no jammer (FOI)

Sector of view 360 deg claim

Frequency coverage UHF, L, S, C, X and Ku bands ELINT module 1L265

Controlled systems 9 EW or air defence complexes tasked simultaneously, claim

Crew 4 combat crew, claim

SEEN WITH 1L265 ELINT module vehicle (1 per complex)

SOURCES 3G AEX17 ARMADA3 ARMADA4 ATP CH97 DLG22 FOI ICDS IEEE22 IZ234 RIA15 RWW SDU13 TOPWAS5 VPK3 WEG19 WP-RU22 WP-RU23 WP-RU24

Borisoglebsk-2 РБ-3015 «Борисоглебск-2»

TR-A

Tactical HF/VHF/UHF radio intercept, direction-finding and jamming complex In service 2013

OPERATED BY motor rifle brigade, EW company [D 2024-02 H]

tank brigade, EW company [D 2024-02 H]

motor rifle division, separate EW battalion [O 2022-01 M] seen with 3rd

Motor Rifle Division (Vislenskaya), separate EW battalion, Belgorod oblast

motor rifle brigade, EW subunit [O 2018 M] seen with 74th Separate Guards

Motor Rifle Brigade

unit not identified, EW units [O 2018 M] seen with 201st Military Base, Tajikistan (R-934BMV stations)

unit not identified [O 2022-03 L] seen with 69th Separate Covering Brigade (R-330BMV captured, Kyiv oblast, 2022-03)

INDICATES Usually indicates the EW company of a motor rifle or tank brigade or division within reach; since 2022 also division EW battalions. Vehicles of the complex may deploy singly rather than as a full set.

RECOGNISE Antenna Antennas on a mast, reported for one R-934BMV or R-378BMV station hit by drone. No text source gives antenna shape or size: INA.

Mast Vehicle-mounted antenna mast; height and stowed position INA in text sources.

Chassis MT-LBu tracked amphibious armoured carrier, modified for the EW equipment. One source says one R-325UMV rides on a Ural truck.

Complex Up to 9 vehicles: R-330KMV command post plus pairs of R-378BMV, R-330BMV, R-934BMV and R-325UMV jammers, with diesel generators. May deploy in part.

NOT TO BE CONFUSED WITH R-330Zh Zhitel p. 24 Ural-43203 wheeled truck (Diabazol complex), not tracked MT-LBu.

RB-341V Leer-3 p. 26 Single vehicle working with up to three Orlan-10 UAVs, not a tracked multi-vehicle set.

1L262 Rtut-BM Also on MT-LBu tracked chassis (3G table, tertiary); no text source found giving a visible difference.

SNAR-10 (Big Fred) p. 20 MT-LBu with turret-mounted radar and parabolic antenna; antenna folds forward on the roof when travelling.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Tracked, amphibious MT-LBu chassis. About 15 min to deploy or pack up the complex on arrival.

Weight **11,000 kg** WEG figure for base MT-LB chassis, not MT-LBu
also 12.6 t WEG2015

L × W × H **6.45 × 2.86 × 1.86 m** WEG figure for base MT-LB chassis

Road speed **60 km/h** maximum, base MT-LB chassis
also 61.5 km/h WEG2015

Road range **500 km** cruising, base MT-LB chassis
also 500 km WEG2015

Swimming speed **5 km/h** maximum

Fording **Amphibious**

Time into action **15 min or less** whole complex, on arrival at site

Time out of action **15 min or less** whole complex

Ground pressure **0.46 kg/cm2** standard track; 0.28 kg/cm2 wide track

Employment distance from line of contact **15 to 30 km** practical, Donbas 2014-15, with Leer-3, Zhitel, R-934UM, observed

Crew, R-330KMV command post **4** 2 of them at automated workstations

Vehicles per complex **up to 9**

Jammer stations controlled **up to 4 pairs** per R-330KMV command post

Frequency coverage **HF, VHF, UHF** intercept and jamming of ground and air radio; also satcom, GPS and cellular terminals

also 3 MHz to 3 GHz 3G

Frequency-hopping signals **up to 300 hops/s** VHF, detection and jamming, 'per open sources', claim

Performance vs R-330 **2x bandwidth, up to 100x faster** HF

Mandat and UHF bands; reported, same lineage

Power supply **own diesel generators or 220/380 V 50 Hz mains**

INA not published

Jamming range, ground radio

INA

Detection range, radio emitters

also 30 km 3G

Armament **INA** no source other than WEG names a

weapon on the EW vehicles

SEEN WITH R-330KMV command post (1 per complex); R-378BMV HF jamming station (2 per complex); R-330BMV VHF jamming station (2 per complex); R-934BMV airborne VHF/UHF radio jamming station (2 per complex); R-325UMV jamming station (2 per complex); ATO-40 service and repair vehicle; Diesel generators; MT-LBu (up to 9 per complex)

R-330B Mandat-B and R-934BMV P-330B; P-934MB «Mangam-B»

WH-AT

Ground VHF/UHF radio and air-communication jammers, tactical level In service 1996

OPERATED BY EW company, EW company of a motor rifle or tank brigade: R-330B squad in the VHF-band jamming platoon; R-934B squad [VHF aircraft] in the radio-communications jamming platoon

[D 2016 H]

unit not identified, R-934B (2024), Borisoglebsk-2 R-934 station (2023-11) and R-934BMV (2023-10, 2024-11, 2025-04) reported hit by Ukrainian units; no Russian holding unit named [R 2025 M]

INDICATES Seen with a brigade or division, it fits the organic EW company. RWW and FOI place these jammers in the EW company of motor rifle and tank brigades. It shows radio and air-link jamming is planned, not the size of the force.

RECOGNISE **Antenna** R-934UM2: one receiving and direction-finding antenna-feeder system plus seven transmitting antenna-feeder systems. Sizes INA.

Mast R-330T/B carry a telescoping mast system that is raised when deployed; R-330U stows its antenna in a rectangular housing. Mast height INA.

Chassis R-934B and R-330B: MT-LBu tracked, or Ural-4320-31 6x6 with K1-4320 body (R-934B); R-330T on KamAZ-43101 or Ural-43203-31 [K2-4320].

Complex Wheeled versions tow a generator trailer [16 kW for R-934B, ED-30 for R-330]; R-934UM2 has a truck with two generators and a shelter.

NOT TO BE CONFUSED WITH R-378BMV HF jammer

[Borisoglebsk-2] p. 23 Same MT-LB carrier in the same nine-vehicle set; no text source gives an external difference. INA.

R-330Zh Zhitel (Diabazol) p. 24 Ural-43203 wheeled carrier, not tracked MT-LB; no external antenna difference given in sources. INA.

SOURCES 3G AP45 ATP CRFS DEFBLOG FOI ICDS MIL6 MIL7 MIL9 MIL10 RWW TOPWAR4 UKRINF VRF3 WEG14 WP-RU13

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Ural-4320 6x6 for R-934B/R-330T/R-934UM2 [truck-class off-road]; tracked MT-LBu R-934B/R-330B. Figures below are chassis-class, not station-verified.

Weight, combat **11,000 kg** R-330B on MT-LBu; WEG figure, unverified

Road speed **85 km/h** R-934B on Ural-43203; WEG chassis figure

Road speed **60 km/h** R-330B on MT-LBu; WEG figure, unverified

Road range **1,000 km** R-934B on Ural-43203; WEG chassis figure

Frequency range, R-934B **100-400 MHz** Export sheet; aircraft links 100-150 and 220-400 MHz, claim

Transmitter power, R-934B **500 W** Export sheet, claim

Jammed links, R-934B **4 at fixed frequencies; 1 with frequency hopping** Export sheet, claim

RMS direction-finding error, R-934B **2 deg** Export sheet figure copied by WEG;

RWW gives not more than 3 deg, claim

Crew, R-934B **3** Export sheet, claim

Frequency range, R-330B **30-100 MHz** Export sheet, R-330B/T; 3G

and APA both from Rosoboronexport, claim

SEEN WITH R-330KMV command post [1 per Borisoglebsk-2 set]; R-934BMV jamming station [2 per Borisoglebsk-2 set]

EW M

R-330Zh Zhitel P-330Ж «Жумель»

WH-AT

Truck-and-trailer jammer and direction finder against satcom, GPS and GSM user equipment In service 2008

OPERATED BY EW company, EW company, jamming platoon (radio, GPS, cellular, radio-controlled fuses), jamming squad, 1 [single station on RWW chart; no count stated] [D 2016 M]

EW brigade, REB-S (anti-aircraft communications and navigation) battalion [D 2018 M]

EW company, EW company of motor rifle or tank division [sources confirm the company exists; Zhitel holding not stated] [D 2018 L]

INDICATES Doctrinally one Zhitel sits in the EW company of a manoeuvre brigade, so a sighting usually points to such a formation nearby. It is also held by the REB-S battalions of district EW brigades, so it can indicate an attached higher-level EW element.

RECOGNISE **Antenna** Four active phased-array transmit antennas, one atop each trailer mast. Truck has a separate DF array: ring of about 8 vertical elements.

Mast Stowed: four telescopic masts upright at trailer corners, just above the box. Deployed: four guyed masts, raised by hand winch.

Chassis, truck Ural-4320-family or KamAZ-43114 6x6 with full-length box body; air-conditioner box at front of body above cab; DF mast at body rear. Houses all operator equipment.

Chassis, trailer Two-axle trailer carrying a tall rectangular box with the antennas and a diesel generator; outrigger jacks at the corners.

NOT TO BE CONFUSED WITH R-934BMV jamming station [Borisoglebsk-2] p. 23 Listed on tracked MT-LB chassis, no trailer; Zhitel is a wheeled 6x6 truck towing a four-mast antenna trailer. RB-301B Borisoglebsk-2 complex [R-330KMV, R-378BMV, R-330BMV, R-934BMV, R-325BMV] p. 23 Group of up to nine tracked MT-LB vehicles; Zhitel is a single wheeled truck plus trailer.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY 6x6 truck, but the towed antenna trailer is fragile; WEG expects it near roads, not cross-country. Masts raised by hand, up to 40 min to deploy.

L × W × H **7.36 × 2.55 × 2.71 m** Truck; trailer not included (unclear in source)

Road speed **85 km/h** maximum, Truck, maximum road

Time into action **40 min or less** Full deployment including masts, claim

Time out of action **40 min or less** Full march order, claim

Jamming range **25 km** maximum effective, Ground user equipment [WEG gives no target type; matches manufacturer's ground-user figure]

Jamming range, airborne users **50 km or more** claimed minimum, User equipment installed on aircraft, claim

Area covered by a pair **About 40 × 20 km** claimed, Two paired stations, jamming ground communications and navigation

Frequency band, intercept and DF **100-2,000 MHz** claim

SOURCES 3G AR23 ATP CRFS FOI ICDS MIL33 ORYX PRO RUSIAD RWW TOPWAR7 TOPWAR8 VRF8 WEG48 WMC8 WMC18 WMC19 WMC20

Palantin «ПАЛАТИНУ»

WH-AT

Operational-tactical HF/VHF comms jamming and SIGINT complex; networks other EW systems In service 2019

OPERATED BY combined arms army, EW units of a Western MD combined arms army in Voronezh Oblast (army not named)

R 2019-04 M

combined arms army, EW troops of a Guards combined arms army, Western MD, Luga range, Leningrad Oblast (Palantin-K, Oct 2019); Feb 2022 exercise in Leningrad Oblast names only the Western MD

R 2022-02 M

unit not identified, EW subunits of the Eastern MD in Zabaykalsky Krai (planned deliveries by end 2020) **R 2020-07 L**

INDICATES First delivered to EW units of a combined arms army; no source read names the holding subunit or echelon. A Palantin suggests an EW node that may network Moskva, Zhitel or Dzyudoist; in April 2026 one was reported covering air defence assets in Zaporizhzhia.

RECOGNISE **Antenna** Antenna devices carried on masts raised to 15 m on deployment. From photograph: compact multi-element antenna head at the top of a slender mast.

Mast Antenna masts of 15 m raised at the site. From photograph: masts stand beside the box body, braced by guy lines while being raised.

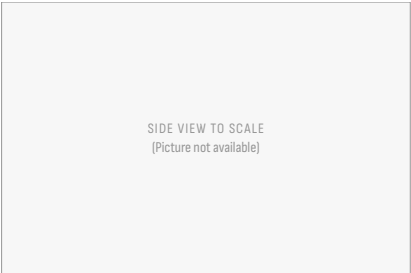
Chassis Four-axle KamAZ truck. From photograph: large box-body shelter with a side ladder. Generators are started on deployment.

Complex Reported as 22 mobile stations spread along the front; each starts generators, raises masts and joins a data network.

NOT TO BE CONFUSED WITH **Murmansk-BN** p. 22 Murmansk-BN uses 32 m telescopic masts holding a large HF antenna net between trucks; Palantin masts are about 15 m.

Borisoglebsk-2 p. 23 Borisoglebsk-2 is on MT-LB tracked carriers; Palantin is on four-axle KamAZ trucks.

R-330Zh Zhitel p. 24 Zhitel (Diabazol) is on the Ural-43203 truck; Palantin is on a four-axle KamAZ.



MOBILITY Truck-mounted on four-axle KamAZ; about 30 min from march to operation including raising 15 m masts.

Time into action about 30 min from the march, including raising 15 m antennas, claim

Chassis four-axle KamAZ truck

Weight INA

Time out of action INA

Jamming range, UAV control links up to 20 km maximum, air target: UAV radio control links, claim also 20 km MIL20

Jamming frontage, radio communications about 1,000 km wide strip claimed, ground target: radio communications and radio reconnaissance; battalion tactical group exercise near Voronezh, Sep 2019, claim

Jamming frequency range 3–30 MHz (HF) and 30–300 MHz (VHF) ground target: radio communications

Jamming targets HF/VHF radio incl. SDR-based, cellular and trunked radio, navigation, UAV control links claim

Reconnaissance function radio-electronic reconnaissance; detects and locates transmitters links Moskva, Zhitel and Dzyudoist EW systems into one network claim

Network function 22 mobile stations Palantin-K, claim

Composition 15 m claim

Antenna mast height INA

Crew INA

SEEN WITH Palantin mobile station (four-axle KamAZ) [22 per complex (reported, Palantin-K)]; Moskva EW complex; R-330Zh Zhitel; MTK-1A Dzyudoist

EW

Leer-2 85Я6 «Aeep-2»

WH-AT

Mobile technical control, radio reconnaissance and jamming vehicle on Tigr-M chassis In service 2012

OPERATED BY airborne division, EW subunit (FOI: VDV EW companies formed from 2016), INA; MoD announced 10 complexes for VDV EW units, Dec 2012 (title only, state claim) [R 2018 M]

INDICATES A Leer-2 most often indicates an Airborne Troops EW subunit; Navy EW centres, the Strategic Rocket Forces and Ground Forces EW units also hold it. It says nothing about the parent unit type.

RECOGNISE **Antenna** All EW antennas sit on a platform on the vehicle roof. Antenna shapes and sizes: INA, no text source.

Mast No separate mast reported. Two telescopic antennas are raised on the roof when static (3G calls them mast antennas). Height INA.

Chassis GAZ Tigr-M 4x4 armoured car: armoured bonnet, bulletproof hull and door windows, electric winch (4,000 kg, 20 m rope). No armament.

NOT TO BE CONFUSED WITH Leer-3 (RB-341V) p. 26 Leer-3 works through Orlan-10 UAVs from a command vehicle; Leer-2 is a single armoured Tigr-M car with roof antennas.

MOBILITY Fords 1.3 m; not amphibious. Chassis figures are Tigr-M, from WEG.

Weight, combat	7,800 kg
L × W × H	5.7 × 2.3 × 2.3 m
Speed, maximum road	140 km/h
Road range	1,000 km cruising

Jamming frequency band	0.02-2.7 GHz	Radio emitters incl. mobile communications; 3G table copied from an Airbase.ru forum image, 2014
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Signal analysis band	0.1 MHz-18 GHz	Radio reconnaissance; direction finding 0.02-18 GHz
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Simultaneous suppression	12 emission sources	Quasi-simultaneous; VDV EW chief, Nov 2012, via forum, claim
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Direction-finding accuracy	3 deg roof antennas, 2 deg mast antennas
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SEEN WITH GAZ Tigr-M (1 per vehicle)

SOURCES 3G AIRBASE AR7 CNA FOI GAGAD ICDS MIL15 ORYX RG2018 VPK2 WEG18 WEG2015 WP-RU20

Leer-3 P5-341B «Aeep-3»

WH-AT

UAV-borne cellular (GSM) jamming, phone location and fake-SMS system using Orlan-10 drones In service 2015

OPERATED BY EW brigade, Leer-3 platoon, platoon operates the complex; number per brigade INA [D 2024 H]

EW battalion, Leer-3 drone/EW company [R 2019 M]

unit not identified, Leer-3 crews [R 2022-11 M]

INDICATES A Leer-3 control truck points to an EW brigade platoon, an army EW battalion company or a division EW battalion. Expect Orlan-10 flights that jam or imitate cell networks, locate phones for artillery and send SMS within about 6 km of the drone.

RECOGNISE **Antenna** Control truck carries a command-telemetry radio antenna system for the UAVs. From photograph: small drum-shaped radome on a thin vertical pole. Size INA.

Mast From photograph: the antenna pole rises from the rear top corner of the box body; no large telescopic mast seen on the exhibited vehicle. Height INA.

Chassis KamAZ-5350 6x6 truck, closed box body with operator workstations. From photograph: bonneted cab, chamfered roof edges, side door, small windows, spare wheel on rear wall.

Complex Up to three Orlan-10 fixed-wing UAVs, catapult-launched and recovered by parachute; wingspan over 3 m.

Complex From photograph: Orlan-10 has a slim fuselage, straight wing and tail fin.

NOT TO BE CONFUSED WITH Leer-2 p. 26 Leer-2 is built on a Tigr-M 4x4 armoured car; the Leer-3 control vehicle is a KamAZ-5350 6x6 truck with a box body.

Orlan-10 Same airframe also flies artillery reconnaissance with UAV companies; the airframe alone does not show the EW role. Look for the Leer-3 control truck or cellular jamming.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Control vehicle is a KamAZ-5350 6x6 truck; complex deploys in about 30 minutes. Leer-3 vehicle weight, dimensions and fording not found.

Road speed	100 km/h	maximum, KamAZ-5350 base truck at maximum permitted mass; not a Leer-3 figure, claim
Road range	1,550 km	minimum at reference consumption, KamAZ-5350 base truck, 420 l tanks, at reference fuel consumption; not a Leer-3 figure, claim
Time into action	30 min	whole complex, claim
Weight	INA	

Jamming range, cellular terminals (ground)	up to 6 km	up to, radius of the zone of blocked GSM terminals around the UAV, claim
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Detection range, GSM base stations	6 km	within, cellular base stations, GSM and possibly 3G/4G, measured from the UAV
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Practical range, Orlan-10 from control vehicle	120 km	up to, radius of UAV employment from the control point, claim
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Endurance, Orlan-10	10 h	maximum, maximum flight time of the Orlan-10 in the Leer set, claim
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SOURCES 3G ATP CNA CRFS DFR FOI ICDS IZ20 IZ22 IZ222 MASH MIL16 ORYX RG15 RG222 RWW UNIAN UZST VRF5 WMC8 WMC9 WP-RU21 not in WEG

Pole-21 Р-340РП «Поле-21»

NA

Networked GNSS jamming posts protecting fixed sites from GNSS-guided weapons and UAVs In service 2016

OPERATED BY combined arms army, Army EW specialists; first used in an exercise with battalion tactical groups, with R-934BMV and Silok-01 (MoD via AIF) [D 2021-04 M]

INDICATES A post on a cell tower or fixed mount suggests a protected fixed site such as a strategic facility or command post. A single R-340RP in the field indicates a GNSS-jamming layer near that position.

RECOGNISE **Antenna** Transmit and receive antennas on cellular communications towers. Small antennas resembling civilian cell antennas, 1 to 3 per post (Ukrainian expert, 2023). Size: INA.

Mast Posts attach to existing cell towers or buildings, or to a stationary ground mount. Height: INA.

Chassis Also shown as a wheeled mobile unit and as a unit deployed in the field. Vehicle type: INA.

NOT TO BE CONFUSED WITH R-330Zh Zhitel p. 24. Zhitel is a vehicle-borne station; Pole-21 posts sit on towers or fixed mounts. FOI notes similarities in function. Focus.ua wrongly calls Pole-21 'Zhitel'.

MOBILITY Posts are fixed on masts or mounts. A wheeled mobile unit is also reported. Weights and dimensions: INA.

Power supply 220 V, 50 Hz; 600 W consumption WEG (Pole-21E)

Operating temperature -40 to 50 °C Pole-21E per WEG

Weight INA

Jamming range, GNSS receivers (ground or air not stated) up to 25 km effective, WEG (Pole-21E); Russian press gives 25 km per jamming module for the domestic Pole-21.

Jamming frequency band 1.176 to 1.602 GHz (GPS, GLONASS, Galileo, BeiDou)

Coverage area of one complex 150 km² WEG; number of posts not stated

Output power per module 300 to 1,000 W variable max radiation pattern, single frequency; Pole-21E

SEEN WITH Jamming posts (up to 100 per complex); jamming modules (1 to 3 per post)

SOURCES 3G AIF4 ATP DPSU FERG FERG2 FOCUS FOI IZRUI IZV KPOST TOPWAR6 UKRINF1 WEG25 WP-UK2

Repellent-1 Репеллент-1

WH-AT

Wheeled counter-UAV EW complex: detects and jams small drone control and GNSS links In service 2016

INDICATES Sightings so far (Donbas 2018, Kherson 2022) name no holding unit type. Its presence shows a counter-UAV EW asset is deployed in that area, not which formation owns it.

RECOGNISE **Antenna** Sensor and jamming head on a collapsible telescopic mast; upper section holds ELINT and jamming equipment and a panoramic camera. Size: INA.

Mast Long collapsible telescopic mast mounted beside the control cabin on the rear of the truck. Height: INA.

Chassis Three-axle truck, reported as MAZ-6317; WEG says Kamaz-6317 8x8 and Armada says Kamaz-6350. Chassis maker varies by customer (Wikipedia).

Complex Single vehicle with rear control cabin (two operator monitors, small-arms and NBC protection). Mobile and stationary versions are offered. Crew 3.

MOBILITY Not amphibious (WEG). Fords 1.0 m; 60% gradient, 0.6 m step (WEG). Chassis type disputed, so figures may differ by chassis.

Weight, combat 23,650 kg WEG figure, Kamaz-6317 chassis

L × W × H 9.4 × 2.7 × 3.35 m

Road speed 85 km/h maximum

Road range 1,000 km maximum

Detection range, UAV control-link signals up to 30 km up to, ground-station command signals; UAV telemetry up to 10 km; target size and altitude not given, claim

Jamming range, small UAV control and telemetry links up to 30 km up to, air target: small UAVs; telemetry and satellite-navigation links; control link up to 10 km, claim

Simultaneous UAV targets up to 12; detect and jam in up to 25 s claimed, brochure claim, claim

SOURCES 3G ARMADA8 ARMYINF2 BCAT IZV2 LIGA OSCE RUSIPL UASV UKRINF4 UKRINF22 WEG32 WP-EN33

Infaua РБ-531Б «Инфауна»

WH-AT-A

Vehicle-mounted radio jammer and IED jammer, with smoke self-screening In service 2012

OPERATED BY unit not identified, dedicated EW units and manoeuvre reconnaissance units [D 2024 L]

INDICATES Presence suggests a VDV or other EW subunit escorting a column or protecting a battalion-level position against radio-controlled IEDs and guided munitions. WEG 2015 places it in VDV EW battalions; no source gives counts.

RECOGNISE **Antenna** Main antenna in a domed housing on the roof, raised on a telescopic mast when deployed. Size: INA. Auxiliary antenna on starboard side.

Mast Telescopic mast raises the roof antenna. Mast height: INA.

Chassis Modified BTR-80 (K1Sh1), 8x8, turret removed. Middle hull larger and higher than BTR-80; four hatches at front, one door at each side centre.

NOT TO BE CONFUSED WITH Kushetka-B command vehicle (BTR-80 chassis) Small turret, and a long rectangular box where Infaua has a camera; a 2018 Horiivka sighting was misidentified as Infaua.

MOBILITY Amphibious 8x8 on BTR-80 derivative; operates on the move and escorts columns.

Weight 13,614 kg WEG: combat weight, listed under BTR-80 chassis

L × W × H 7.65 × 2.90 × 2.46 m

Road speed 80 km/h maximum

Road range 600 km maximum

Jamming range, target not stated 3 km disruption range, WEG 'Disruption Range'; target and conditions not given

Smoke screen distance 130 m up to, aerosol against laser and TV-guided weapons

Jamming band, communications 20–200 MHz frequency range as given by WEG, WEG 2015 and RWW; sub-bands not given

Output power 60 W

SEEN WITH K1Sh1 chassis (BTR-80 derivative) (1 per vehicle); Lesoschek portable EW system

SOURCES 3G AR4 ATP DFRLAB FOI GS8 ICDS RDP2 RG RWW VRF4 WEG16 WEG2015 WP-RU18

PMP heavy folding pontoon bridge park ПМП

WH-AT

Truck-carried folding pontoon park for 60-t floating bridges and 40-170 t rafts In service 1961

OPERATED BY motor rifle division, engineer battalion, pontoon bridge company (also in tank divisions), half set: 16 river and 2 shore links and 1 service vehicle on KrAZ-214; 6 BMK-90/150 boats per company [D 1991 M](#)

engineer-sapper brigade, pontoon bridge battalion of the army or front engineer brigade (FM 100-2-3 'Engineer Brigade, Army or Front'), one full set: 32 river and 4 shore links on KrAZ-255, 15 BMK-150 boats [D 1991 M](#)

pontoon-bridge regiment, two pontoon bridge battalions (army or front/district level), three full sets: 96 river and 12 shore links on KrAZ-255 [D 1991 M](#)

motor rifle brigade, engineer battalion, pontoon bridge company, one pontoon park; RWW calls it 'PP-61' but its figures (268 m at 60 t; 90- and 120-t bridges) match PP-91, not PMP. 6 BMK boats, 7 PTS-2 [D 2016 L](#)

unit not identified, INA; Oryx lists 47 PMP(-M) river links and 3 shore links lost [R 2026-09 M](#)

INDICATES A half set (16 river links, 2 shore links, about 6 boats) matches a division engineer battalion's pontoon company. Over 100 link carriers, or three full sets, matches an army or front pontoon-bridge regiment. Either points to a planned bridged river crossing.

RECOGNISE Chassis 6x6 truck carriers: KrAZ-214, later KrAZ-255B. Ural-4320 and KamAZ-53501 carriers also seen. Separate trucks carry river links, shore links and boats.

Working equipment Roller bed on the truck; a folding jib and a winch behind the cab recover the link. Truck backs to the water, brakes hard and the link slides off the rollers.

Stowed River link: four hinged steel pontoons folded into one tall block filling the truck bed, 6.75 m long. Shore link is shorter, 5.6 m, with a ramp end.

Stowed, boats BMK-90/150 tug-motor boats ride on their own truck carriers, 6 per half set and 12 per full set; PMP-M sets have 16.

Deployed Link unfolds on hitting the water to a flat deck 6.75 m long, about 8 m wide. Links pin into a continuous steel ribbon, 6.5 m roadway, held in line by boats.

Deployed, 20-t For 20-t loads each link is split into half-links, giving a narrower 3.29 m roadway and a longer bridge.

NOT TO BE CONFUSED WITH PP-91 / PP-2005 (PP-91M) pontoon parks Larger links (7.36 x 8.28 m) on KrAZ-260G/Ural-53236 or, for PP-2005, KamAZ-63501 8x8. Truck alone does not settle it: VPK lists Ural-4320-30 for PP-91 too.

TMM-3 truck-launched scissors bridge [p. 29](#) Same KrAZ-255B chassis, but carries a folded 10.5 m scissors span with a launching girder and folding trestle legs, not a pontoon block.

PMM-2 self-propelled ferry [p. 29](#) Tracked, boat-bowed hull with seven road wheels and two box pontoons stacked on top; one vehicle forms a ferry.

VARIANTS PMP-M Same folding river links; shore link has a straightened deck; hydrodynamic shield kit. Carried on KrAZ-255B; KamAZ-53501 carriers also recorded. Bridge usable in currents up to 3 m/s against 2 m/s for PMP; set lists 16 boats instead of 12. **PMP on Ural-4320 carriers** PMP links on Ural-4320 6x6 trucks in place of KrAZ-255B. None known; same links. **PP-91** Bigger river link, 7.36 m long and 8.28 m wide, on KrAZ-260G (Ural-53236) carriers; BMK-225 boats. 60-, 90- and 120-t bridges and double-width roadway; lengths per set differ between sources [see open questions]. **PP-2005 (PP-91M, PP-2005M)** PP-91-type links on KamAZ-63501 8x8 carriers. Boats: BMK-MT with PP-2005M (VPK); BMK-3 seen with PP-91M (AIF, 2025). Bridges to 120 t; rafts 90-360 t (PP-91) or 90/190/380 t (PP-2005M per VPK). In a Group Centre pontoon-bridge company, 2025-02.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Carriers must back to the water's edge to launch and recover links. PMP bridge limited to currents of 2 m/s, PMP-M to 3 m/s.

River link weight **6,790 kg** one river link
Shore link weight **7,252 kg** one shore (ramp) link
Time into action, bridge **7 m of bridge per minute** ideal conditions

Time out of action, per link **a few minutes** retrieval of one link by winch and jib

River link, unfolded **6.75 m long, 8 m wide**

Weight, carrier with link **INA**

Road speed **INA** maximum

Road range **INA** cruising

Fording **INA**

Bridge, full set, 60-t **227 m** maximum, 32 river + 4 shore links, 6.5 m roadway

Bridge, full set, 20-t **382 m** maximum, split half-links, 3.29 m roadway

Maximum current **2 m/s** PMP

Bridge, half set, 60-t **119 m** 16 river + 2 shore links

Bridge, half set, 20-t **191 m** 16 river + 2 shore links, split

Load capacity, bridge **60 t** full-width bridge; 20 t on split-link bridge

Roadway width **6.5 m** 60-t bridge

Rafts **40, 60, 80, 110, 170 t** 2, 3, 4, 5+1 shore, 8+1 shore links

Raft assembly, 40-t **8 min** 2 river links, working party 6

Tank speed on bridge **up to 30 km/h** tanks on 60-t bridge

Crew per carrier **3** WEG figure, in the block copied from the PMM-2 page; WEG 2015 KrAZ-255B page gives 3 in cab

Working party, full bridge **approx. 65** full bridge set

SEEN WITH River link on KrAZ-214 or KrAZ-255B [or Ural-4320, KamAZ-53501] carrier (32 per full set, 16 per half set); Shore (ramp) link on carrier truck (4 per full set, 2 per half set); Roadway mats (vystilki) on trucks (2 per full set); BMK-90/150 tug-motor boats on truck carriers (12 per full set (16 for PMP-M), 6 per half set); PTS-2 (or K-61) tracked amphibious transporter (7 in a brigade engineer battalion; 28 in an army engineer brigade (1991)); GSP tracked self-propelled ferry (12 half-ferries per division engineer battalion; 24 in an army engineer brigade (1991)); PMM-2 self-propelled amphibious ferry (replacing GSP); TMM-3 truck-launched bridge (4 per engineer battalion)

TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge

TMM-3

WH-AT

Brigding: TMM-3 truck scissors bridge; MTU-72 tank-chassis bridge layer In service 1970

OPERATED BY engineer-sapper battalion, brigade engineer battalion; sets almost always attached to the movement support detachment (OOD), 4 TMM-3 truck-launched bridge sets per brigade engineer battalion, each laying a 10 m span; together a 42 m bridge

[D 2024-02 M](#)

engineer-sapper regiment, engineer and bridge subunit (press-release wording) [O 2020-11 M](#)

INDICATES TMM-3 sets are held by the brigade engineer battalion and almost always attached to its movement support detachment, which bridges minor gaps ahead of the main body. Four trucks together suggest a route or a crossing of up to 42 m being prepared.

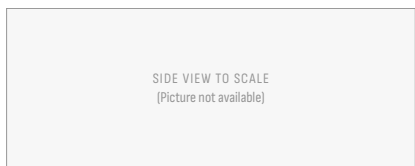
RECOGNISE Chassis TMM-3: KrAZ-255B 6x6 soft-skin truck; spare wheel on a steel frame above the cab. MTU-72: turretless T-72 hull: six road wheels, three return rollers, exhaust left rear.

Working equipment TMM-3: hydraulic slewing mast with forked end and winch lifts the folded span. MTU-72: single bridge launched from the hull, hydraulic front blade. Cantilever type.

Stowed TMM-3: each 10.5 m span folds to 5.2 m; spans ride on the truck bed. MTU-72: bridge folded 9.42 m long on a hull 11.64 m long, 3.38 m high.

NOT TO BE CONFUSED WITH MTU-90 T-90 chassis; RWW gives a 25 m bridge, 50 t class, versus 20 m on MTU-72; held in tank brigades per RWW.

MTU-20 and MT-55A T-54/55-family chassis (MT-55A is on T-55) rather than T-72 chassis.



SIDE VIEW TO SCALE
(Picture not available)

MOBILITY TMM-3 is an unarmoured 6x6 truck (WH-AT). MTU-72 is tracked (TR), armoured. TMM-3 span: gap to 40 m, 3 m deep.

Weight, TMM-3 truck with bridge	19,500 kg
Weight, MTU-72	40,000 kg
L × W × H	9.3 × 3.2 × 3.15 m
Road speed, TMM-3	71 km/h

Obstacle, TMM-3 bridge set up to 40 m wide, 3 m deep Set of four spans; ATP gives 42 m total length

Load class, TMM-3 60 t Wheeled and tracked loads; tracked vehicles at 15 km/h on bridge

Bridge geometry, TMM-3 4 spans of 10.5 m; roadway 3.8 m Spans can be connected to 10.5, 21, 31.5 or 42 m (RWW); ATP gives 10 m per set and 42 m together

Crew, TMM-3 1 + 1 in cab (driver, passenger) per truck

SOURCES ARMYINF4 ATP IISS25 JANESRG JFRC KOLESA MIL27 ORYX RWW TRENCH WEG39 WEG2015 WP-EN42 WP-EN43 WP-RU55 WP-RU56 WSN3

PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry

PTC-4

TR-A

Amphibious carrier for cargo and vehicles (PTS-4); self-propelled river ferry (PMM-2) In service 2007

OPERATED BY motor rifle brigade, engineer battalion, 7 PTS-2 per engineer battalion (RWW names PTS-2, not PTS-4) [D 2016 M](#)

tank brigade, engineer battalion, 7 PTS-2 per engineer battalion (RWW names PTS-2, not PTS-4) [D 2016 M](#)

unit not identified, INA (PMM-2 held by Russian Army; no count given) [R 2024 L](#)

INDICATES PTS-2 amphibians are a brigade engineer battalion asset in RWW, so PTS-type carriers suggest engineer river-crossing support. IISS25 lists PMM-2 without unit type; no unit is tied to PTS-4 here.

RECOGNISE Chassis PTS-4: tracked amphibious hull with forward armoured cab; T-80 tracks and torsion bars, T-72 engine and gearbox. Road wheel count INA.

Working equipment PTS-4: open rear cargo area, platform 8.28 x 3.3 m, one remote-controlled 12.7 mm machine gun. PMM-2: boat-like bow, PTS-2-style cab.

Stowed PMM-2: MT-T-type chassis, seven road wheels and four return rollers per WEG; two large box pontoons carried one above the other on the hull.

Deployed PMM-2: pontoons hinge out to each side, with wave deflectors and ramps; deck 10 m long, 4.2 m wide; two 5 m loading ramps; two propellers lowered aft.

NOT TO BE CONFUSED WITH PTS-2 / PTS-3 Older, on MT-T chassis with T-64 components; PTS-3 has higher sides. PTS-4 uses T-80 and T-72 units and has a more heavily armoured cab.

PMM-2 vs PTS-2 PMM-2 has similar cab and bow but carries two stacked box pontoons on top of the hull.



SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Amphibious tracked. PMM-2 fording 1.2 m, vertical step 2.8 m (WEG15; implausible, see open_questions). PTS-4 vertical step 0.65 m, trench 2.5 m (WEG).

Weight, PTS-4	33.1 t
Weight, PMM-2	36 t
L × W × H	8.3 × 3.3 × 3.2 m
Road speed, PTS-4	60 km/h

Payload on land, PTS-4	12 t
Payload on water, PTS-4	18 t
Payload, PMM-2 single ferry	42.5 t
Payload, PMM-2 two joined	85 t
Armament, PTS-4	one 12.7 mm remote machine gun, 400 rounds

SOURCES AR12 AR13 IISS25 RWW WEG28 WEG2015 WP-EN29 WP-RU36 WP-RU37 WP-RU38 WWW2 WWW12

IMR-2 / IMR-2M / IMR-3M combat engineer vehicle IMR-2

TR

Tank-based heavy engineer vehicle for obstacle and minefield clearing

In service 1980

OPERATED BY engineer-sapper battalion, mine clearing platoon (brigade engineer battalion), 2 IMR-3M per platoon D 2016 M

INDICATES An IMR-3M in a column suggests the mine clearing platoon of a brigade engineer battalion; its elements may join the movement support detachment (OOD) ahead of the main body to clear routes and minefield lanes. Separate engineer brigades also hold it.

RECOGNISE Chassis Tank chassis: T-72 for IMR-2 (six road wheels a side, torsion bar); T-90 for IMR-3M. No gun; superstructure carries a crane.

Working equipment Wide dozer blade at the front; telescopic boom with grab or bucket on a rotating superstructure where a turret would be.

Stowed Blade raised onto the hull roof; boom drawn back and turned. A KMT track-width mine plough may be fitted.

NOT TO BE CONFUSED WITH BAT-2 Also a tracked bulldozer with adjustable blade and boom crane; RWW gives 7.3 m boom reach and 700 hp against about 8.4 m and 840 hp for IMR.

SOURCES ATP IIS25 RWW TASS2 TSAMTO WEG15 WEG2015 WP-EN14 WP-EN15 WP-EN16 WP-RU17

MOBILITY Fords 1.2 m unprepared; 5 m with snorkel equipment (Russian Wikipedia). Not amphibious. Heavy: 44 to 50 t.

Weight, IMR-2 **44.3 t** combat weight

Weight, IMR-3M **49.5 t** combat weight with KMT-R3 plough

L × W × H **9.55 × 3.735 × 3.68 m**

Road speed, IMR-2 **50 km/h** maximum

Crew **2** (commander-operator, driver-mechanic)

Boom lifting capacity **2 t**

Boom reach **8.4 m** IMR-2M

Dozer capacity **200 to 250 m³/h** IMR-2 earthmoving

SEEN WITH UR-77 mine clearing vehicle (2 per mine clearing platoon); BAT-2 combat engineer vehicle (4 per brigade engineer battalion); KMT-R3 track-width mine plough (1 per IMR-3M)

VARIANTS IMR T-54/55 tank chassis; articulating dozer blade and telescoping crane.

UR-77 Meteorit YP-77 «Memeopum»

NATO: M1979

TR-A

Tracked mine-clearing vehicle firing rocket-towed explosive line charges

In service 1978

OPERATED BY engineer-sapper battalion, mine clearing platoon (organic engineer battalion of a maneuver brigade), 2 per platoon D 2016 M

INDICATES Doctrine places two per engineer mine clearing platoon, so one suggests engineers preparing a breach or route clearing. In Ukraine it has also been fired as improvised short-range artillery.

RECOGNISE Chassis Tracked hull derived from the 2S1 Gvozdika, itself built on MT-LB parts; driver and engine compartment front left on the 2S1.

Working equipment Turret-like superstructure with launch ramps at top rear of hull. RWW gives two launch ramps, TWZ two charges; JANESRG says three rockets.

Stowed Charges carried on board: hull holds two launchers and compartments for the line charges (UZP-77 or UZ-67).

NOT TO BE CONFUSED WITH 2S1 Gvozdika Same chassis; the 2S1 has a fully enclosed turret with a 122 mm howitzer, the UR-77 a launcher superstructure.

SOURCES ESPR GS18 IIS25 JANESRG MILUA2 RBTH RWW TW28 TW29 WEG44 WEG2015 WP-EN15 WP-EN47 WP-RU60 WWW14

MOBILITY Tracked, 2S1-derived, amphibious: WEG gives 5 km/h swimming, GlobalSecurity 4.5 km/h. Fording depth and ground pressure INA. Reload after firing takes 30 to 40 minutes.

Weight, combat **15.5 t**

L × W × H **8.4 × 2.8 × 3.1 m**

Road speed **60 km/h**

Road range **500 km** cruising

Range, UZP-77 line charge **200 to 500 m** not stated, charge launch distance, UZP-77; UZ-67 charge 200 to 350 m (WEG)

Crew **2**

Cleared lane length, UZP-77 **80 to 90 m** anti-tank minefield; UZ-67 charge 75 to 80 m

Cleared lane width **up to 6 m** one line charge

SEEN WITH UZP-77 / UZ-67 line charge (2 per vehicle); IMR-3M combat engineer vehicle (2 per mine clearing platoon); 2S1 Gvozdika-derived chassis (1 per vehicle)

ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer ИСАМ «Земледелец»

WH-AT

Remote minelaying by 122 mm rockets (ISDM) and tracked mechanical minelaying (GMZ-3) In service 2020

OPERATED BY engineer-sapper battalion, obstacle platoon (brigade engineer battalion per RWW; division engineer battalion per WEG), 3

GMZ-3 per obstacle platoon **D 2016 M**

engineer-sapper brigade **D 2024 L**

engineer-sapper regiment **D 2024 L**

INDICATES An ISDM points to an engineer brigade or army engineer regiment remote-mining subunit preparing minefields 5–15 km out. A group of three GMZ-3 suggests a brigade engineer obstacle platoon, likely a mobile obstacle detachment.

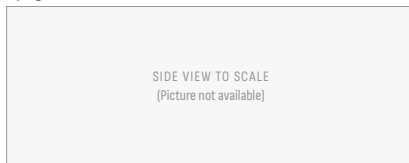
RECOGNISE Chassis ISDM: four-axle KamAZ-6560 8x8 truck; cab armoured per Za Rulem (WEG: armour none). The transloader uses the same chassis.

Chassis GMZ-3: tracked, on the SA-4 Ganef chassis; seven road wheels, front drive sprocket, rear idler, four return rollers. Driver front left, engine to his right.

Working equipment ISDM: launcher on rear deck carrying two box containers, each of 25 tubes in five rows of five. Transloader carries a small crane (blog source).

Working equipment GMZ-3: operator compartment at rear; mines leave through ports in the rear hull plate onto a conveyor and plough. PKT MG; three 81 mm smoke launchers each side.

NOT TO BE CONFUSED WITH BM-21 Grad p. 4 Ural 6x6 truck with 40 open tubes; ISDM is an 8x8 KamAZ with two 25-tube box containers. UMZ ZiL-131 6x6 truck with six firing modules of 30 tubes (180 tubes) scattering mines 30–60 m from the truck.



MOBILITY ISDM: 8x8 truck, not amphibious, fords 1.8 m per WEG. GMZ-3: tracked, not amphibious, 28.5 t; lays surface mines at up to 16 km/h, buries at 6 km/h.

Weight, GMZ-3 **28.5 t** combat

L × W × H **10.3 × 2.5 × 3.27 m**

Road speed, ISDM **90 km/h** maximum, KamAZ-6560 chassis

Road speed, GMZ-3 **60 km/h** maximum

Range, ISDM mine rocket **15 km** maximum, cassette rocket with AP or AT mines, claim

Minimum range, ISDM mine rocket **5 km** cassette rocket with AP or AT mines, claim

Laying speed, GMZ-3 **16 km/h surface; 6 km/h buried** maximum

Range, 762 mm PKT (GMZ-3) **2,000 m** max effective, day

Rocket calibre, ISDM **122 mm** claim

Rockets per launcher, ISDM **50 (2 containers x 25)**

SEEN WITH ISDM transport-loading vehicle (KamAZ-6560 8x8; small crane per blog source); KamAZ-6560 (1 per launcher)

SOURCES ARMINF ATP IISS25 IZV222 JANES2 M360 MPER2 ORYX OVC RWW TI2 WEG47 WEG2015 WP-EN51 WP-EN52 WWW15 ZR16 ZR24

R-149 command and staff vehicles P-149

WH-AT-A

Tactical command and staff vehicles with HF/VHF radios and data terminals (ESU TZ) In service 1990

OPERATED BY motor rifle regiment, R-149MA3  2022-09 M 
command brigade, R-149AKSh-1  2022-03 L 
motor rifle division, R-149AKSh-1  2022-03 L 

INDICATES An R-149MA1 or R-149AKSh-1 suggests a brigade, regiment, division or army control-brigade command post nearby. An R-149MA3 is meant for company to squad command and was lost with motor rifle and tank regiments.

RECOGNISE Cab R-149AKSh-1: KamAZ-5350 6x6 truck cab. BTR-based R-149MA1, MA3 and BMR: no cab; commander and driver sit at front of an 8x8 armoured hull.

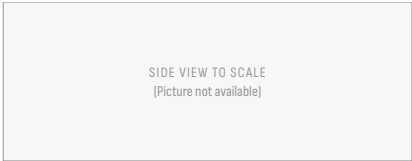
Body AKSh-1: van body split into electronics compartment (3 workstations) and officer compartment (5). MA1 and BMR: K1Sh1 hull, larger than BTR-80, unarmed turret.

Axles AKSh-1: three axles. BTR-based: four large tyred wheels a side; hatch/door between 2nd and 3rd wheels, upper part opens to right, lower folds down (BTR-80 base).

Details R-149MA3: WEG lists a 14.5 mm KPVT turret (BTR-80 sheet text); one lost MA3 was turretless. R-149BMR carries an ASh-4 telescopic mast, AZI NVIS HF and discone antennas.

NOT TO BE CONFUSED WITH BTR-80 APC R-149MA1 and BMR use the K1Sh1 hull, larger than BTR-80 with an unarmed turret; WEG lists the MA3 with the standard armed turret.

BMM-80 ambulance and 1V152 on K1Sh1 ^{p. 33} WEG 2015: the Kushetka-B/R-149BMR 'shares the same configuration as BMM medical vehicles and the 1V152/Kapustnik-B ACRV'.



SIDE VIEW TO SCALE
(Picture not available)

MOBILITY BTR-80-based versions swim (WEG). R-149AKSh-1 on KamAZ-5350 is not amphibious; fords 1.75 m (WEG).

Weight, R-149MA1	15,000-15,050 kg	maximum, claim
Weight, R-149BMR	13.6 t	combat
L × W × H	7.65 × 2.9 × 2.41 m	
Road speed, BTR-based versions	80 km/h	maximum

Range, VHF R-168-100UE-2, R-149MA1	25 km	maximum, on the move, vehicle antenna, claim
Range, VHF R-168-100UE-2 on mast, R-149MA1	55 km	maximum, halted, mast antenna, claim
Range, HF R-168-100KBE, R-149MA1	350 km	maximum, halted, mast antenna; 50 km (20 km moving) on ASh-4 whip, claim
Range, HF air-ground Prima, R-149MA1	450 km	maximum, HF to aircraft, claim
Range, VHF air-ground Prima, R-149MA1	120 km	maximum, VHF to aircraft, claim

SEEN WITH BTR-80 / K1Sh1 (GAZ-59032) chassis (1 per R-149MA1, MA3 or BMR)

SOURCES ACONF2 AR14 AR15 ARMYINF ATP ELSIG GAGA JANESRG KI2 MIL24 ORYX RUSARMY3 UNIAN3 WEG31 WEG2015 WP-EN32 WP-RU46 WP-RU47 WSPOT3 WSPOT4 WSPOT5 WSPOT6 WSPOT7 WSPOT8 WSPOT9 WSPOT10

R-145BM Chayka command and staff vehicle P-1455M «Чайка»

WH-AT-A

Turretless radio command and staff vehicle for mobile command posts In service 1960

OPERATED BY anti-tank battalion, ACV listed as 'BTR-60PU/BRDM-2'; battery placement unclear in OCR, 5 per battalion (BTR-60PU or BRDM-2)  1991 L 

INDICATES Built for radio links at mobile command posts, so it points to a command post or communications node. A Soviet-era source places it at tank, motor rifle and artillery regiment and formation HQs; the current echelon is not fixed.

RECOGNISE Cab BTR-60 hull (BTR-60PA base) with no turret; two compartments, command section forward and radio section aft.

Body Collapsible frame antenna on left side of roof; cylindrical box (TZK) on the roof; characteristic 'horns' on left and right upper edges of front armour.

Axles 8x8 wheeled, amphibious, individual torsion-bar suspension.

NOT TO BE CONFUSED WITH BTR-60PU-12 (9S482) air defence command vehicle ^{p. 35} Telescopic mast on right of front roof, generator on the turret-ring blanking plate, large stowage box on right of hull; turretless like R-145BM.

SOURCES ACONF ARINF FM32 IIS25 JANESRG ORYX RUSARMY2 UNIAN2 WEG30 WEG2015 WP-EN30 WP-EN31 WP-EN32 WP-RU43 WP-RU44 WP-RU45

MOBILITY Amphibious 8x8, light armour; swims (no fording depth given). Petrol engines.

Weight, combat	10.3 t
L × W × H	7.56 × 2.83 × 2.31 m
Road speed	80 km/h
Road range	500 km
Swimming speed	10 km/h

Range, R-130M zenith antenna	75 km	HF, halted or moving, symmetrical zenith-radiation antenna
Range, R-130M dipole	350 km	HF, halted only, symmetrical dipole
Range, R-123MT 4 m whip	20 km	VHF, halted or moving
Range, R-111 3.4 m whip	35 km	VHF, halted or moving
Crew	2	

VARIANTS BTR-60PUM Not stated in source. Russian late-2000s refit of BTR-60PU with new engines adapted from the BTR-80 series.

BMP-1KSh command and staff vehicle БМП-1КШ «Помок-2»

NATO: M1978

TR

Command and staff vehicle for regimental and battalion commanders In service 1972

OPERATED BY motor rifle regiment, regimental command post, 1 in regimental HQ of a BMP-equipped motor rifle regiment (FM 100-2-3 table, OCR scrambled) [D 1991 M]

tank regiment, regimental command post, INA (FM 100-2-3 table reads 0 for the tank regiment HQ) [D 1990 M]

motor rifle regiment, 501st signal company [O 1986 L]

INDICATES A BMP-1KSh marks a regimental command post; one source also gives one per battalion. Sources describe it for motor rifle and tank regiments. Oryx records 15 Russian losses 2022-26 but names no unit.

RECOGNISE Hull BMP-1 hull: pointed nose, almost horizontal ribbed glacis, driver front left, engine louvers roof right, two bulged rear doors. KSh keeps only two roof hatches.

Armament No gun or missile rail; turret fixed; one 7.62 mm PK fired from the rear-door port. Most firing ports blocked. Tube case for mast parts on right rear.

Turret Fixed turret cap carries the telescopic antenna mast (10 m per WEG and VALKA) in place of the gun; only two roof hatches behind the turret.

Roof and rear Box-shaped generator set centre rear of hull roof; four elevatable whip antennas at rear, two each side.

Suspension BMP-1 running gear: six road wheels each side, front sprocket, rear idler, three return rollers, skirts over upper track.

NOT TO BE CONFUSED WITH BMP-1K Keeps the 73 mm gun and rotating turret; K1/K2 have a semi-telescopic antenna, K3 a 10 m mast (WEG). KSh has a fixed turret with the mast in place of the gun.

PRP-3 artillery reconnaissance vehicle Enlarged two-man turret set to the rear with a folding rectangular radar antenna.

SOURCES FM23 GS7 ISS25 JANESRG ORYX VALKA2 VALKACZ WEG8 WEG2015 WP-RU11 WP-RU12

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Amphibious, swims. Radios run on the move from an engine-driven generator; a portable generator set is used at halts (RUWK2).

Weight, combat	13 t
L × W × H	6.74 × 2.94 × 2.15 m
Road speed	65 km/h
Road range	550 km
Swimming speed	7 km/h
Fording	Amphibious, swims
Ground pressure	0.60 kg/cm2

Crew	3 crew plus 4 staff officers
Armament	1 x 7.62 mm PK, 2,000 rounds carried machine gun, fired through the rear-door port (RUWK2); WEG lists it as PKT

Antenna mast height	10 m AMU Hawkeye telescopic mast, raised
Radios	2 × R-111, R-123MT, R-130M
Navigation	TNA-3 gyroscopic navigation device

VARIANTS BMP-1KShM Extra power sources on top of the troop compartment (VALKA). MP-31 INA.

1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16) 1В12 «Машина-С»

NATO: M1974

TR-A

Tracked command, observation and fire direction vehicles, SP howitzer battalion In service 1972

OPERATED BY self-propelled artillery battalion, each battery: one 1V13 (battery FDC) and one 1V14 (battery COP); battalion HQ: one 1V15 (battalion COP) and one 1V16 (battalion FDC), 8 per battalion: 3 1V13, 3 1V14, 1 1V15, 1 1V16 [D 2016 H]

self-propelled artillery regiment, howitzer battalion with 2S19M1, at least 3 1V13 and 3 1V14 seen with one battalion [R 2021-09 L]

INDICATES A 1V13 marks a battery firing position, where it stands behind the guns. A 1V14 or 1V15 marks a battery or battalion command observation post, usually with the supported manoeuvre unit. A full set of eight serves one SP howitzer battalion.

RECOGNISE Hull High box-like MT-LBu hull with steep glacis; straight vertical rear with a single door. Three or four antennas on the hull roof.

Armament 1V13 and 1V16: 12.7 mm DShKMT machine gun on the turret. 1V14 and 1V15: 7.62 mm PKMB machine gun on a mount on the turret roof.

Turret Flat, round rotating turret shaped as a truncated cone on the rear half of the hull. On 1V14 and 1V15 it carries a laser rangefinder and optical sights.

Suspension Seven road wheels each side, no return rollers. The MT-LB it derives from has one road wheel fewer.

NOT TO BE CONFUSED WITH MT-LB Six road wheels, lower hull and a small machine-gun turret at the front over the driver's compartment instead of a turret at the rear.

ARK-1M Rys p. 19 Same MT-LBu chassis but carries a parabolic radar antenna.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Amphibious, propelled by its tracks in water.

Weight	14.8 t 1V13 and 1V14, combat weight
L × W × H	7.475 × 2.85 × 2.425 m hull
Road speed	62 km/h maximum, claim
Road range	500 km maximum, claim

Detection range, 1V14 optical reconnaissance	up to 10,000 m maximum, day
Detection range, 1V14 optical reconnaissance (night)	up to 2,500 m maximum, night

Range, laser target illumination (1V14)	up to 7,000 m maximum, for laser-guided (corrected) munitions
Radio range	20-25 km stated, 1V13 and 1V14, vehicle radios
Radio range, R-111	220-250 km stated, 1V15 and 1V16, vehicle radio

SEEN WITH 1V13 battery fire direction vehicle (3 per battalion set)

SOURCES AG1 ARMF13 ARMF14 ARMF15 ARMF16 FAS1 JANESRG JREW02 MILINF ORYX RWW VRF1 WEG2015 WP-RU1 not in WEG

PRP-4A Argus / PRP-4M Deytery ПРП-4А «Арзус»

TR-A

Mobile artillery reconnaissance post on BMP-1 chassis (target location, fire adjustment) In service 1988

OPERATED BY motor rifle brigade, fire control and artillery reconnaissance battery, artillery spotting platoon D 2016 M

INDICATES *Doctrinally a brigade- or regiment-level artillery reconnaissance asset feeding a fire direction centre. In defence it sits behind first-echelon companies; in offence it moves with forward reconnaissance detachments up to 20 km ahead of the main body.*

RECOGNISE **Hull** BMP-1 infantry fighting vehicle chassis, amphibious; 4 crew.

Armament One 7.62 mm PKTM machine gun and four smoke dischargers only. PRP-4M is reported with a fake gun mantlet and offset barrel, resembling a BMP-2.

Turret PRP-4/4M/4MU: retractable radar; 4M antenna mount on roof behind radar hatch; 4MU left optical flap hinged at top. Argus: WEG says mast-mounted radar.

NOT TO BE CONFUSED WITH BMP-2 PRP-4M with a fake gun mantlet has no real 30 mm rg, only a 7.62 mm machine gun.

MOBILITY Amphibious on BMP-1 chassis; swim speed 7 km/h (WEG). Armour reported to stop 12.7 mm rounds all round (WEG).

Weight, combat	13.8 t
L × W × H	6.74 × 2.94 × 2.15 m
Road speed	65 km/h maximum, paved road
Road range	550 km paved road

Detection range, radar (moving tank-type target)	12 km up to, moving tank-type target
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Detection range, radar (person)	7 km up to, single soldier
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Detection range, stationary tank, engine off (sensor not stated)	2 km no less than, stationary tank-type target, engine not running; ARMYR and RWW name no sensor, WEG lists it under radar, claim
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SEEN WITH 1V16M battalion fire direction vehicle; 1RL232-2M SNAR-10M1 battlefield surveillance radar

SOURCES AR11 ATP GS7 JANESRG JREW02 MIL23 ORYX RWW WEG27 WEG2015 WP-EN28 WP-RU35

Polyana-D4 9C52 «Поляна-Д4»

WH-AT

Automated command post of an Army air defence missile brigade [S-300V, Buk, Tor] In service 1986

OPERATED BY air defence missile brigade, brigade command post (combat control post MP06 and command-staff vehicle MP02)

D 2021-12 H

air defence missile brigade, brigade command post; vehicle captioned as 9S52M Polyana-D4M O 2010-12 M

combined arms army, army air defence and aviation command post (SKP PVO i aviatsii) R 2021-12 M

INDICATES *Indicates the command post of an Army air defence missile brigade [S-300V/V4 or Buk], or an army or district AD command post. Up to four SAM battalions, a radar post and a point-defence CP report to a brigade Polyana.*

RECOGNISE **Cab** MP06 combat control post on BAZ-6950 8x8: forward fibreglass two-door cab, three reverse-raked windscreens, oval vent hatch in centre of flat front wall.

Body Engine compartment behind cab. SKN-6950 'Rodinka' metal-frame van body with wide chamfered sides was built for this chassis for C2 complexes; MP06 body not described.

Axles Four axles in two widely spaced pairs [base 1.5 × 3.8 × 1.5 m]; front pair steers. Large single tyres 1300x530-533.

Details Complex adds MP02 command-staff van on Ural-375 or Ural-43203 6x6 with KP4 trailer, MP45 maintenance van and two generator trucks (KamAZ-4310 on D4M).

NOT TO BE CONFUSED WITH 9S457 S-300V battalion command post p. 13 Tracked MT-T chassis shared with other S-300V vehicles, with telescoping datalink mast; Polyana vehicles are wheeled trucks. Baikal-1 air defence command post Carried on MAZ-7910 8x8 with integrated FL-94 telescoping mast and 5S15 turbine generator; Polyana MP06 is on BAZ-6950 with separate generator trucks.

SOURCES APA4 APA9 ATP IJSS25 KZ2019 KZ2021 NASHAVT RDP4 RUSAR1 RUSARMM VPK6 VRF7 WEG26 WP-EN26 WP-EN27 WP-RU34

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Wheeled truck complex with generator trucks; needs a road or firm track. Deploys or packs up in about 20 to 35 min. Fording 1.2 m (Ural) and 1.3 m (BAZ) chassis figures.

Weight, gross, Ural-375D chassis (MP02)	13.2 t chassis figure; MP02 as fitted INA
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Weight, gross, BAZ-6950 chassis (MP06)	29.4 t chassis figure; MP06 as fitted INA
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L × W × H	7.36 × 2.67 × 2.68 m cargo truck
Road speed, Ural-375D	75 km/h maximum

SAM battalions controlled	up to 4 9S52 and 9S52M; S-300V (9S457) or Buk (9S470) battalion CPs
Air targets tracked simultaneously	up to 250 9S52M1

Targets received	up to 1,000 per 10 s 9S52M1, claim
Linked objects, total	up to 14 9S52M1; AEW aircraft, subordinate SAM unit CPs, higher and cooperating CPs, claim

Superior CP interfaces

SEEN WITH MP02 command-staff vehicle (KShM) (1)

PU-12M7 9C482M7 PY-12M7

WH-AT-A

Air defence battery command post: fuses radar data and cues SHORAD fire units In service 1977

OPERATED BY motor rifle regiment, air defence missile-artillery battery; also regimental air defence chief's post (ARMF: PU-12 family 1977-95; ORUZH: undated PU-12M7 sheet), 1 per battery (PU-12 family)

[D 1995 M](#)

tank regiment, air defence missile-artillery battery; also regimental air defence chief's post (ARMF: PU-12 family 1977-95; ORUZH: undated PU-12M7 sheet), 1 per battery (PU-12 family) [D 1995 M](#)

motor rifle regiment, regiment on BTR, Soviet 1980s establishment, 3 PU-12 per regiment [D 1989 L](#)

INDICATES A PU-12-series or Ranzhir vehicle marks the command post of a short-range air defence battery (Strela-10, Tunguska, Tor, Osa, Shilka or Iglu teams, up to 4-f fire units) or a regimental air defence chief's post nearby.

RECOGNISE **Body** BTR-80 based K1Sh1 hull: raised centre section and extra roof hatches; no turret (from photograph).

Mast Telescopic mast at hull rear, raised for work; dome-shaped head on top and a box halfway up (from photograph). WEG: passive antennas mid-mast, day/night camera on top.

Axles Four axles, 8x8, front two axles steer; single waterjet at hull rear; amphibious (BTR-80 base).

Details Roof-mounted box sensor with two lenses on a pan-tilt head (from photograph); BTR-80 side door between second and third axles.

NOT TO BE CONFUSED WITH PPRU-1M / PPRU-M1 Also on MT-LBu but carries its own 9S80 Dog Ear radar; 9S737 Ranzhir takes data from external radars such as P-19 or 9S18M1.

BTR-60PU-12 p. 32 BTR-60 hull with large stowage box on right side, generator and telescopic mast; PU-12M6/M7 use the BTR-80 hull.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY PU-12M7 is amphibious on the BTR-80 base. 9S737 and 9S737M are tracked; 9S737M fords 1.0 m.

Weight, PU-12M7

13.6 t

Weight, 9S737 (WEG figure)

25000 kg WEG lists a Tata LTK 2518 truck chassis; see open_questions

Weight, 9S737M

31 t GM-5965.05 tracked chassis; RUSA2 gives 'not more than 31,000 kg'

L × W × H

7.7 × 2.9 × 2.41 m

Coordinates measurement envelope, PU-12M7 (connected radars)

up to 100 km maximum, coordinates measurement envelope from connected radars; PU-12M7 has no radar of its own, claim

Data link range, PU-12M7 by radio

up to 40 km maximum, radio link to higher and subordinate posts, claim

Data link range, PU-12M7 by field cable

up to 15 km maximum, field telephone cable, claim

Data link range, 9S737 to higher post

up to 30 km maximum, radio, claim

Data link range, 9S737 to fire units

up to 5 km maximum, radio, claim

SOURCES APA4 ARMF JANESRG ORUZH ORYX ROE2 RUSA1 RUSA2 VPK7 WEG29 WMC12 WMC13 WP-RU39 WP-RU40 WP-RU41 WP-RU42 WP-RU43

Barnaul-T «Барнаул-Т»

TR-A

Automated command and control set for tactical short-range air defence, battery to brigade In service 2015

OPERATED BY motor rifle brigade, brigade AD subunit; BO quotes a 'missile division' (battalion) commander [O 2021-12 M](#)

airborne division, air defence missile regiment, one full divisional set [O 2018-06 M](#)

airborne division, air defence missile regiment [O 2018 L](#)

INDICATES A 9S932-1 marks a short-range air defence battery command post. Strela-10, Tor, Osa, Tunguska or MANPADS fire units it controls are likely nearby; in VDV units the kit belongs to the division's AD regiment.

RECOGNISE **Antenna** 1L122-1E rectangular L-band antenna array, 1.2 × 0.8 m (maker's figure), on the top rear of the hull; folded down in transit.

Mast INA: no text source describes a mast on the 9S932-1.

Chassis Modified MT-LBu tracked amphibious carrier (9S932-1). Airborne version on BTR-MD/BTR-MDM tracked carrier.

Complex Tracked 9S932-1 at battery, 9S931 on MT-LBu, 9S931-1 on KamAZ 6x6 with trailer at brigade; 9S933 and 9S935 are portable MANPADS kits.

NOT TO BE CONFUSED WITH PPRU-1M / PPRU-M1 Also an MT-LBu AD battery command post, but carries a 9S80-series X-band radar, not the 1.2 × 0.8 m 1L122 L-band array.

1L122-2E Self-propelled Garmon radar on MT-LBu with its antenna on a lifting boom; antenna 1.9 × 1.8 m per TW, but GS gives 1.2 × 0.8 m for both versions.

9S737 Ranzhir p. 35 Older AD battery command post on MT-LBu-derived chassis; check for the 1L122 array on the rear hull.

SIDE VIEW TO SCALE
(Picture not available)

MOBILITY Amphibious tracked MT-LBu; swims 5 km/h maximum. The BTR-MDM (VDV) version is parachute-droppable.

Weight, combat

11,000 kg MT-LBu chassis figure; 9S932-1 weight INA

L × W × H

6.45 × 2.86 × 1.86 m MT-LBu chassis

Road speed

60 km/h maximum, maximum

Road range

500 km cruising, cruising

Detection range, 1L122-1E

40 km maximum (1-40 km), aircraft, helicopters, cruise missiles, UAVs at up to 700 m/s; export 1L122-1E brochure figure, claim

Minimum detection range, 1L122-1E

1 km export 1L122-1E, claim

Detection altitude, 1L122-1E

10 km maximum, export 1L122-1E; 'at least 10 km', claim

Data link range, 9S932-1

20 km communication range, stationary; 10 km on the move (9S933: 3 km / 1 km)

Radar band, 1L122-1E

L-band claim

Elevation coverage, 1L122-1E

-5 to +45 deg export 1L122-1E, claim

SEEN WITH 1L122-1E Garmon (1 per 9S932-1); 9S931-1 MP-K

SOURCES APA4 BO DB GS6 KZ MIL4 ORYX SDU TOPWAR3 TW22 VALKA WEG7 WMC2 WP-RU10

Table 1. Master list of designations

Every system, vehicle, component and munition in this guide's lexicon, 809 in all, sorted by designation (digits sort as numbers). NATO reporting names and codes and US designations are set in **BLUE CAPITALS**, for example **STONE**, **SS-26** or **M1974**. NATO designations and reporting names can also be looked up in Table 1b. **(UA)** marks a type reported in Russian use in Ukraine since 2014. A **bold** page number leads to a full entry; a *grey italic* one to the entry of the system it belongs to. Look up other names and Cyrillic forms in Tables 1b and 1c.

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
1B44		1B44	meteorological station - of 9K58 Smerch	6
1B75 1B75E	Penicillin	«Пенициллин»	acoustic-thermal locating system (UA)	20
1K113 3F5, 3VF4	Smelchak	1K113 «Смелычак», 3Ф5	guided mortar bomb - of 2S4 Tyulpan (UA)	9
1K148	Yastreb-AV	1K148 «Ястреб-АВ»	counter-battery radar (UA)	
1L13-3 1L13	Nebo-SV BOX SPRING	1Л13, «Небо-СВ»	air surveillance radar	
1L119	Nebo-SVU	«Небо-СВУ» 1Л119	air surveillance radar (UA)	
1L122	Garmon, Garmony	1Л122 «Гармонь»	portable air surveillance radar (UA)	
1L125	Niobium-SV		multifunction radar (UA)	
1L219 Zoopark-1	Zoopark-1	1Л219 «Зоопарк-1», 1Л219М	counter-battery radar (UA)	19
1L222	Avtobaza-M	«Автомобаз»	ELINT system	
1L259		1Л259	counter-battery radar (UA)	19
1L260 Zoopark-1M	Zoopark-1M	1Л260, «Зоопарк-1М»	counter-battery radar (UA)	19
1L261			counter-battery radar (UA)	19
1L262E SPR-2M	Rtut-BM, Mercury-BM	1Л262З «Ртуть-БМ»	proximity-fuze jammer (UA)	
1L265		1Л265	ELINT system	22
1L266		1Л266	EW command post	22
1L267	Moskva-1	«Москва-1»	EW command post (UA)	22
1L269 Krasukha-20	Krasukha-2	1Л269 «Красуха-2»	radar jammer (UA)	21
1L271	Aistenok, Aistyonok	1Л271 «Аистенок»	portable counter-battery radar (UA)	20
1L277	Sobolyatnik		portable ground surveillance radar (UA)	
1RL144M 1RL144M1, 1RL144, 1RL138	HOT SHOT	1РЛ144М1	fire-control radar (UA)	17
1RL239 ARK-1, ARK-1M	Rys SMALL FRED (disputed)	1РЛ239 / АРК-1 / «Рысь»	counter-battery radar (UA)	19
1RL257	Krasukha-4, Krasukha-4S	1РЛ257 «Красуха-4»	radar jammer (UA)	21
1T12-2M		1Т12-2М	topographic survey vehicle - of 9K58 Smerch	6
1V13 1V13M, M1974/1		1В13	artillery battery fire control centre (UA)	33
1V14 1V15, 1V15M, 1V16			artillery command vehicle (UA)	33
1V18 1V19, M1979-2B	Klyon-1, Klen-1, Klen-2		artillery command vehicle (UA)	
1V110			artillery command vehicle - of 9K51 Grad <i>legacy</i> (UA)	4
1V119	Spektr	1В119	artillery command vehicle (UA)	
1V126 1V126M, 1V152, 1V153, 1V152M	Kapustnik-B, Kapustnik-BM	1В126 «Капустник-Б»	artillery fire control system - of 9K59 Prima	
1V152		1В152	artillery command vehicle	
1V1003			artillery command vehicle (UA)	
2A36 M1976	Giatsint-B	2А36 «Гиацинт-Б»	152 mm towed gun (UA)	10
2A45M 2A45MR	Sprut-B		125 mm towed anti-tank gun <i>legacy</i>	
2A65 M1987	Msta-B	2А65 «Мста-Б»	152 mm towed howitzer (UA)	10
2B9 2B9M	Vasilek	2В9 «Василёк»	82 mm automatic gun-mortar (UA)	
2B11		2В11	120 mm mortar - of 2S12 Sani (UA)	
2B14	Podnos	2В14 «Поднос»	82 mm mortar	
2B16	Nona-K	2В16 «Нона-К»	120 mm towed gun-mortar (UA)	
2B17M	Tornado-G	2В17М	122 mm multiple rocket launcher (UA)	4
2B26	Grad-K	2В26	122 mm multiple rocket launcher - of 9K51 Grad (UA)	4
2F77 2F77M		2Ф77, 2Ф77М	transloader - of 2K22 Tunguska (UA)	17
2K22 2K22M, 2K22M1, SA-19	Tunguska, Tunguska-M1 GRISON	2К22, 2К22М «Тунгуска-М»	gun-missile air defence system (UA)	17
2S1 M1974	Gvozdika	2С1 «Гвоздика»	122 mm self-propelled howitzer (UA)	33
2S3 Object 303, M1973	Akatsiya	2С3 «Акация»	152 mm self-propelled howitzer (UA)	
2S3M 2S3M1, 2S3M2	Akatsiya	2С3М	152 mm self-propelled howitzer - of 2S3 Akatsiya (UA)	
2S4 Object 305, M1975	Tyulpan	2С4 «Тюльпан»	240 mm self-propelled mortar (UA)	9
2S5 Object 307, M1981	Giatsint-S	2С5 «Гиацинт-С»	152 mm self-propelled gun (UA)	7

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
2S5M		2C5M	152 mm self-propelled gun · of 2S5 Gatsint-S <i>legacy</i>	7
2S6 2S6M, 2S6M1		2C6, 2C6M	gun-missile air defence vehicle UA	17
2S7 Object 216, M1975	Pion	2C7 «Пион»	203 mm self-propelled gun UA	
2S7M	Malva	2C7M «Малва»	203 mm self-propelled gun UA	9
2S9 2S9-1	Nona-S	2C9 «Нона-С»	120 mm self-propelled gun-mortar UA	
2S12 2S12A	Sani	2C12 «Сани», 2C12A	120 mm mortar UA	
2S19 Object 316, M1990	Msta-S FARM	2C19 «Мсма-С»	152 mm self-propelled howitzer UA	
2S19M1	Msta-S	2C19M1	152 mm self-propelled howitzer UA	8
2S19M2 2S33	Msta-SM	2C19M2, 2C33 «Мсма-СМ»	152 mm self-propelled howitzer UA	8
2S23	Nona-SVK	2C23 «Нона-СВК»	120 mm self-propelled gun-mortar UA	
2S25	Sprut-SD	2C25	125 mm self-propelled anti-tank gun	
2S31	Vena	2C31 «Вена»	120 mm self-propelled gun-mortar UA	
2S34 2S1M	Khosta, Chosta	2C34 «Хоста»	120 mm self-propelled gun-mortar UA	
2S35	Koalitsiya-SV	2C35 «Коалиция-СВ»	152 mm self-propelled howitzer UA	8
2S38 ZAK-57	Derivatsiya-PVO	2C38, 3AK-57	57 mm self-propelled anti-aircraft gun <i>trial</i> UA	
2S40	Floks	2C40, «Флокс»	120 mm self-propelled gun-mortar UA	
2S43	Malva	2C43 «Мальва»	152 mm wheeled self-propelled howitzer UA	8
3BK13		3BK13	artillery round · of 2S1 Gvozdika <i>legacy</i>	33
3BK19		3BK19	artillery round	
3F2 3VF2		3Ф2	mortar bomb · of 2S4 Tyulpan	9
3K55 K-300P, K300P, SSC-5	Bastion, Bastion-P STOOGEE	3K55, Бастуион	coastal missile system UA	
3K60 SSC-6	Bal, Bal-E SENIGHT	3K60, Бал	coastal missile system UA	
3M14 3M14T, 3M14K, SS-N-30A, SS-N-27	Kalibr SAGARIS	Калибр, 3М-14	cruise missile UA	
3M16		3M16	122 mm cluster rocket · of 9K51 Grad <i>legacy</i>	4
3-O-8 3VO11		3-O-8	mortar bomb · of 2S4 Tyulpan UA	9
3-O-13 3VO13, 3VO14		3-O-13	artillery round UA	
3-O-23 3VO28, 3VO29, 3VO30		3-O-23	artillery round	
3OF24		3OF24	artillery round · of 2S1 Gvozdika	33
3OF25 3VOF32, 3VOF33		3OF25	artillery round <i>legacy</i>	
3OF29 3VOF39, 3VOF40		3OF29	artillery round · of 2S5 Gatsint-S	7
3OF38 3VOF63, 3VOF66	Santimetr	3OF38	guided artillery round	
3OF39 3VOF64, 3VOF93, 2K25	Krasnopol	3OF39, «Краснополь»	guided artillery round UA	
3OF39M 3VOF64M, 2K25M	Krasnopol-M1	3OF39M	guided artillery round · of 3OF39 Krasnopol UA	
3OF43 3VOF34, 3VOF42		3OF43	artillery round · of 2S7 Pion	
3OF44 3VOF35		3OF44	artillery round · of 2S7 Pion	
3OF45 3OF45-1, 3VOF58, 3VOF72, 3VOF73		3OF45	artillery round · of 2S19 Msta-S	
3OF49 3VOF49		3OF49	artillery round	
3OF50		3OF50	artillery round	
3OF56		3OF56	artillery round · of 2S1 Gvozdika	33
3OF59 3VOF86, 3VOF87		3OF59	artillery round · of 2S5 Gatsint-S	7
3OF61 3VOF91		3OF61	artillery round	
3OF64 3VOF96, 3VOF97, 3VOF98		3OF64	artillery round	
3OF69 Kitolov-2	Kitolov-2	«Кумолов-2»	guided mortar bomb UA	
3OF69M Kitolov-2M	Kitolov-2M	«Кумолов-2М»	guided mortar bomb · of 3OF69 Kitolov-2 UA	
3OF75	Santimetr-M	3OF75	guided artillery round	
3RB30 3VRB36, 3VRB37, 3VRB38		3PE30	communications jammer shell	
3S55 K-340P		K-340П	missile transporter erector launcher · of 3K55 Bastion UA	
5N63S 3ON6	FLAP LID B	5H63C	fire-control radar · of S-300PS S-300PS <i>legacy</i> UA	
5N84A Oborona-14	Oborona-14 TALL KING C	5H84A «Оборона»	early warning radar UA	
5P85SD 5P85SM, 5P85S, 5P85D		5П85С, 5П85Д	SAM launcher · of S-300PS S-300PS <i>legacy</i> UA	
5P85SM2-01		5П85СМ2-01	SAM launcher UA	12
5P85T2		5П85Т2	SAM launcher UA	12
5V55R 5V55K, 5V55RM, V-500R		5Б55Р, 5Б55К	surface-to-air missile · of S-300PS S-300PS <i>legacy</i> UA	
6P50 6P49, 6P57, 6P58, 6P59	Kord	Kopg	12.7 mm heavy machine gun UA	

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
9A33B 9A33BM3		9A33B	SAM launcher - of 9K33 Osa UA	17
9A35		9A35	SAM launcher - of 9K35 Strela-10 UA	17
9A39 9A39M1		9A39, 9A39M1	SAM loader-launcher - of 9K37 Buk stored UA	14
9A39M1-2			SAM loader-launcher - of 9K37M1-2 Buk-M1-2 UA	14
9A51	Prima	9A51	122 mm multiple rocket launcher - of 9K59 Prima legacy	
9A52 BM-30	Smerch	9A52	300 mm multiple rocket launcher legacy UA	6
9A52-2 BM-30	Smerch-M	9A52-2	300 mm multiple rocket launcher UA	6
9A52-2T		9A52-2T	300 mm multiple rocket launcher - of 9K58 Smerch export	6
9A52-4	Kama	9A52-4	300 mm multiple rocket launcher trial	6
9A53	Uragan-1M	9A53	220/300 mm multiple rocket launcher UA	5
9A54 9A53-S	Tornado-S	9A54	300 mm multiple rocket launcher UA	6
9A82		9A82	SAM launcher - of S-300V S-300V	13
9A83 9A83-1, 9A83M2		9A83	SAM launcher - of S-300V S-300V UA	13
9A83M2			SAM launcher - of S-300V4 S-300V4 UA	13
9A84		9A84	SAM loader-launcher - of S-300V S-300V UA	13
9A84-2			SAM loader-launcher - of S-300V4 S-300V4 UA	13
9A85 9A85-2		9A85	SAM loader-launcher - of S-300V S-300V UA	13
9A85-2			transloader - of S-300V4 S-300V4 UA	13
9A310 9A310M1, 9A38		9A310, 9A38	SAM launcher - of 9K37 Buk stored UA	14
9A310M1-2		9A310M1-2	SAM launcher - of 9K37M1-2 Buk-M1-2 UA	14
9A316		9A316	SAM loader-launcher - of 9K317 Buk-M2 UA	14
9A317		9A317	SAM launcher - of 9K317 Buk-M2 UA	14
9A317M		9A317M	SAM launcher - of 9K317M Buk-M3 UA	14
9A330		9A330	SAM launcher legacy UA	15
9A331 9A331-1		9A331, 9A331-1	SAM launcher UA	15
9A331M 9A331M2		9A331M2	SAM launcher UA	15
9A4172 9K121, AT-16	Vikhr SCALLION		air-launched anti-tank guided missile UA	
9K11 9M14, AT-3	Malyutka, Malyutka-2 SAGGER	9K11, 9M14	anti-tank guided missile system legacy	
9K32 9K32M, SA-7	Strela-2 GRAIL	9K32, «Смерелa-2»	MANPADS legacy	
9K33 9K33M2, 9K33M3, SA-8	Osa, Osa-AK, Osa-AKM GECKO	9K33, 9K33M2	short-range SAM system UA	17
9K34 SA-14	Strela-3 GREMLIN	9K34, «Смерелa-3»	MANPADS	
9K35 9K35M3, SA-13	Strela-10, Strela-10M3 GOPHER	9K35 «Смерелa-10», 9K35M3 «Смерелa-10M3»	short-range SAM system UA	17
9K37 9K37M1, SA-11	Buk, Buk-M1 GADFLY	9K37, 9K37M1 «Бук-M1»	medium-range SAM system stored UA	14
9K37M1-2 SA-11	Buk-M1-2 GADFLY	9K37M1-2 «Бук-M1-2»	medium-range SAM system UA	14
9K38 SA-18	Igla GROUSE	9K38, «Игла»	MANPADS UA	
9K51 M-21, M1964	Grad	9K51 «Град»	122 mm multiple rocket launcher system UA	4
9K51M	Tornado-G	9K51M «Торнадо-Г»	122 mm multiple rocket launcher system UA	4
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BMD-4M		БМД-4М	airborne fighting vehicle · of BMD-4 (UA)	
BMK-130M BMK-150			towing and motor boat (UA)	
BMK-460		БМК-460	towing and motor boat (UA)	
BMK-MO			towing and motor boat (UA)	
BMK-MT			towing and motor boat (UA)	
BMO-T Object 564		БМО-Т, Объект 564	armoured personnel carrier (UA)	
BMP-1 Object 765, BMP Model 1970		БМП-1, Объект 765	infantry fighting vehicle (UA)	
BMP-1AM	Basurmanin	БМП-1АМ, Басурманин	infantry fighting vehicle · of BMP-1 (UA)	
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BMP-1U	Shkval	БМП-1У, Шквал	infantry fighting vehicle · of BMP-1 legacy (UA)	
BMP-2 Object 675		БМП-2, Объект 675	infantry fighting vehicle (UA)	
BMP-2D		БМП-2Д	infantry fighting vehicle · of BMP-2 (UA)	
BMP-2M	Berezhok	БМП-2М, Бережок	infantry fighting vehicle · of BMP-2 (UA)	
BMP-3 Object 688		БМП-3, Объект 688	infantry fighting vehicle (UA)	
BMP-3F		БМП-3Ф	infantry fighting vehicle · of BMP-3	
BMP-3M		БМП-3М	infantry fighting vehicle · of BMP-3	
BMPT	Terminator	БМПТ, Терминатор	tank support fighting vehicle (UA)	
BMPT-72 Object 183	Terminator-2, Terminator	БМПТ-72, Терминатор-2	tank support fighting vehicle · of BMPT Terminator (UA)	
BMR-3M BMR-3MA, Object 197	Vepr	БМР-3М, Вепрь	mine-clearing vehicle (UA)	
BRDM-2 GAZ-41-08		БРДМ-2, ГАЗ-41-08	armoured reconnaissance vehicle (UA)	
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BREM-4	BREM-Ch		armoured recovery vehicle UA	
BREM-64			armoured recovery vehicle	
BREM-D	BREhM-D	БРЭМ-Д	armoured recovery vehicle UA	
BREM-K		БРЭМ-К	armoured recovery vehicle UA	
BREM-L	Beglyanka, Beglianka	БРЭМ-Л	armoured recovery vehicle UA	
BRM-1K BRM-1KM, Object 676, BMP M1976/2		БРМ-1К, БРМ-1	armoured reconnaissance vehicle UA	
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BTR-80A		БТР-80А	armoured personnel carrier - of BTR-80	
BTR-82		БТР-82	armoured personnel carrier - of BTR-80 UA	
BTR-82A BTR-82AM		БТР-82А, БТР-82АМ	armoured personnel carrier - of BTR-80 UA	
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BTR-90 GAZ-5923	Rostok	БТР-90, ГАЗ-5923	armoured personnel carrier UA	
BTR-D		БТР-Д	armoured personnel carrier UA	
BTR-MDM	Rakushka	БТР-МДМ, Пакушка	armoured personnel carrier UA	
BTR-T			heavy armoured personnel carrier legacy	
BTS-4A		БТС-4А	armoured recovery vehicle UA	
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D-20		Д-20	152 mm towed gun-howitzer UA	
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Eleron T-16			reconnaissance UAV UA	
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Fateh-360	Fath-360		short-range ballistic missile	
Forpost	Forpost, Searcher II	Форпост	reconnaissance UAV UA	
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GAZ-3307		ГАЗ-3307	cargo truck 4x2 UA	
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GAZ-3344-20	Aleut		articulated tracked carrier UA	
GAZ-39371	Vodnik		armoured utility vehicle UA	
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Geran-2 Shahed-136, Shahed 136	Geran-2	Герань-2	one-way attack drone UA	
Geran-3	Geran-3		one-way attack drone <i>trial</i> UA	
Gerbera	Gerbera	Гербера	one-way attack drone UA	
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Groza.04.K	Groza-04K	Гроза.04.K	UAV jammer UA	
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KamAZ-4386	Typhoon-VDV		mine-resistant ambush-protected vehicle UA	
KamAZ-5350	Mustang	КАМАЗ-5350	cargo truck 6x6 UA	
KamAZ-6350		КАМАЗ-6350	cargo truck 8x8 UA	
KamAZ-6560		КАМАЗ-6560	heavy truck chassis 8x8	
KamAZ-43269 BPM-97	Vystrel	КАМАЗ-43269, БПМ-97	armoured car UA	
KamAZ-53501		КАМАЗ-53501	cargo truck 6x6 UA	
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KamAZ-53949	Linza	КАМАЗ-53949, Линза	armoured ambulance UA	
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Kapustnik-BM	Kapustnik-BM	Капустник-БМ	artillery fire-control system · of 9K51M Tornado-G	4
Kh-22 AS-4	Burya KITCHEN	Х-22 «Буря»	air-launched anti-ship missile UA	
Kh-32		Х-32	air-launched cruise missile UA	
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Kh-59 Kh-59M, AS-13, AS-18	Ovod KINGBOLT, KAZOO	Х-59 «Овод»	air-launched cruise missile UA	
Kh-69		Х-69	air-launched cruise missile UA	
Kh-101 AS-23A	KODIAK	Х-101	air-launched cruise missile UA	
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KMT-7			mine roller	
KMT-8			mine plough	
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Korsar	Korsar	Корсар	reconnaissance UAV UA	
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KT-L			recovery vehicle UA	
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Kurganets-25	Kurganets-25	Курганец-25	infantry fighting vehicle <i>trial</i>	
KZKT-7428	Rusich		tank transporter <i>legacy</i>	
Ladoga	Ladoga, VTS Ladoga		heavy command and reconnaissance vehicle UA	
Lancet launcher	Lancet launcher		UAV control station - of Izdeliye 51 Lancet UA	
LPD-801			UAV jammer UA	
M-30 M-1938		M-30	122 mm towed howitzer <i>stored</i> UA	
M-46		M-46	130 mm towed gun UA	
M-160 M1949, 52-M-853		M-160	160 mm mortar <i>stored</i>	
M-1938 PM-38		ПМ-38	120 mm mortar <i>stored</i>	
M1977			armoured recovery vehicle	5
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MAZ-63172			tractor truck UA	
MAZ-74101	Oplot-T		tractor truck UA	
MAZ-531605			cargo truck UA	
MDK-3		МДК-3	trench digger UA	
Merlin-VR	Merlin-VR		reconnaissance UAV UA	
MKTK-1	Plavsk	«Плавск»	radio monitoring station	
MKTK-1A	Dzyudoist, Judoist	«Дзюдоист»	SIGINT system UA	
ML-20 M-1937		MA-20	152 mm towed gun-howitzer <i>stored</i> UA	
MLZh-VF-VT	MLZh-VF-VT		railway pontoon bridge UA	
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MO.1.01.04M			220 mm thermobaric rocket, extended range - of TOS-1A Sointsepek UA	7
MO.1.01.04M2			220 mm thermobaric rocket, heavier warhead - of TOS-2 Tosochka	7
Mohajer-6			reconnaissance and strike UAV UA	
Molniya-1 Molniya-2	Molniya		FPV drone UA	
MON-50		МОН-50	directional anti-personnel mine UA	
MON-90		МОН-90	directional anti-personnel mine UA	
MON-100			directional anti-personnel mine UA	
MON-200			directional anti-personnel mine UA	
MP-32M1 MP32M1			command post vehicle - of 9S729M1 Slepok-1 UA	
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MRT0	MRT0	MPT0	maintenance vehicle - of 9K720 Iskander-M	11
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MT-55A			bridgelayers	
MT-LB		MT-ЛБ	tracked armoured transporter UA	
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MT-LBM 6MB MT-LBM-6MB, Object 6MB		Объект 6МБ	tracked armoured transporter - of MT-LB UA	
MT-LBV		MT-ЛБВ	tracked armoured transporter - of MT-LB <i>legacy</i> UA	
MT-LBVM MT-LBVMK, MT-LB VM1K, MT-LBVMK		MT-ЛБВМ, MT-ЛБВМК	tracked armoured transporter - of MT-LB UA	
MTK-2 MTK			mine-clearing vehicle UA	
MT0-AT			repair vehicle UA	
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Nebo-T	Nebo-T		air surveillance radar	
NSV NSVT, NSVS	Utyos	НСВ, Умёс	12.7 mm heavy machine gun UA	

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Oreshnik RS-SS-28 MOD	Oreshnik	Орешник	intermediate-range ballistic missile UA	
Orion	Orion, Inokhodets	Орион, Иноходец	reconnaissance and strike UAV UA	
Orlan-10		Орлан-10	reconnaissance UAV UA	
Orlan-30		Орлан-30	reconnaissance UAV UA	
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P-18 1RL131, P-18M	Terek SPOON REST D	П-18 «Терек», 1РА131	early warning radar UA	
P-19 1RL134	Dunai FLAT FACE B	П-19 «Дунай», 1РА134	air surveillance radar	
P-35 P-37, 1RL110, 1RL139	Saturn, Mech BAR LOCK	П-35 «Сатурн»	early warning radar UA	
P-240 P-240I-4	Pereselenets, Pereselenetz		digital communications vehicle UA	
P-260-U P-260			signals vehicle · of Redut-2US UA	
P-800 3M55, SS-N-26	Oniks STROBILE	П-800 «Оникс»	anti-ship cruise missile · of 3K55 Bastion UA	
PAK-200 PAK-200M			field kitchen UA	
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PAZ-32053			bus UA	
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PFM-1 PFM-1S	Lepestok	ПФМ-1, ПФМ-1С	scatterable anti-personnel mine UA	
PKM			man-portable minelaying kit	
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Predel-E	Predel-E		coastal defence radar UA	
Prokhod-1	Prokhod-1		remote-controlled mine-clearing system UA	
PRP-3 1Zh3, Object 767	Val SMALL FRED (disputed)	ПРП-3 «Вал»	artillery reconnaissance vehicle UA	
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R-166 R-166-0.5, R-166S	Artek	P-166, Артек	radio station UA	
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RB-341V	Leer-3	PБ-341В «Лер-3»	cellular jammer UA	26
RB-531B	Infaua	PБ-531Б «Инфауна»	counter-IED jammer UA	27
RB-636AM2 RB-636	Svet-KU	PБ636 «Свет-КУ»	SIGINT system UA	
RB-636 Svet-VSG	Svet-VSG		SIGINT system · of RB-636AM2 Svet-KU	
REBovets Valley	REBovets Valley		UAV jammer <i>trial</i>	
Redut-2US			signal and communications system UA	
REM-KL			recovery vehicle UA	
Repellent-1	Repellent-1	Репеллент-1	UAV jammer UA	27
Repeynik	Repeynik		portable counter-UAS radar UA	
Rezonans-N	Rezonans-N, Resonans-N, Rezonans-NE		early warning radar	
RKhM-4		PXM-4	CBRN reconnaissance vehicle UA	
RKhM-6	Povozka	PXM-6	CBRN reconnaissance vehicle UA	
RKhM-8		PXM-8	CBRN reconnaissance vehicle	
RKhM-9		PXM-9	CBRN reconnaissance vehicle	
RLM-D		PAM-Д	UHF radar module · of 55Zh6M Nebo-M UA	18
RLM-M		PAM-M	VHF radar module · of 55Zh6M Nebo-M UA	18
RLM-S		PAM-C	S-band radar module · of 55Zh6M Nebo-M UA	18
RM-G			armoured recovery vehicle	
RP-377L RP-377LA	Lorandit, Lorandit-M		communications jammer UA	
RP-377UVM1 RP-377UVM2, RP-377UM2	Lesochek		counter-IED jammer UA	
RPG-7 RPG-7V, RPG-7V2, 6G3		РПГ-7	anti-tank rocket launcher UA	

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RPG-22	Netto	РПГ-22, Неммо	72.5 mm disposable anti-tank launcher <i>legacy</i> UA	
RPG-26	Aglen	РПГ-26, Азлень	disposable anti-tank rocket launcher UA	
RPG-27	Tavolga	Таволга	105 mm disposable anti-tank launcher UA	
RPG-28	Klyukva	РПГ-28, Калюкба	125 mm disposable anti-tank launcher UA	
RPG-29 TKB-0175	Vampir	РПГ-29 «Вампир», ТKB-0175	105 mm anti-tank rocket launcher	
RPG-30	Kryuk	РПГ-30, Крюк	105 mm disposable anti-tank launcher	
RPG-32	Barkas, Hashim, Nashshab	РПГ-32	105 mm reusable anti-tank launcher	
RPMK-1 1B44, 1B44-1	Ulybka, Ulibka-M		meteorological station UA	
RPO-A RPO-M, RPO	Shmel	РПО-А «Шмель»	thermobaric rocket launcher UA	
Rys	Rys, Iveco LMV	Рысь	armoured utility vehicle UA	19
S-60 AZP-57		C-60, АЗП-57	57 mm anti-aircraft gun UA	
S-300PM S-300PM1, S-300PM2, S-300PMU1, S-300PMU2, SA-20	S-300PM, Favorit GARGOYLE	C-300PM, C-300PMY1	long-range SAM system UA	
S-300PS 75R6, S-300PMU, SA-10	S-300PS GRUMBLE	C-300PS, 75P6	long-range SAM system <i>legacy</i> UA	
S-300V 9K81, SA-12A, SA-12B	S-300V, Antey-300 GLADIATOR, GIANT	C-300B, 9K81	long-range SAM system UA	13
S-300V4 S-300VMD, SA-23	S-300V4	C-300B4	long-range SAM system UA	13
S-350 50R6A	S-350, Vityaz	C-350 «Витязь», 50P6A	medium-range SAM system UA	13
S-400 40R6, SA-21	S-400, Triumph GROWLER	C-400 «Триумф», 40P6	long-range SAM system UA	12
S-500 55R6M	S-500, Prometey, Triumphator-M	C-500 «Прометей», 55P6M	long-range SAM and missile defence system <i>trial</i> UA	
SA-14RM/SM			encryption command post UA	
Saniya	Saniya, Sania	«Саня»	UAV jammer UA	
Sapfir	Sapfir		UAV jammer UA	
Sarmat-2 Sarmat-3, LSTC-1944	Sarmat, STS Sarmat		armoured utility vehicle UA	
SBA-60K2	Bulat	«Булат»	armoured personnel carrier	
SBR-5M	Fara-VR		portable battlefield surveillance radar UA	
SERP-VS6D	Serp	СЕРП-ВС6Д	UAV jammer UA	
Shipovnik-Aero	Shipovnik-Aero	«Шиповник-Аэро»	UAV jammer UA	
Silok-01	Silok	«Силок-01»	UAV jammer UA	
SNAR-10 1RL232, 1RL-127-1	Leopard BIG FRED	СНАР-10 «Леопард», 1PA232 «Леопард»	battlefield surveillance radar UA	20
SNAR-10M SNAR-10M1, 1RL232-1M	Pantera	СНАР-10M, 1PA232-1M «Пантера»	battlefield surveillance radar UA	20
Sopka-2 12A6	Sopka-2		air surveillance radar	
Sozvezdie	Sozvezdie		tactical command and control system <i>legacy</i>	
SPG-9 SPG-9M, SPG-9D	Kopye	СПГ-9, Копьё	73 mm recoilless gun UA	
SPN-4 SPN-4M, 1RL248-4	HEART ACHE		radar jammer	
Strizh-3	Strizh-3		UAV jammer UA	
Stupor	Stupor		UAV jammer UA	
Supercam S350			reconnaissance UAV UA	
Syuzhet-MF	Syuzhet-MF		meteorological station <i>trial</i>	
T-12 2A19		T-12, 2A19	100 mm anti-tank gun <i>stored</i>	
T-14 Object 148	Armata	T-14, Объект 148	main battle tank · of Armata <i>trial</i>	
T-15	Armata	T-15	infantry fighting vehicle · of Armata <i>trial</i>	
T-54 Object 137		T-54, Объект 137	main battle tank <i>legacy</i> UA	
T-54-T			armoured recovery vehicle <i>legacy</i>	
T-55		T-55	main battle tank <i>stored</i> UA	
T-55A		T-55A	main battle tank · of T-55 <i>stored</i> UA	
T-55AM T-55AMV		T-55AM, T-55AMB	main battle tank · of T-55 <i>legacy</i>	
T-62 Object 166		T-62, Объект 166	main battle tank <i>legacy</i> UA	
T-62M		T-62M	main battle tank · of T-62 UA	
T-62MV		T-62MB	main battle tank · of T-62 UA	
T-64A Object 434		T-64A, Объект 434	main battle tank UA	
T-64B Object 447, T-64BV		T-64S, Объект 447	main battle tank UA	
T-64BV			main battle tank · of T-64B UA	
T-64BVK			armoured command vehicle · of T-64B <i>legacy</i> UA	
T-72 T-72 Ural, Object 172M	Ural	T-72, T-72 «Урал»	main battle tank UA	

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T-72A		T-72A	main battle tank · of T-72 Ural UA	
T-72AV		T-72AB	main battle tank · of T-72A UA	
T-72B Object 184		T-72B, Объект 184	main battle tank · of T-72 Ural UA	
T-72B1 T-72B1K		T-72B1, T-72B1K	main battle tank · of T-72B UA	
T-72B3		T-72B3	main battle tank · of T-72B UA	
T-72B3A			main battle tank · of T-72B3 <i>legacy</i> UA	
T-72B3M		T-72B3M	main battle tank · of T-72B3 UA	
T-72BA		T-72BA	main battle tank · of T-72B UA	
T-72BM		T-72BM	main battle tank · of T-72B <i>legacy</i> UA	
T-72M1 T-72M		T-72M1, T-72M	main battle tank · of T-72 Ural <i>legacy</i>	
T-80B Object 219R		T-80B, Объект 219P	main battle tank <i>stored</i> UA	
T-80BV		T-80BВ	main battle tank · of T-80B UA	
T-80BVK		T-80BВК	armoured command vehicle · of T-80BV <i>legacy</i> UA	
T-80BVM		T-80BВМ	main battle tank · of T-80BV UA	
T-80U Object 219AS		T-80У, Объект 219AC	main battle tank UA	
T-80UE-1		T-80УЕ-1	main battle tank · of T-80U <i>legacy</i> UA	
T-80UK		T-80УК	armoured command vehicle · of T-80U <i>legacy</i> UA	
T-80UM2		T-80УМ2	main battle tank · of T-80U <i>legacy</i> UA	
T-90 Object 188		T-90, Объект 188	main battle tank UA	
T-90A		T-90A	main battle tank · of T-90 UA	
T-90AK		T-90AK	armoured command vehicle · of T-90A UA	
T-90M Proryv-3	Proryv-3	T-90M, Прорыв-3	main battle tank · of T-90A UA	
T-90S T-90SK		T-90C, T-90CK	main battle tank · of T-90 UA	
Tachyon	Tachyon, Takhion	Тахион	reconnaissance UAV UA	
TBS-M3		ТБС-М3	220 mm thermobaric rocket, long range · of TOS-2 Tosochka UA	7
TDA-2K TDA-2M		ТДА-2К, ТДА-2М	smoke generator vehicle UA	
TDA-3			smoke generator vehicle UA	
TDM-2K			smoke generator vehicle UA	
Tetrazdr	Tetrazdr		UAV jammer <i>trial</i>	
Tirada-2S Tirada-2.3, RB-371A	Tirada-2	«Турпага-2С», «Турпага 2.3»	satellite communications jammer UA	
TM-46		TM-46	anti-tank blast mine <i>legacy</i>	
TM-57		TM-57	anti-tank blast mine <i>legacy</i>	
TM-62 TM-62M		TM-62	anti-tank blast mine UA	
TM-83		TM-83	side-attack anti-tank mine	
TMM-3 TMM		TMM-3	truck-mounted bridgelayer UA	29
TMM-6		TMM-6	8x8 heavy bridgelayer UA	29
TMS-65 TMS-65M, TMS-65U		TMC-65	decontamination vehicle	
Tobol	Tobol	«Тобол»	satellite communications jammer UA	
Torn	Torn, Torn-MDM		SIGINT system UA	
Toros	Toros		armoured utility vehicle	
TOS-1 Object 634	Buratino	ТОС-1 «Буратино»	220 mm heavy flamethrower system <i>legacy</i> UA	7
TOS-1A Object 634B	Solntsepek	ТОС-1А «Солнце́пек»	220 mm heavy flamethrower system UA	7
TOS-2	Tosochka	ТОС-2 «Тосочка»	220 mm heavy flamethrower system UA	7
TOS-3	Drakon	ТОС-3 «Дракон»	heavy flamethrower system, tracked <i>trial</i> UA	7
TS-3132	Gusar		armoured utility vehicle UA	
TZ-74			decontamination and smoke vehicle <i>legacy</i>	
TZM-T		ТЗМ-Т	rocket transloader · of TOS-1A Solntsepek UA	7
UAZ-452	Bukhanka	УАЗ-452, Буханка	utility van UA	
UAZ-469 UAZ-3151	Kozlik	УАЗ-469	utility vehicle UA	
UAZ-469RKh UAZ-469RKhB			CBRN reconnaissance and decontamination vehicle UA	
UAZ-39094	Fermer		utility pickup UA	
UMPB D-30SN			jet-powered glide bomb UA	
UMPK		УМПК	glide bomb guidance kit UA	
UMZ		УМЗ	truck-mounted minelayer UA	
UMZ-G UMZ-K, UMZ-T	Kleshch-G, Kleshch-G		universal tracked remote minelayer <i>trial</i>	
UR-67		УР-67	mine-clearing line charge <i>legacy</i> UA	

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UR-77 Object 256, M1979	Meteorit	УР-77	mine-clearing line charge vehicle (UA)	30
UR-83P		УР-83П	portable mine-clearing line charge	
Ural-375D Ural-375		Урал-375Д	cargo truck 6x6 <i>legacy</i> (UA)	
Ural-4320 Ural-43203		Урал-4320	cargo truck 6x6 (UA)	
Ural-4320-7810	Motovoz-M		cargo truck (UA)	
Ural-6370 Ural-6370-0121-30			cargo truck 6x6 (UA)	
Ural-43206		Урал-43206	cargo truck 4x4 (UA)	
Ural-63095	Typhoon-U	Урал-63095, Тайфун-У	mine-resistant ambush-protected vehicle	
Ural-63706-0122 Ural-63706-0011	Tornado-U		cargo truck 6x6 (UA)	
Ural-432009	Ural-VV	Урал-432009, Урал-BB	armoured truck (UA)	
Ural-532301			cargo truck with armoured cab (UA)	
Ural-542301			tank transporter (UA)	
Uran-6		Уран-6	remote-controlled mine-clearing robot (UA)	
Volna Kupol Garant	Volna Kupol Garant		satellite communications jammer (UA)	
Volnorez	Volnorez	«Волнорез»	UAV jammer (UA)	
VPK-7829 K-16, K-17	Bumerang, Boomerang	Бумеранг	armoured personnel carrier <i>trial</i>	
VPK-233136 SBM	Tigr SBM	ВПК-233136	armoured car of GAZ-233014 Tigr (UA)	
VSM-1		BCM-1	helicopter remote mining system	
VT-72A			armoured recovery vehicle	
Z-STS	Akhmat, Remdiesel Z-STS Akhmat		armoured utility vehicle (UA)	
ZALA 421-16 Z-16			reconnaissance UAV (UA)	
Zaslon-REB	Zaslon-REB		cellular jammer	
Zastava	Zastava	Застава	reconnaissance UAV (UA)	
Zavarukha	Zavarukha		portable acoustic reconnaissance system (UA)	
ZIL-131		ЗИЛ-131	cargo truck 6x6 <i>legacy</i> (UA)	
ZIL-157		ЗИЛ-157	cargo truck 6x6 <i>legacy</i>	
ZSU-23-4 2A6	Shilka	3CY-23-4, 2A6	23 mm self-propelled anti-aircraft gun <i>stored</i> (UA)	
ZU-23-2 ZU-23, 2A13		3Y-23, 3Y-23-2	23 mm anti-aircraft gun (UA)	

Table 1b. Other names and designations

Names, NATO names and codes (in **BLUE CAPITALS**), and other designations, each pointing (→) to its row in Table 1. Page numbers as in Table 1. English translations of Russian names are in *italics*.

NAME OR DESIGNATION	PAGE	NAME OR DESIGNATION	PAGE	NAME OR DESIGNATION	PAGE
1B44 → RPKM-1		1V152 → 1V126		2S3M1 → 2S3M	
1B44-1 → RPKM-1		1V152M → 1V126		2S3M2 → 2S3M	
1B75E → 1B75	20	1V153 → 1V126		2S6M → 2S6	17
1L13 → 1L13-3		1V153M → 1V126		2S6M1 → 2S6	17
1RL110 → P-35		1V168 → ASUNO-S		2S9-1 → 2S9	
1RL131 → P-18		1Zh3 → PRP-3		2S12A → 2S12	
1RL133 → PSNR-5	34	2A6 → ZSU-23-4		2S33 → 2S19M2	8
1RL133-3 → Kreda-1E		2A13 → ZU-23-2		3F5 → 1K113	9
1RL134 → P-19		2A18 → D-30		3M14K → 3M14	
1RL138 → 1RL144M	17	2A18M → D-30		3M14T → 3M14	
1RL139 → P-35		2A19 → T-12		3M55 → P-800	
1RL144 → 1RL144M	17	2A29 → MT-12		3OF45-1 → 3OF45	
1RL144M1 → 1RL144M	17	2A29R → MT-12R		3OF43B → OF-843B	
1RL232 → SNAR-10	20	2A45MR → 2A45M		3UBK10 → 9K116	
1RL232-1M → SNAR-10M	20	2B5 → BM-21	4	3VF2 → 3F2	9
1RL248-4 → SPN-4		2B9M → 2B9		3VF4 → 1K113	9
1V13M → 1V13	33	2B17 → BM-21-1	4	3VO11 → 3-O-8	9
1V15 → 1V14	33	2B17-1 → BM-21-1	4	3VO13 → 3-O-13	
1V15M → 1V14	33	2B17M → 9K51M	4	3VO14 → 3-O-13	
1V16 → 1V14	33	2F77M → 2F77	17	3VO28 → 3-O-23	
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1V121 → PRP-4	33	2K22M1 → 2K22	17	3VO30 → 3-O-23	
1V125 → PRP-4	33	2K25 → 3OF39		3VOF32 → 3OF25	
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3VOF63 → 30F38	
3VOF64 → 30F39	
3VOF64M → 30F39M	
3VOF66 → 30F38	
3VOF72 → 30F45	
3VOF73 → 30F45	
3VOF86 → 30F59	7
3VOF87 → 30F59	7
3VOF91 → 30F61	
3VOF93 → 30F39	
3VOF96 → 30F64	
3VOF97 → 30F64	
3VOF98 → 30F64	
3VRB36 → 3RB30	
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5V55K → 5V55R	
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8K14 → 9K72	
9A33BM3 → 9A33B	17
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T-55AMV → T-55AM	
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Tornado-G → 2B17M	4
Tornado-G → 9K51M	4
Tornado-S → 9A54	6
Tornado-S → 9K515	6
Tornado-U → Ural-63706-0122	
Tosochka → TOS-2	7
Tray (English) → 2B14	
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Tulip (English) → 2S4	9
Tunguska → 2K22	17
Tunguska-M1 → 2K22	17
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Typhoon-K → KamAZ-63968	
Typhoon-K 4x4 → KamAZ-53949	
Typhoon-K 6x6 → KamAZ-63968	
Typhoon-U → Ural-63095	
Typhoon-VDV → KamAZ-4386	
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UNSh → K1Sh1	
Uragan → 9K57	5
Uragan → 9P140	5
Uragan-1 → 9K57-1	5
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Ural-6370-0121-30 → Ural-6370	
Ural-43203 → Ural-4320	
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Ural-VV → Ural-432009	
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1В119 → 1V119		2С3М → 2S3М		55Ж6МЕ → 55Zh6М	18
1В126 «Капустник-Б» → 1V126		2С40 → 2S40		55Ж6М «Небо-М» → 55Zh6М	18
1В13 → 1V13	33	2С43 «Мальва» → 2S43	8	55К6 → 55K6	12
1В152 → 1V152		2С4 «Тюльпан» → 2S4	9	55К6Е → 55K6	12
1К113 «Смелычак» → 1K113	9	2С5 «Гуацинт-С» → 2S5	7	55Р6М → S-500	
1К148 «Ястреб-АВ» → 1K148		2С5М → 2S5М	7	5736Е → 57Е6	16
1Л122 «Гармонь» → 1L122		2С6 → 2S6	17	5В55К → 5V55R	
1Л13 → 1L13-3		2С6М → 2S6	17	5В55Р → 5V55R	
1Л219 «Зоонарк-1» → 1L219	19	2С6М1 → 2S6	17	5Н63С → 5N63S	
1Л219М → 1L219	19	2С7М «Малка» → 2S7М	9	5Н84А «Оборона» → 5N84A	
1Л259 → 1L259	19	2С7 «Пион» → 2S7		5Н85Д → 5P85SD	
1Л260 → 1L260	19	2С9 «Нона-С» → 2S9		5Н85С → 5P85SD	
1Л2623 «Ртуть-БМ» → 1L262E		2Ф77 → 2F77	17	5Н85СМ2-01 → 5P85SM2-01	12
1Л265 → 1L265	22	2Ф77М → 2F77	17	5Н85Т2 → 5P85T2	12
1Л266 → 1L266	22	35Н6 «Каста» → 35N6		75Р6 → S-300PS	
1Л269 «Красуха-2» → 1L269	21	39Н6Е «Каста-2Е2» → 39N6		85Я6 «Яер-2» → 85Ya6	26
1Л271 «Аистенок» → 1L271	20	39Н6 «Каста-2-2» → 39N6		91Н6Е → 91N6A	12
1РА131 → P-18		35К13 → 3BK13	33	92Н6Е → 92N6A	12
1РА133-3 «Крего-1» → Kredo-1E		35К19 → 3BK19		96К6 → 96K6	16
1РА134 → P-19		3ВФ4 → 1K113	9	96А6Е → 96L6	18
1РА144М1 → 1RL144М	17	3К55 → 3K55		9А310 → 9A310	14
1РА232-1М «Пантера» → SNAR-10М	20	3К60 → 3K60		9А310М1-2 → 9A310M1-2	14
1РА232 «Леопард» → SNAR-10	20	3М-14 → 3M14		9А316 → 9A316	14
1РА239 / АРК-1 / «Рысь» → 1RL239	19	3М16 → 3M16	4	9А317 → 9A317	14
1РА257 «Красуха-4» → 1RL257	21	3-0-13 → 3-0-13		9А317М → 9A317M	14
1Т12-2М → 1T12-2M	6	3-0-23 → 3-0-23		9А330 → 9A330	15
2А13 → 2U-23-2		3-0-8 → 3-0-8	9	9А331 → 9A331	15
2А18М → D-30		30Ф24 → 30F24	33	9А331-1 → 9A331	15
2А19 → T-12		30Ф25 → 30F25		9А331М2 → 9A331M	15
2А29 → МТ-12		30Ф29 → 30F29	7	9А335 → 9A33B	17
2А36 «Гуацинт-Б» → 2A36	10	30Ф38 → 30F38		9А35 → 9A35	17
2А6 → ZSU-23-4		30Ф39 → 30F39		9А38 → 9A310	14
2А65 «Мста-Б» → 2A65	10	30Ф39М → 30F39M		9А39 → 9A39	14
2Б11 → 2B11		30Ф43 → 30F43		9А39М1 → 9A39	14
2Б14 «Поднос» → 2B14		30Ф44 → 30F44		9А51 → 9A51	
2Б16 «Нона-К» → 2B16		30Ф45 → 30F45		9А52 → 9A52	6
2Б17 → ВМ-21-1	4	30Ф49 → 30F49		9А52-2 → 9A52-2	6
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2Б26 → 2B26	4	30Ф56 → 30F56	33	9А52-4 → 9A52-4	6
2Б5 → ВМ-21	4	30Ф59 → 30F59	7	9А53 → 9A53	5
2Б9 «Василёк» → 2B9		30Ф61 → 30F61		9А54 → 9A54	6
2К22 → 2K22	17	30Ф64 → 30F64		9А82 → 9A82	13
2К22М1 «Тунгуска-М1» → 2K22	17	30Ф75 → 30F75		9А83 → 9A83	13
2К22М «Тунгуска-М» → 2K22	17	3РБ30 → 3RB30		9А84 → 9A84	13
2С12А → 2S12		3Ф2 → 3F2	9	9А85 → 9A85	13
2С12 «Сани» → 2S12		3Ф5 → 1K113	9	9К11 → 9K11	
2С19М1 → 2S19M1	8	40Н6Е → 40N6	12	9К111 → 9K111	
2С19М2 → 2S19M2	8	40Р6 → S-400	12	9К111-1 → 9K111-1	
2С19 «Мста-С» → 2S19		48Н6ДМ → 48N6	12	9К114 → 9K114	
2С1 «Гвоздика» → 2S1	33	48Н6Е → 48N6	12	9К115 → 9K115	
2С23 «Нона-СВК» → 2S23		48Н6Е2 → 48N6	12	9К115-1 → 9K115-1	
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2С31 «Вена» → 2S31		50К6А → 50K6A	13	9К116 → 9K116	
2С33 «Мста-СМ» → 2S19M2	8	50Н6А → 50N6A	13	9К119М → 9K119	
2С34 «Хоста» → 2S34		50П6А → 50P6A	13	9К123 → 9K123	
2С35 «Колалиция-СВ» → 2S35	8	50Р6А → S-350	13	9К123-1 → 9K123	
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9K35 «Стрела-10» → 9K35	17
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Зльбрус → 9K72	

Table 2. Unit type to equipment

Tag gives basis, date and confidence. See page 3 for explanations.

UNIT TYPE	EQUIPMENT	HOLDING SUBUNIT AND NUMBER	TAG	PAGE
aerospace forces air defence regiment	96L6 all-altitude acquisition radar	3rd SAM battalion. INA (photographed deployed and stowed)	O 2010-02 M	18
	96L6 all-altitude acquisition radar	S-400 'battery' in Armada's usage; APA export diagram lists it as a battery component. 1 per S-400 battery (Armada, term undefined; APA lists 96L6E as a battery component without a count); S-300PMU2 count INA	R 2014 M	18
	Buk-M1 / Buk-M2 / Buk-M3	4 Aerospace Forces regiments with Buk-M1-2, Buk-M2 and S-300V; about 80 Buk-M1-2/Buk-M2 in Aerospace Forces holdings	R 2024 M	14
	Pantsir-S1	Pantsir vehicles deployed with each S-400 battalion (Armada); Lenta: batteries attached to S-400 regiments. 3 per S-400 battalion (Armada, 2024)	D 2024 H	16
	Pantsir-S1	Pantsir-S battery. 6 Pantsir-S delivered to Eastern MD air defence units on Kamchatka, Dec 2015; the regiment was reported due one battery	O 2015-12 M	16
	Pantsir-S1	Pantsir-S1 gun-missile battalion; each battery covers one S-400 battalion	O 2020-02 M	16
air defence missile battalion	Pantsir-S1	INA per regiment; IISS lists 9 SAM regts with Pantsir-S1 in the 45th, 4 in the 14th, 3 in the 4th and 4 in the 11th AF&AD Army	R 2024 M	16
	S-400 Triumf	IISS: 13 Aerospace Forces regts with S-400 and Pantsir-S1, plus 2 mixed with S-300	R 2024 M	12
	S-350 Vityaz	Battalion set: 1 command post 50K6A, 2-3 multifunction radars 50NGA, 12 launchers 50P6A. 12 launchers (12 missiles each) per battalion, per Russian MoD statement	R 2020 M	13
air defence missile brigade	S-400 Triumf	IISS: 2 SAM bn in Belarus; 1 SAM bty in Syria	R 2024 M	12
	Buk-M1 / Buk-M2 / Buk-M3	4 air defence missile battalions, each 3 batteries; brigade command post Polyana-D4. Buk-M1: 6 TELARs, 3 loader-launchers, 1 CP, 1 search radar per battalion; 24 TELARs per 4-battalion brigade (JIT: 53rd Bde had 3 operational battalions in 2014)	D 2016 H	14
	Buk-M1 / Buk-M2 / Buk-M3	battalion TELAR. INA per battalion; brigade had 3 operational battalions in 2014 (JIT)	O 2014-07 M	14
	Buk-M1 / Buk-M2 / Buk-M3	air defence missile battalion (brigade set of Buk-M3). Buk-M3: up to 6 TELARs or 9S36M radars and up to 12 launchers per battalion	R 2019 M	14
	Buk-M1 / Buk-M2 / Buk-M3	air defence missile battalion	R 2018 L	14
	Polyana-D4	brigade command post (combat control post MP06 and command-staff vehicle MP02)	D 2021-12 H	34
	Polyana-D4	brigade command post; vehicle captioned as 9S52M Polyana-D4M	O 2010-12 M	34
	S-300V4	air defence missile battalion (up to 4 per brigade, under Polyana-D4M1 brigade CP); battalion CP 9S457M controls up to 4 batteries; WEG: an upper-level 9S52 Polyana-D4 command post may integrate S-300V with Buk into one brigade. S-300V battery: 1 9S32, 4 9A83, 2 9A82, 1 9A84, 2 9A85; S-300V4 battery counts INA	D 2017 M	13
	S-300V4	INA. IISS lists S-300V and S-300V4 in Army holdings without a count and counts 16 SAM brigades, but does not say which hold them; Izvestia reports S-300V4 brigades under the OSK	R 2024 M	13
	Pantsir-S1	30 held by the Navy (IISS p. 187)	R 2024 M	16
air defence missile regiment	Pantsir-S1	INA (1 SAM regt with S-400 and Pantsir-S1)	R 2024 M	16
	S-400 Triumf	about 12 launchers reported at the unit site; 4 launchers and 96L6 and 92N6 radars reported destroyed	O 2025-11 M	12
	S-400 Triumf	air defence missile battalion (division). usually 2 battalions of 8 launchers each (16 per regiment); 2012 press report relating MoD press service	R 2012 M	12
	S-400 Triumf	Navy: regiments under fleet air defence divisions. IISS: 64 S-400 in Navy; 3 SAM regts: 2 Baltic Fleet, 1 Pacific Fleet (naval aviation)	R 2024 M	12
	S-400 Triumf	Navy: regiments under fleet air defence divisions. IISS: 64 S-400 in Navy; 3 SAM regts: 2 Baltic Fleet, 1 Pacific Fleet (naval aviation)	R 2024 M	12
airborne division	Barnaul-T	air defence missile regiment. one full divisional set	O 2018-06 M	35
	Barnaul-T	air defence missile regiment	O 2018 L	35
	Leer-2	EW subunit (FOI: VDV EW companies formed from 2016). INA; MoD announced 10 complexes for VDV EW units, Dec 2012 (title only, state claim)	R 2018 M	26
anti-tank battalion	R-145BM Chayka command and staff vehicle	ACV listed as 'BTR-60PU/BRDM-2'; battery placement unclear in OCR. 5 per battalion (BTR-60PU or BRDM-2)	D 1991 L	32
army corps	Iskander-M	corps missile brigade. 1 missile brigade (12 launchers)	O 2021 M	11
artillery brigade	2A36 Giatsint-B 152 mm towed gun	gun battalion (152 mm) of the army artillery brigade; 2A36 or 2S5 or M-46. 36 per brigade (two 18-tube gun battalions) or 48 (four 24-tube battalions)	D 1991 M	10

UNIT TYPE	EQUIPMENT	HOLDING SUBUNIT AND NUMBER	TAG	PAGE
combined arms army	2A65 Msta-B	artillery battalions [several per brigade; 2S19 or 2A65]	D 2019 M	10
	2S35 Koalitsiya-SV	self-propelled artillery battalions. INA; planned replacement of the brigade's 2S19 by 2S35, reported 2017, not carried out by late 2024	R 2019 L	8
	2S5 Giatsint-S	gun battalion (152 mm). 18 per battalion [24 in forward armies]; 2S5 or 2A36 or M-46	D 1991 M	7
	2S7 Pion / 2S7M Malka	2S7 battalion (I217, 2017); 2S7M Malka (mil.ru via RDP18). 1 battalion; 12 x 2S7 (RUWIKI, 2019)	O 2018-04 M	9
	2S7 Pion / 2S7M Malka	12 x 2S7 each (RUWIKI, 2019)	O 2019 L	9
	Uragan-1M	MRL battalion [Uragan 'being succeeded' by Uragan-M1; no holding confirmed]	R 2019-10 L	5
	1B75 Penicillin	1L271 in rocket, missile and tube artillery units	R 2017 L	20
	Iskander-M	army missile brigade. 1 missile brigade (12 launchers) per army	D 2024 H	11
	Iskander-M	army missile brigade. 12 launchers per brigade	O 2023 M	11
	Palantin	EW units of a Western MD combined arms army in Voronezh Oblast [army not named]	R 2019-04 M	25
command brigade	Palantin	EW troops of a Guards combined arms army, Western MD, Luga range, Leningrad Oblast (Palantin-K, Oct 2019); Feb 2022 exercise in Leningrad Oblast names only the Western MD	R 2022-02 M	25
	Pole-21	Army EW specialists; first used in an exercise with battalion tactical groups, with R-934BMV and Silok-01 (MoD via Aif)	O 2021-04 M	27
	Polyana-D4	army air defence and aviation command post (SKP PVO i aviatsii)	R 2021-12 M	34
	R-149 command and staff vehicles	R-149AKSh-1	O 2022-03 L	32
engineer-sapper battalion	IMR-2 / IMR-2M / IMR-3M combat engineer vehicle	mine clearing platoon [brigade engineer battalion]. 2 IMR-3M per platoon	D 2016 M	30
	ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer	obstacle platoon [brigade engineer battalion per RWW; division engineer battalion per WEG]. 3 GMZ-3 per obstacle platoon	D 2016 M	31
	TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge	brigade engineer battalion; sets almost always attached to the movement support detachment (OOD). 4 TMM-3 truck-launched bridge sets per brigade engineer battalion, each laying a 10 m span; together a 42 m bridge	D 2024-02 M	29
	UR-77 Meteorit	mine clearing platoon [organic engineer battalion of a maneuver brigade]. 2 per platoon	D 2016 M	30
	ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer		D 2024 L	31
	PMP heavy folding pontoon bridge park	pontoon bridge battalion of the army or front engineer brigade (FM 100-2-3 'Engineer Brigade, Army or Front'). one full set: 32 river and 4 shore links on KRAZ-255, 15 BMK-150 boats	D 1991 M	28
	ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer		D 2024 L	31
	TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge	engineer and bridge subunit (press-release wording)	O 2020-11 M	29
	Krasukha-2 / Krasukha-4	independent REB-S battalion of an Air and Air Defence Army (VKS); FOI infers, does not name Krasukha there	R 2018 L	21
	Leer-3	Leer-3 drone/EW company	R 2019 M	26
EW brigade	Krasukha-2 / Krasukha-4	REB-S (counter-air) EW battalion. INA; usually one Krasukha-2 and one Krasukha-4 per brigade (reported 2022)	D 2018 H	21
	Krasukha-2 / Krasukha-4	part of 1 Krasukha-4 complex (command-post container found near Makariv); Armada estimates 4 complete Krasukha systems in theatre	R 2022-03 M	21
	Leer-3	Leer-3 platoon. platoon operates the complex; number per brigade INA	D 2024 H	26
	Moskva-1	REB-S (counter-air EW) battalion, reported	D 2018 H	22
	Moskva-1	INA (ARM22 lists one Moskva-1 among five brigade systems)	R 2023-09 L	22
	Murmansk-BN	Probably a company of the REB-N (anti-ground) battalion (FOI assessment)	D 2018 H	22
	Murmansk-BN	At least 1 complex per brigade [Armada]; 2 antenna groups observed at 16th EW bde, vehicle count at 19th suggests 2 (PAS imagery, 2019)	O 2019 M	22
	R-330Zh Zhitel	REB-S (anti-aircraft communications and navigation) battalion	D 2018 M	24
	R-330B Mandat-B and R-934BMV	EW company of a motor rifle or tank brigade: R-330B squad in the VHF-band jamming platoon; R-934B squad (VHF aircraft) in the radio-communications jamming platoon	D 2016 H	24

UNIT TYPE	EQUIPMENT	HOLDING SUBUNIT AND NUMBER	TAG	PAGE
	R-330Zh Zhitel	EW company; jamming platoon (radio, GPS, cellular, radio-controlled fuses), jamming squad. 1 (single station on RWW chart; no count stated)	D 2016 M	24
	R-330Zh Zhitel	EW company of motor rifle or tank division (sources confirm the company exists; Zhitel holding not stated)	D 2018 L	24
high-power artillery brigade	2S4 Tyulpan	2S4 battalion (ATP figure 5-18 shows two 2S4 groups; subunit name not given). ATP fig. 5-18: groups of 6 and 4 2S4 (OCR uncertain); 8 reported in 45th brigade (2017)	D 2024 H	9
	2S4 Tyulpan	heavy mortar battalion (240-mm); brigade allocated to a front (Soviet RVGK)	D 1991 M	9
	2S7 Pion / 2S7M Malka	2S7 battalions (45th bde: two battalions of eight, 2017). ATP fig. 5-18: 24 x 2S7 per brigade, 8 x 2S7 battalion block (OCR uncertain); 16 reported in 45th bde, 2017	D 2024-02 H	9
	2S7 Pion / 2S7M Malka	16 x 2S7 Pion (with 8 x 2S4 Tyulpan)	O 2017-12 M	9
military district	S-400 Triumf	SAM regiments of the district's Air Force and Air Defence Army. IISS late 2024, SAM regiments listed with S-400 in each district's Air Force and Air Defence Army: 45th (Leningrad MD) 5, plus 4 listed with S-300PS, S-300PM1 and S-400; 14th (Central MD) 4; 4th (Southern MD) 3; 11th (Eastern MD) 4 listed with S-300PS and S-400, plus 1 with S-300V and S-400; 6th (Moscow MD) none (Moscow area: 5 regiments under Aerospace Defence Command). Launchers per district INA	D 2024 H	12
missile battalion	Iskander-M	4 launchers	O 2008 M	11
missile brigade	Iskander-M	3 missile battalions of 2 batteries each; 2 launchers per battery. 12 launchers and 12 transloaders per brigade; 4 of each per battalion	D 2021 H	11
motor rifle brigade	2S19 Msta-S	two self-propelled artillery battalions, each of three batteries of 6 howitzers; ATP notes 2S3 may be held instead. 18 per battalion (6 per battery), 36 per brigade; ATP: some battalions up to 24 systems	D 2024 M	8
	Barnaul-T	brigade AD subunit; BO quotes a 'missile division' (battalion) commander	O 2021-12 M	35
	BM-21 Grad 122 mm MRL	MRL battalion; 6 launchers per battery. 18 per battalion	D 2016 M	4
	Borisoglebsk-2	EW company	D 2024-02 H	23
	Borisoglebsk-2	EW subunit	O 2018 M	23
	PMP heavy folding pontoon bridge park	engineer battalion, pontoon bridge company, one pontoon park; RWW calls it 'PP-61' but its figures (268 m at 60 t; 90- and 120-t bridges) match PP-91, not PMP. 6 BMK boats, 7 PTS-2	D 2016 L	28
	PRP-4A Argus / PRP-4M Deytery	fire control and artillery reconnaissance battery, artillery spotting platoon	D 2016 M	34
	PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry	engineer battalion. 7 PTS-2 per engineer battalion (RWW names PTS-2, not PTS-4)	D 2016 M	29
	SNAR-10 / SNAR-10M1 Leopard	fire control and artillery reconnaissance battery, radar platoon	D 2016 M	20
	Strela-10 and Osa short-range SAM	air defence missile-artillery battalion, missile battery (SA-13). 6 launchers per battery	D 2024 M	17
	Tor	air defence missile battalion, three batteries. 12 launcher vehicles per battalion (4 per battery)	D 2024 M	15
	Tor	air defence missile battalion (Tor-M2DT); brigade link by location and retraining report only	R 2019-11 L	15
	Tornado-G	brigade artillery, MRL battalion. INA for Tornado-G; it replaces the BM-21, 18 per brigade MRL battalion (RWW p.100); 18 delivered to one brigade 2012 (RBASE)	D 2019 M	4
	Tornado-G	18	O 2012-04 M	4
	Tunguska	air defence missile-artillery battalion, Tunguska battery. 6 launcher vehicles per battery	D 2016 M	17
	Zoopark-1 / Zoopark-1M	fire control and artillery reconnaissance battery, radar platoon (with SNAR-10M1)	D 2016 M	19
motor rifle division	Borisoglebsk-2	separate EW battalion	O 2022-01 M	23
	PMP heavy folding pontoon bridge park	engineer battalion, pontoon bridge company (also in tank divisions). half set: 16 river and 2 shore links and 1 service vehicle on KrAZ-214; 6 BMK-90/150 boats per company	D 1991 M	28
	R-149 command and staff vehicles	R-149AKSh-1	O 2022-03 L	32
	Tor	air defence missile regiment. 16 per regiment (4 batteries of 4, bmpd); ATP: two battalions; ru.wikipedia: Tor-M2 'divisional' set of 3 batteries, 12 vehicles	D 2024 H	15
	Tor	air defence missile regiment. one regimental set of Tor-M2 (16 combat vehicles)	O 2017-11 M	15

UNIT TYPE	EQUIPMENT	HOLDING SUBUNIT AND NUMBER	TAG	PAGE
motor rifle regiment	Tornado-G	artillery regiment, MRL battalion (replacing Grad battalion)	D 2019 H	4
	Zoopark-1 / Zoopark-1M		D 2024 L	19
	BMP-1KSh command and staff vehicle	regimental command post. 1 in regimental HQ of a BMP-equipped motor rifle regiment (FM 100-2-3 table, OCR scrambled)	D 1991 M	33
	BMP-1KSh command and staff vehicle	501st signal company	O 1986 L	33
	PU-12M7	air defence missile-artillery battery; also regimental air defence chief's post (ARMF: PU-12 family 1977-95; ORUZH: undated PU-12M7 sheet). 1 per battery (PU-12 family)	D 1995 M	35
	PU-12M7	regiment on BTR, Soviet 1980s establishment. 3 PU-12 per regiment	D 1989 L	35
ponton-bridge regiment	R-149 command and staff vehicles	R-149MA3	O 2022-09 M	32
	PMP heavy folding pontoon bridge park	two pontoon bridge battalions (army or front/district level), three full sets: 96 river and 12 shore links on KrAZ-255	D 1991 M	28
RKhBZ brigade	TOS-1A Solntsepek	flamethrower battalion of 3 batteries, each 6 BM-1 and 6 TZM-T; RWW places these battalions in NBC defence brigades and regiments and does not state the echelon. 18 per battalion (6 per battery)	D 2016 M	7
	TOS-1A Solntsepek	TOS battalion of 3 companies of 6 launchers (ATP para 5-146). 18 per battalion	D 2024 M	7
rocket artillery brigade	BM-27 Uragan	rocket launcher battalions. 54 per brigade (3 battalions of 18)	D 1991 M	5
	BM-30 Smerch	three groups of 4 x 9A52-2 in ATP fig. 5-16; whether each group is a battalion or a battery is not readable from the text. 12 x 9A52-2 shown in ATP fig. 5-16 (3 groups of 4)	D 2024 M	6
rocket artillery regiment	BM-30 Smerch	7 delivered April 2017, 5 more announced	O 2017-04 M	6
	BM-30 Smerch		O 2026-09 M	6
	BM-27 Uragan	three rocket launcher battalions. 54 per regiment (3 battalions of 18)	D 1991 M	5
	self-propelled artillery battalion	battalion fire control network, conceptual ATP fig. 5-3 (1L271)	D 2024 M	20
self-propelled artillery regiment	1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16)	each battery: one 1V13 (battery FDC) and one 1V14 (battery COP); battalion HQ: one 1V15 (battalion COP) and one 1V16 (battalion FDC). 8 per battalion: 3 1V13, 3 1V14, 1 1V15, 1 1V16	D 2016 H	33
	2S35 Koalitsiya-SV	self-propelled howitzer battery. 6 per battery, 18 per battalion	D 2016 M	8
	1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16)	howitzer battalion with 2S19M1. at least 3 1V13 and 3 1V14 seen with one battalion	R 2021-09 L	33
	2S35 Koalitsiya-SV	8 2S35 on troop trials	R 2020-05 M	8
	BM-27 Uragan	Uragan rocket artillery battalion. 8 per battalion	R 2017-12 M	5
	Uragan-1M	rocket artillery battalion (planned replacement for 9K57 Uragan, MoD statement; no holding confirmed)	R 2017-12 L	5
tank army	Iskander-M	army missile brigade. 1 missile brigade (12 launchers)	O 2021 M	11
tank brigade	2S19 Msta-S	two howitzer (self-propelled artillery) battalions, three batteries of 6. 18 per battalion, 36 per brigade	D 2016 M	8
	Borisoglebsk-2	EW company	D 2024-02 H	23
	PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry	engineer battalion. 7 PTS-2 per engineer battalion (RWW names PTS-2, not PTS-4)	D 2016 M	29
	Tor	air defence missile battalion, three batteries. 12 launcher vehicles per battalion (4 per battery)	D 2024 M	15
	Tunguska	anti-aircraft missile-artillery battalion, SA-19 battery; ATP 4-161 gives this battalion to tank brigades, but the battery figure is for brigades generally. 6 per battery (generic brigade figure: RWW 'maneuver brigade', ATP fig. 4-19)	D 2024 L	17
tank division	Zoopark-1 / Zoopark-1M	fire control and artillery reconnaissance battery, radar platoon	D 2016 M	19
	1B75 Penicillin	1L271; holding subunit not stated	O 2018-01 M	20
	Tor	air defence missile regiment. one regimental set of Tor-M2 (16 combat vehicles)	O 2017-03 M	15
tank regiment	Tornado-G	artillery regiment, MRL battalion	D 2019 H	4
	BMP-1KSh command and staff vehicle	regimental command post. INA (FM 100-2-3 table reads 0 for the tank regiment HQ)	D 1990 M	33
	PU-12M7	air defence missile-artillery battery; also regimental air defence chief's post (ARMF: PU-12 family 1977-95; ORUZH: undated PU-12M7 sheet). 1 per battery (PU-12 family)	D 1995 M	35
unit not identified	2S5 Giatsint-S	Airborne Forces artillery; unit not named. 18 (IIS 2024 figures)	R 2024 M	7
	ARK-1M Rys	division's reconnaissance battery (WEG wording, identical to Wikipedia; Soviet-era)	R 2024 L	19

UNIT TYPE	EQUIPMENT	HOLDING SUBUNIT AND NUMBER	TAG	PAGE
	Borisoglebsk-2	EW units	O 2018 M	23
	Borisoglebsk-2		O 2022-03 L	23
	Infaua	dedicated EW units and manoeuvre reconnaissance units	D 2024 L	27
	Krasukha-2 / Krasukha-4	1 Krasukha-4 reported hit	R 2023-11 L	21
	Krasukha-2 / Krasukha-4	1 Krasukha-2 reported	R 2018-08 L	21
	Leer-3	Leer-3 crews	R 2022-11 M	26
	Moskva-1	1 claimed destroyed	R 2022-09 L	22
	Murmansk-BN	Radio-jamming node (yжел paguonomex); its 1st section named at the 471st (KZ). 2 complexes at the Kamchatka centre (KZ, reported Jan 2016; FOI: delivered 2014)	O 2019 H	22
	Nebo-M	4 Nebo-M and Nebo-UM combined	R 2015 L	18
	Nebo-M	air defence and radio-technical crews with a division-level command post (MoD release; unit not named)	R 2017-09 L	18
	Palantin	EW subunits of the Eastern MD in Zabaykalsky Krai (planned deliveries by end 2020)	R 2020-07 L	25
	PMP heavy folding pontoon bridge park	INA; Oryx lists 47 PMP(-M) river links and 3 shore links lost	R 2026-09 M	28
	PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry	INA (PMM-2 held by Russian Army; no count given)	R 2024 L	29
	R-330B Mandat-B and R-934BMV	R-934B (2024), Borisoglebsk-2 R-934 station (2023-11) and R-934BMV (2023-10, 2024-11, 2025-04) reported hit by Ukrainian units; no Russian holding unit named	R 2025 M	24
	S-350 Vityaz	training centre of the Anti-aircraft Missile Forces, Zhukov Military Academy of Aerospace Defence. first S-350 set (crews and instructors training)	R 2020-02 M	13

Table 3a. Reach of indirect fire

Guns, howitzers, mortars and rocket launchers, including mine-laying rockets. Practical rows give observed employment ranges.				
SYSTEM, MUNITION AND CONDITION	RANGE	MINIMUM	TERM	PAGE
BM-30 Smerch 9M544 and 9M549 guided Tornado-S guided cluster rockets, inertial and GLONASS; Tornado-S guided rockets; crew statement CEP 5-10 m (inertial and GLONASS; official claim)	120 km	20-40 km	maximum (claim), minimum (claim)	6
Uragan-1M 9M544 guided 300 mm 9M544 guided rocket; claim in Russian press (ZR gives 120 km as the system maximum without naming the rocket)	120 km		claimed	5
BM-30 Smerch Employment range, Tornado-S guided rockets against a command post (2026-08) and a fortified strongpoint (2026-09); MoD figures via state media	about 100 km		practical (claim)	6
BM-30 Smerch 9M528 HE-frag unguided extended-range rocket, separable HE-frag warhead	90 km		maximum (claim)	6
2S35 Koalitsiya-SV 2S35 guided or extended-range shell precision-guided or rocket-assisted projectile; munition not named	80 km		maximum (claim)	8
BM-30 Smerch 9M55K cluster unguided, range-corrected rocket with fragmentation submunitions CEP 150 m (inertial range and attitude correction, no terminal guidance; also given for 9M528)	70 km	20 km	maximum, minimum	6
BM-30 Smerch 9M55F HE-frag unguided, separable HE-frag warhead	70 km	20 km	maximum, minimum	6
BM-30 Smerch Employment range, Smerch unguided rockets, Ukraine 2022, used at or near maximum range	35-70 km		practical	6
Uragan-1M 9M55F HE-frag 300 mm 9M55F HE-frag rocket with trajectory correction	70 km	20 km	maximum, minimum	5
BM-27 Uragan Employment range, observed 9K57 and 9K58 Smerch together, Donbas, May-June 2022; RUSI does not separate the two	35-70 km		practical	5
2S7 Pion / 2S7M Malka HE rocket-assisted RAP projectile, 103 kg	47.5 km		maximum	9
2S35 Koalitsiya-SV 2S35 standard HE-frag standard (unassisted) rounds	40 km		maximum	8
BM-21 Grad 122 mm MRL 9M521 HE-frag 9M521 HE-frag, export round for 9K51M Tornado-G; in 9K51 only with a manual electronic-fuze setter (RUWP); not adopted by the Russian Army (MISSIL)	40 km	15 km	maximum (claim), minimum (claim)	4
Tornado-G 9M521 HE-frag 9M521 extended-range HE-frag rocket; attribution to 9M521 from RUWIKI (Karpenko), WEG, ZR and WWV1 give 40 km as system maximum	40 km		maximum	4
2S7 Pion / 2S7M Malka HE-frag Conventional HE-frag, 110 kg projectile	37.5 km		maximum	9
Tornado-G 9M522 HE-frag 9M522 rocket, separable HE-frag warhead	37.5 km		maximum	4
2S19 Msta-S rocket-assisted HE rocket-assisted projectile (HERA/RAP), type not named	36 km		range	8
Uragan-1M 9M27F HE-frag 220 mm 9M27F unguided HE-frag rocket	35 km	10 km	maximum effective, minimum effective	5
Uragan-1M 9M27K cluster 220 mm 9M27K rocket, 30 9N210 fragmentation submunitions	35 km		maximum effective	5
BM-27 Uragan 9M27F HE-frag 9M27F HE-frag rocket	35 km	10 km	maximum, minimum	5
BM-27 Uragan 9M27K cluster 9M27K rocket, 30 9N210 HE-frag submunitions	35 km	10 km	maximum, minimum	5
BM-27 Uragan 9M27K2 AT mines 9M27K2 rocket, 24 PTM-1 anti-tank mines	35 km	10 km	maximum, minimum	5
BM-27 Uragan 9M27K3 AP mines 9M27K3 rocket, PFM-1 anti-personnel mines	35 km	10 km	maximum, minimum	5
2A36 Giatsint-B 152 mm towed gun rocket-assisted 3VOF41 active-rocket projectile 3OF30	33.5 km		maximum	10
2S5 Giatsint-S rocket-assisted HE-frag 3OF30 rocket-assisted HE-frag	33 km		maximum	7
2S5 Giatsint-S HE-frag 3OF59 HE-frag 3OF59, improved aerodynamic shape, full charge	30.5 km		maximum	7
2A65 Msta-B OF61 HE-frag base bleed OF61 (3OF61) HE-frag with base-bleed gas generator, round 3VOF91	29 km		maximum (claim)	10
2S19 Msta-S HE-frag base bleed 3OF61 HE-frag base bleed (round 3VOF91)	29 km		maximum	8

SYSTEM, MUNITION AND CONDITION	RANGE	MINIMUM	TERM	PAGE
2A36 Giatsint-B 152 mm towed gun HE-frag 3VOF39 HE-frag 3OF29, full charge; ammunition not stated; listed with HE-frag maximum	28.5 km	8.6 km	maximum, minimum	10
2S5 Giatsint-S HE-frag 3OF29 HE-frag 3OF29, full charge; HE-frag, indirect fire	28.4 km	8,600 m	maximum, minimum	7
2A65 Msta-B OF45 HE-frag OF45 (3OF45) HE-frag projectile; OF45 HE-frag projectile	24.7 km	6.5 km	maximum (claim), minimum (claim)	10
2S19 Msta-S HE-frag 3OF45 (OF-45) HE-frag, full charge	24.7 km		maximum	8
2S35 Koalitsiya-SV ZS43 HE-frag 2A64 gun; munition not stated by WEG or WPMALRU (Rostec: HE-frag more than 24 km)	24.5 km		maximum	8
2S5 Giatsint-S HE-frag 3OF29 reduced charge HE-frag 3OF29 with reduced charge 4Zh48	21.5 km		maximum	7
BM-21 Grad 122 mm MRL 9M22U HE-frag 9M22U (M-21OF) HE-fragmentation rocket	20.4 km	5 km	maximum, minimum	4
Tornado-G 9M22U HE-frag legacy Grad 9M22U HE-frag rocket	20.4 km	5 km	maximum effective, minimum effective	4
2S5 Giatsint-S Krasnopol guided 3OF39 laser-guided HE-frag Krasnopol	20 km		maximum	7
2S19 Msta-S guided Krasnopol laser-guided projectile; laser designation by an observer required	20 km		maximum	8
Tornado-G 9M538 HE-frag 9M538 rocket, enhanced monobloc HE-frag warhead	20 km		maximum (claim)	4
Tornado-G 9M539 HE-frag 9M539 rocket, separable HE-frag warhead; Tecmash statement at IDEX 2019; 9M539 rocket; Tecmash statement at IDEX 2019	20 km	5 km	maximum (claim), minimum (claim)	4
Tornado-G 9M541 cluster 9M541 rocket, cluster warhead with HEAT-frag submunitions	20 km		maximum	4
2S4 Tyulpan 308 rocket-assisted cluster mine cluster mine, 14 submunitions	19.3 km	7.1 km	maximum, minimum	9
2S4 Tyulpan Sayda incendiary rocket-assisted mine incendiary (napalm) rocket-assisted mine; incendiary rocket-assisted mine	19 km	0.7 km	maximum, minimum	9
2S4 Tyulpan 3F2 rocket-assisted HE mine 228 kg rocket-assisted HE mine; WEG and GS do not name the round, 3F2 per RUWIKI	18 km	0.8 km	maximum, minimum	9
BM-21 Grad 122 mm MRL 9M28F HE-frag 9M28F HE-fragmentation rocket	15 km	1.5 km	maximum, minimum effective	4
Tornado-G 9M28F HE-frag legacy Grad 9M28F HE-frag rocket	15 km	1.5 km	maximum effective, minimum effective	4
ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer ISDM mine rocket cassette rocket with AP or AT mines	15 km	5 km	maximum (claim), minimum (claim)	31
BM-27 Uragan 9M51 thermobaric 9M51 thermobaric rocket	13 km	5 km	maximum, minimum	5
2S4 Tyulpan 53-F-864 HE mine HE mine 130.7 kg, full charge; HE mine, smallest charge	9.65 km	0.8 km	maximum, minimum	9
2S4 Tyulpan Smel'chak-M laser-guided mine laser-guided mine, 134 kg	9.5 km	0.9 km	maximum, minimum	9
2S4 Tyulpan 3F5 Smel'chak laser-guided mine laser-guided mine, target designated by 1D15 from 0.2-5 km CEP 80-90% within a 2-3 m circle (semi-active laser homing, target lit in final 400-800 m of flight)	9.2 km	3.6 km	maximum, minimum	9
TOS-1A Solntsepek thermobaric rocket MO.1.01.04M thermobaric rocket, 3.7 m, 217 kg	6,000 m		maximum	7
TOS-1A Solntsepek incendiary rocket MO.1.01.04M.OP incendiary rocket, 3.7 m, 204 kg	6,000 m		maximum effective	7
TOS-1A Solntsepek original short rocket MO.1.01.04 thermobaric rocket, 3.3 m, 173 kg, fired from TOS-1A	2,700 m		maximum	7
TOS-1A Solntsepek minimum range MO.1.01.04M rocket		600 m	minimum	7

Table 3b. Reach of long-range missiles and one-way attack drones

Ballistic and cruise missiles and one-way attack drones. CEP, where known, is printed under the munition.				
SYSTEM, MUNITION AND CONDITION	RANGE	MINIMUM	TERM	PAGE
Iskander-M 9M723 ballistic missile 9M723, domestic Iskander-M; From the MIS Iskander-E (export) data table, max 280 km; domestic 9M723 minimum not confirmed CEP 10 to 30 m (With electro-optical terminal seeker)	up to 500 km	50 km	maximum, minimum	11
Iskander-M 9M728 (R-500) cruise missile 9M728 cruise missile, Iskander-K	500 km		official maximum (claim)	11

Table 3c. Reach of direct fire

Guns and machine guns laid on the target, and line charges.				
SYSTEM, MUNITION AND CONDITION	RANGE	MINIMUM	TERM	PAGE
2A65 Msta-B Direct fire range ATP Table E-2 and RWW table; projectile not stated; the same table's maximum-range row is misalligned	2,130 m		direct fire	10
2S19 Msta-S secondary MG 12.7x108 mm, ground and air targets	2,000 m ground; 1,500 m air		effective	8
R-149 command and staff vehicles 14.5 mm API-T (KPVT), R-149MA3 API-T, day; armament per WEG only	2,000 m		effective	32
ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer 7.62 mm PKT (GMZ-3) day	2,000 m		max effective	31
2A36 Giatsint-B 152 mm towed gun direct fire ammunition and target not stated in source	1,530 m		direct fire	10
2S5 Giatsint-S HEAT BP-540 HEAT BP-540, direct fire	1,000 m		direct fire	7
UR-77 Meteorit UZP-77 line charge charge launch distance, UZP-77; UZ-67 charge 200 to 350 m (WEG)	200 to 500 m		not stated	30

Table 3d. Reach of air defence weapons

Missiles and guns. Range is horizontal; altitude is the vertical reach. Figures are against the target named in the condition.					
SYSTEM, MUNITION AND CONDITION	RANGE	MINIMUM	ALTITUDE	TERM	PAGE
S-300V4 9M82MD air targets incl. AEW aircraft; S-300V4 new long-range missile	400 km			claimed maximum (claim)	13
S-400 Triumf 40N6 aerodynamic targets (STRASAM); APA brochure figure is for export 40N6E, 215 nmi; 40N6; the 27 km often quoted is reported for the 48N6DM missile specifically	400 km		30-35 km	maximum (claim)	12
S-400 Triumf 48N6DM (48N6E3) aerodynamic targets; RUSIAD names export 48N6E3 (Army Technology lineage); TASS 2018 state_claim: 48N6DM up to 250 km	250 km			maximum (claim)	12
S-300V4 9M82M aircraft targets; S-300VM	200 km		25 m-30 km	maximum (claim), maximum and minimum (claim)	13
S-350 Vityaz 9M96D extended-range domestic 9M96D; target type not stated by TWZ	about 121 km (75 miles)			range	13
S-350 Vityaz 9M96E2 aerodynamic target; export missile	120 km	1 km	5 m to 30 km	maximum (claim), minimum (claim)	13
S-400 Triumf 9M96E2 export 9M96E2; aerodynamic targets; APA gives 64.8 nmi; export 9M96E2; APA gives 15 ft to 100 kft	120 km		0.005-30 km	range (claim)	12
S-300V4 9M82 aircraft and jammer targets; S-300V; aircraft targets; S-300V	100 km	13 km	1-30 km	maximum, minimum (claim), maximum and minimum	13
S-300V4 9M83 aircraft targets; S-300V	75 km	6 km	250 m-25 km	maximum, minimum, maximum and minimum	13
Buk-M1 / Buk-M2 / Buk-M3 9M317M (Buk-M3) air target	2.5-70 km		35 km	maximum effective, maximum	14
S-350 Vityaz S-350E system aerodynamic targets; export exhibit figure, also given by Russian MoD for the domestic S-350 (state claim); aerodynamic targets; export system brochure	1.5 to 60 km		10 m to 30 km	maximum (claim)	13
S-400 Triumf 48N6 (ballistic targets) ballistic missile targets, 48N6 series	60 km			range	12
Buk-M1 / Buk-M2 / Buk-M3 9M317 (Buk-M2) aircraft; air target	3-50 km		15 m-25 km	engagement zone, intercept altitude	14
Pantsir-S1 Pantsir-SM missile Pantsir-SM with new missile; KBP chief designer, 2016	40 km			claimed (claim)	16
S-300V4 9M83/9M82 vs ballistic missile ballistic targets; S-300V and S-300VM; ballistic targets; S-300V (S-300VM: 1-30 km)	40 km		1-25 km	maximum, maximum and minimum (claim)	13
S-350 Vityaz 9M96E aerodynamic target; export missile	40 km	1 km	5 m to 20 km	maximum (claim), minimum (claim)	13
S-400 Triumf 9M96E export 9M96E; aerodynamic targets; APA gives 21.6 nmi; export 9M96E; APA gives 15 ft to 66 kft	40 km		0.005-20 km	range (claim)	12
Buk-M1 / Buk-M2 / Buk-M3 9M38M1 (Buk-M1) air target	3.25-34 km		30 m-22 km	maximum, intercept altitude	14
Pantsir-S1 Pantsir-S1M missile export Pantsir-S1M; new missile, designation not given by source; High Precision Systems, 2019	30 km			claimed (claim)	16
S-350 Vityaz S-350E system vs ballistic ballistic targets	1.5 to 30 km		2 to 25 km	maximum (claim)	13
S-350 Vityaz 9M96 basic domestic 9M96; target type not stated by TWZ	about 29 km (18 miles)			range	13
S-400 Triumf S-400 (missile not stated) reported engagement altitude band			0.1-27 km	reported	12
Buk-M1 / Buk-M2 / Buk-M3 9M317 vs ballistic and cruise missiles ALCM-type cruise missile at 30 m (MISM2; ROE gives 100 m, unchecked); Lance-type tactical ballistic missile 15-20 km (MISM2)	20 km			maximum (claim)	14
Pantsir-S1 57E6 missile aircraft-type targets; KBP figures for export 57E6-E; KBP figures for export 57E6-E; 5 m only in optical mode over water	20 km	1.2 km	5 m to 15 km	maximum (claim), minimum (claim)	16
Tor 9M338K Tor-M2 domestic, air targets	16 km		10 m to 10 km	maximum (claim)	15

SYSTEM, MUNITION AND CONDITION	RANGE	MINIMUM	ALTITUDE	TERM	PAGE
Strela-10 and Osa short-range SAM 9M33M3 Osa-AKM missile; Osa-AKM missile; WEG figure	15 km		4 km	operational range, maximum flight altitude	17
Tor 9M330/9M331 aircraft and PGM targets, Tor/Tor-M1	12 km	1.5 km	10 m to 6 km	maximum, minimum	15
Tor 9M331 (Tor-M2E) Tor-M2E export; target 300 m/s; Tor-M2E export; target 300 m/s (10 m to 6 km at 600 m/s)	1 to 12 km		10 m to 10 km	maximum (claim)	15
S-350 Vityaz 9M100 IR-homing short-range missile; fielding unconfirmed	10 km			maximum (claim)	13
Tunguska 9M311-M1 missile Tunguska-M1, air target; Tunguska-M1, air target; APA gives 2.5 km for 2K22 and 2K22M too; air target; ceiling for 9M311 family	10 km	2.5 km	3.5 km	maximum, minimum	17
Pantsir-S1 TKB-1055 small missile anti-drone round, up to 48 per vehicle in cannon-less versions	7 km			claimed (claim)	16
Strela-10 and Osa short-range SAM 9M37M/9M333 Strela-10 engagement zone, all variants; target closing at up to 415 m/s or receding at 310 m/s; Strela-10; 9M333 minimum 10 m	0.8 to 5 km		25 m to 3.5 km	engagement envelope	17
Pantsir-S1 30 mm 2A38M gun air and ground targets; air targets	4 km	200 m	0 to 3 km	maximum (claim), minimum (claim)	16
Tunguska 30 mm 2A38M guns air target	4 km	200 m	3 km	maximum, minimum	17

Table 3e. Reach of sensors

Detection, tracking and location ranges, grouped by what the sensor is for. Target type and size are part of the figure: a radar that sees a bomber at 600 km sees a small drone much closer.

REACH	SYSTEM, DATUM AND CONDITION	TARGET	TERM	PAGE
Ground surveillance				
up to 40 km	SNAR-10 / SNAR-10M1 Leopard Detection range, SNAR-10M1 radar. tanks and vehicles; Russian MoD figure; instrumented range 0.2–40 km	tank, vehicle	maximum (claim)	20
30 km	SNAR-10 / SNAR-10M1 Leopard Detection range, SNAR-10 radar. moving ships, without MTI	ship	maximum (claim)	20
20 km	1B75 Penicillin Detection range, 1L271. tank-type moving ground target	tank	maximum (claim)	20
16 km	SNAR-10 / SNAR-10M1 Leopard Detection range, SNAR-10 radar. moving vehicles, without MTI; 10 km with MTI	vehicle	maximum (claim)	20
15 km	SNAR-10 / SNAR-10M1 Leopard Detection range, SNAR-10M1 radar. moving person; developer figure	person	claimed (claim)	20
12 km	PRP-4A Argus / PRP-4M Deytery Detection range, radar (moving tank-type target). moving tank-type target	tank	up to	34
up to 10,000 m	1V12 artillery fire control complex [1V13, 1V14, 1V15, 1V16] Detection range, 1V14 optical reconnaissance. day	not stated	maximum	33
10 km	PRP-4A Argus / PRP-4M Deytery Recognition range, laser rangefinder. up to three tank-type targets; large targets incl. buildings 25 km; WEG names the 1D14–1 periscopic rangefinder	tank	up to (claim)	34
10 km	PRP-4A Argus / PRP-4M Deytery Detection range, PSNR-5M radar on PRP-4M. moving tank, PRP-4M application; Jane's 2001 export data, confidence reported	tank	range (claim)	34
8 km	PRP-4A Argus / PRP-4M Deytery Detection range, optical-electronic (tank-type target, day). tank-type target, daytime	tank	up to	34
7 km	PRP-4A Argus / PRP-4M Deytery Detection range, radar (person). single soldier	person	up to	34
3 km	PRP-4A Argus / PRP-4M Deytery Detection range, optical-electronic (tank-type target, night). tank-type target, night	tank	up to	34
up to 2,500 m	1V12 artillery fire control complex [1V13, 1V14, 1V15, 1V16] Detection range, 1V14 optical reconnaissance (night). night	not stated	maximum	33
2 km	PRP-4A Argus / PRP-4M Deytery Detection range, stationary tank, engine off (sensor not stated). stationary tank-type target, engine not running; ARMYR and RWW name no sensor; WEG lists it under radar	tank	no less than (claim)	34
750 m	1B75 Penicillin Minimum range, 1L271. mortar firing positions; 200 m for moving ground targets	mortar	minimum (claim)	20
Counter-battery				
65 km	Zoopark-1 / Zoopark-1M Detection range, tactical missile, 1L260. Launch positions of Lance / ATACMS type missiles	tactical missile	up to	19
45 km	Zoopark-1 / Zoopark-1M Detection range, rocket artillery, 1L260. MLRS firing position (DOGS, unverified, specifies M270 type)	rocket launcher, artillery	up to	19
40 km	Zoopark-1 / Zoopark-1M Detection range, tactical missile, 1L219M. Tactical missile launch position	tactical missile	detection of launch position	19
mortar 22, artillery 20, MLRS 35, tactical missile 40 km	Zoopark-1 / Zoopark-1M Fire-correction range, 1L219M. Adjusting friendly fire, by weapon type	not stated	fire correction range	19
30 km	ARK-1M Rys Detection range, ARK-1M vs battlefield missiles. Missile (WEG)	not stated	unspecified	19
30 km	Zoopark-1 / Zoopark-1M Detection range, rocket artillery, 1L219M. MLRS firing position	rocket launcher, artillery	detection of launch position	19
23–27 km	Zoopark-1 / Zoopark-1M Detection range, tube artillery, 1L260. 155–203 mm gun firing positions (23 km for 155 mm, 27 km for 203 mm)	artillery	up to	19
Iskander 80, Smerch 70, Uragan 50, Grad 40, 203 mm 32, 152 mm 27 km	Zoopark-1 / Zoopark-1M Own-fire tracking range, 1L260. Tracking friendly fire, by weapon	not stated	up to	19

REACH	SYSTEM, DATUM AND CONDITION	TARGET	TERM	PAGE
25 km	ARK-1M Rys Detection range, ARK-1M vs MRL rockets. Rocket (WEG); mode not stated	rocket launcher	unspecified	19
up to 25 km	1B75 Penicillin Detection range, 1B75. radius; artillery, mortar and MLRS firing positions, sound and thermal	rocket launcher, mortar, artillery	maximum (claim)	20
20 km	Zoopark-1 / Zoopark-1M Detection range, mortar, 1L219M. Mortar firing position; calibre not stated	mortar	detection of launch position	19
20 km	Zoopark-1 / Zoopark-1M Detection range, mortar, 1L260. Mortar firing position; calibre not stated. ATP lists this figure under '1L219M Zoopark-1M'	mortar	range as listed by ATP (no qualifier)	19
15 km	1B75 Penicillin Detection range, 1L271 (fire adjustment). by shell bursts; by trajectory 5 km	shell bursts	maximum (claim)	20
15 km	Zoopark-1 / Zoopark-1M Detection range, tube artillery, 1L219M. Field artillery firing position	artillery	detection of launch position	19
13 km	ARK-1M Rys Detection range, ARK-1M vs mortars. Mortar (WEG); mode not stated	mortar	unspecified	19
10 km	SNAR-10 / SNAR-10M1 Leopard Detection range, SNAR-10 radar. shell impact (100 mm), without MTI	shell bursts	maximum (claim)	20
10 km	SNAR-10 / SNAR-10M1 Leopard Detection range, SNAR-10M1 radar. shell burst on land; 20 km on water; developer figure	shell bursts	claimed (claim)	20
8 km	ARK-1M Rys Detection range, ARK-1M vs artillery (howitzers). Artillery (WEG), howitzers (JREW02); target and mode not stated by WEG	artillery	unspecified	19
5 km	1B75 Penicillin Detection range, 1L271 (mortar positions). mortar firing positions, 81-120 mm	mortar	maximum (claim)	20
SIGINT and DF				
400 km	Moskva-1 Detection range, airborne emitters. passive; emitting aircraft and cruise missiles; 360 deg sector	cruise missile, emitters	up to (claim)	22
up to 30 km	Repellent-1 Detection range, UAV control-link signals. ground-station command signals; UAV telemetry up to 10 km; target size and altitude not given	UAV, emitters	up to (claim)	27
6 km	Leer-3 Detection range, GSM base stations. cellular base stations, GSM and possibly 3G/4G, measured from the UAV	UAV, cell base station	within	26
Laser designation				
up to 7,000 m	1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16) Range, laser target illumination (1V14). for laser-guided (corrected) munitions	not stated	maximum	33
Air tracking and fire control				
120 km	Buk-M1 / Buk-M2 / Buk-M3 Detection range, 9A317 / 9A317M TELAR radar (M2/M3). RCS 1-2 m2 at 3 km altitude and 18-20 km vs same target at 10-15 m (M2); M3: RCS 1-2 m2, altitude not stated	RCS 1-2 m ²	detection range	14
80 km	Buk-M1 / Buk-M2 / Buk-M3 Detection range, 9S35 TELAR radar (M1). RCS 2 m2; 100 km vs RCS 3 m2	RCS 2 m ²	detection range	14
80 km	Strela-10 and Osa short-range SAM Detection range, Dog Ear (PPRU-1M) battery radar. Strela-10 battery acquisition radar, F/G-band; target type not stated; tracking 35 km (JREW02)	not stated	acquisition range	17
36 km	Pantsir-S1 Detection range, 1RS1 search radar. 2 m2 target; S-band; up to 20 targets tracked	RCS 2 m ²	maximum (claim)	16
28 km	Pantsir-S1 Detection range, 1RS2-1 tracking radar. target size not stated; WEG and Armada	not stated	maximum (claim)	16
17-26 km	Pantsir-S1 Detection range, optical post. F-16 autotracking, 10 km visibility; 13-15 km for HARM missile	fighter, anti-radiation missile	auto-tracking (claim)	16
20 km	Strela-10 and Osa short-range SAM Detection range, Land Roll tracking radar. Osa; target type not stated in source	not stated	tracking range	17
20 km	Tor Detection range, TV sight. target not stated	not stated	maximum	15
15 km	Tor Detection range, tracking radar (Tor/Tor-M1). 60-degree sector; target and variant not stated	not stated	maximum	15
13 km	Tunguska Detection range, 1RL144M tracking radar. J-band tracker	not stated	maximum	17
Air surveillance				
600 km	Nebo-M Detection range, Nebo-M. complex; target type, size and altitude not stated	not stated	maximum (claim)	18
600 km	Nebo-M Altitude coverage, Nebo-M. complex, ballistic targets implied; not stated	ballistic missile	maximum (claim)	18

REACH	SYSTEM, DATUM AND CONDITION	TARGET	TERM	PAGE
580-600 km	S-400 Triumf Detection range, 91N6E. target RCS >10 m ² (strategic bomber)	RCS 10 m ² , bomber	detection/ tracking	12
330 km	S-300V4 Detection range, 9S15 Bill Board. target size not stated by WEG	not stated	maximum	13
300 km	96L6 all-altitude acquisition radar Detection range. Range band 5 to 300 km for detected targets; target size and altitude not stated	not stated	instrumented (claim)	18
300 km	S-400 Triumf Detection range, 96L6E. export 96L6E in S-300PMU family, target not stated; JREW-p1877, PDF p. 1877	not stated	max (claim)	12
200 km	Nebo-M Detection range, Podlet-K1. main mode, 10-200 km; target size not stated	not stated	maximum (claim)	18
175 km	S-300V4 Detection range, 9S19 High Screen. ballistic missile search; target size not stated	ballistic missile	maximum	13
160 km	Buk-M1 / Buk-M2 / Buk-M3 Detection range, 9S18M1-3 / 9S18M3 search radar (M2/M3). RCS 1-2 m ² ; Buk-M2 9S18M1-3	RCS 1-2 m ²	detection range	14
100-150 km	Buk-M1 / Buk-M2 / Buk-M3 Detection range, 9S18M1 search radar (M1). target size and altitude not stated	not stated	detection range	14
150 km	S-300V4 Detection range, 9S32 Grill Pan. 2 m ² RCS target, manual mode (140 km automatic)	RCS 2 m ²	maximum acquisition	13
120 km	Buk-M1 / Buk-M2 / Buk-M3 Detection range, 9S36 mast radar (M2). RCS 1-2 m ² at 3 km altitude; 30-35 km vs same target at 10-15 m	RCS 1-2 m ²	detection range	14
100 km	96L6 all-altitude acquisition radar Maximum target altitude. All-altitude mode, aerodynamic targets	aircraft	maximum (claim)	18
75 km	Pantsir-S1 Detection range, Pantsir-SM radar. target size not stated	not stated	claimed (claim)	16
70 km	Buk-M1 / Buk-M2 / Buk-M3 Detection range, 9S36M mast radar (M3). cruise missile flying at 5 m	cruise missile	detection range	14
40 km	Barnaul-T Detection range, 1L122-1E. aircraft, helicopters, cruise missiles, UAVs at up to 700 m/s; export 1L122-1E brochure figure	cruise missile, helicopter, UAV	maximum (1-40 km) (claim)	35
32 km	Tor Detection range, surveillance radar (Tor-M2). air targets, size not stated	aircraft	detection (claim)	15
30 km	Strela-10 and Osa short-range SAM Detection range, Land Roll surveillance radar. Osa; target type not stated in source	not stated	acquisition range	17
25-27 km	Tor Detection range, surveillance radar (Tor/Tor-M1). aircraft; WEG does not state the variant	aircraft	maximum	15
18 km	Tunguska Detection range, 1RL144 surveillance radar. E-band; low-flying search mode for targets at about 15 m (JREW02)	not stated	maximum	17
9-15 km	Tor Detection range, surveillance radar (Tor/Tor-M1), UAV. UAVs	UAV	maximum	15
12 km	Tor Detection range, surveillance radar (Tor/Tor-M1), helicopter. helicopters	helicopter	maximum	15
10 km	Barnaul-T Detection altitude, 1L122-1E. export 1L122-1E; 'at least 10 km'	not stated	maximum (claim)	35
10 km	Nebo-M Minimum range, Podlet-K1. air targets, lower limit of stated 10-200 km detection zone	aircraft	minimum (claim)	18
10 km	Nebo-M Altitude coverage, Podlet-K1. maximum detection altitude	not stated	maximum (claim)	18
0.45 to 10 km	Strela-10 and Osa short-range SAM Detection range, 9S86 Snap Shot. Strela-10 range-only radar; target type not stated; minimum 0.45 km	not stated	maximum	17
1 km	Barnaul-T Minimum detection range, 1L122-1E. export 1L122-1E	not stated	minimum (claim)	35
Other				
up to 100 km	PU-12M7 Coordinates measurement envelope, PU-12M7 (connected radars). coordinates measurement envelope from connected radars; PU-12M7 has no radar of its own	not stated	maximum (claim)	35

Table 3f. Reach of communications

Radio and data-link ranges. Ranges depend on antenna, terrain and whether the vehicle is halted.			
REACH	SYSTEM, DATUM AND CONDITION	TERM	PAGE
450 km	R-149 command and staff vehicles Range, HF air-ground Prima, R-149MA1. HF to aircraft	maximum (claim)	32
350 km	R-145BM Chayka command and staff vehicle Range, R-130M dipole. HF, halted only, symmetrical dipole	range	32
350 km	R-149 command and staff vehicles Range, HF R-168-100KBE, R-149MA1. halted, mast antenna; 50 km (20 km moving) on ASH-4 whip	maximum (claim)	32
350 km	R-149 command and staff vehicles Range, HF, R-149BMR. HF band 1.5-30 MHz	maximum (claim)	32
15-300 km	R-149 command and staff vehicles Range, HF R-168-5KV, R-149AKSh-1. 1.5-30 MHz, 2 sets	maximum	32
220-250 km	1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16) Radio range, R-111. 1V15 and 1V16, vehicle radio	stated	33
130 km	1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16) Radio range, R-130MT. 1V15 and 1V16, HF vehicle radio	stated	33
120 km	R-149 command and staff vehicles Range, VHF air-ground Prima, R-149MA1. VHF to aircraft	maximum (claim)	32
75 km	R-145BM Chayka command and staff vehicle Range, R-130M zenith antenna. HF, halted or moving, symmetrical zenith-radiation antenna	range	32
60 km	R-149 command and staff vehicles Range, VHF, R-149BMR. VHF band 30-108 MHz	maximum (claim)	32
25-60 km	R-149 command and staff vehicles Range, VHF R-168-100U-2, R-149AKSh-1. 30-108 MHz, 2 sets	maximum	32
55 km	R-149 command and staff vehicles Range, VHF R-168-100UE-2 on mast, R-149MA1. halted, mast antenna	maximum (claim)	32
40 km	1B75 Penicillin Radio range, 1B75. to external subscribers	maximum (claim)	20
up to 40 km	PU-12M7 Data link range, PU-12M7 by radio. radio link to higher and subordinate posts	maximum (claim)	35
25-40 km	R-149 command and staff vehicles Range, VHF R-168-25U-2, R-149MA3. 30-108 MHz, 2 sets	maximum	32
35 km	R-145BM Chayka command and staff vehicle Range, R-111 3.4 m whip. VHF, halted or moving	range	32
up to 30 km	PU-12M7 Data link range, 9S737 to higher post. radio	maximum (claim)	35
20-25 km	1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16) Radio range. 1V13 and 1V14, vehicle radios	stated	33
25 km	R-149 command and staff vehicles Range, VHF R-168-100UE-2, R-149MA1. on the move, vehicle antenna	maximum (claim)	32
20 km	Barnaul-T Data link range, 9S932-1. stationary; 10 km on the move (9S933: 3 km / 1 km)	communication range	35
20 km	R-145BM Chayka command and staff vehicle Range, R-123MT 4 m whip. VHF, halted or moving	range	32
6-20 km	R-149 command and staff vehicles Range, UHF R-168-MRA, R-149MA3. 1.5-1.75 GHz data radio	maximum	32
up to 15 km	PU-12M7 Data link range, PU-12M7 by field cable. field telephone cable	maximum (claim)	35
6 km	PRP-4A Argus / PRP-4M Deytery Dismounted observation post, distance from vehicle. dismounted observation post	maximum	34
up to 5 km	PU-12M7 Data link range, 9S737 to fire units. radio	maximum (claim)	35

Table 3g. Reach of electronic attack

Jamming ranges. Most are manufacturer claims.			
REACH	SYSTEM, DATUM AND CONDITION	TERM	PAGE
5,000 km	Murmansk-BN Jamming range, HF communications. HF (shortwave) radio links, skywave; ship, aircraft and ground emitters	effective (Armada's term) (claim)	22
about 1,000 km wide strip	Palantín Jamming frontage, radio communications. ground target: radio communications and radio reconnaissance; battalion tactical group exercise near Voronezh, Sep 2019	claimed (claim)	25
150–300 km	Krasukha-2 / Krasukha-4 Krasukha-4; jamming range. air and ground-based radar targets; target size and altitude not stated	maximum (claim)	21
up to 250 km	Krasukha-2 / Krasukha-4 Krasukha-2; jamming range. air target: AEW aircraft radar; target altitude not stated	up to	21
250 km	Moskva-1 Jamming range, airborne S-band radar. air targets; ARM23 says the 1L267 'can engage targets' per unnamed sources; Moskva-1 itself has no jammer (FOI)	up to	22
50 km or more	R-330Zh Zhitel Jamming range, airborne users. User equipment installed on aircraft	claimed minimum (claim)	24
up to 30 km	Repellent-1 Jamming range, small UAV control and telemetry links. air target: small UAVs; telemetry and satellite-navigation links; control link up to 10 km	up to (claim)	27
up to 25 km	Pole-21 Jamming range, GNSS receivers (ground or air not stated). WEG (Pole-21E); Russian press gives 25 km per jamming module for the domestic Pole-21	effective	27
25 km	R-330Zh Zhitel Jamming range. Ground user equipment (WEG gives no target type; matches manufacturer's ground-user figure)	maximum effective	24
22 km	2S19 Msta-S Range, jammer round. 3RB30 radio jamming projectile; jamming radius 700 m	range (claim)	8
up to 20 km	Palantín Jamming range, UAV control links. air target: UAV radio control links	maximum (claim)	25
up to 6 km	Leer-3 Jamming range, cellular terminals (ground). radius of the zone of blocked GSM terminals around the UAV	up to (claim)	26
3 km	Infana Jamming range, target not stated. WEG 'Disruption Range'; target and conditions not given	disruption range	27

Table 3h. Reach of UAVs

How far from its control point a UAV operates, and at what altitude.			
REACH	SYSTEM, DATUM AND CONDITION	TERM	PAGE
120 km	Leer-3 Practical range, Orlan-10 from control vehicle. radius of UAV employment from the control point	up to (claim)	26
up to 5,000 m	Leer-3 Altitude, Orlan-10. operating altitude	up to (claim)	26

Table 3i. Deployment distances

Where the system has been seen deployed or employed relative to the line of contact, and how far its elements spread. These are observations and reports, not rules; they indicate what may lie at a given depth.

REACH	SYSTEM, DATUM AND CONDITION	TERM	PAGE
up to 100 km	S-400 Triumph Fire unit distance from command post. 98Zh6E fire unit from 55K6E, export composition	maximum (claim)	12
15 to 30 km	Borisoglebsk-2 Employment distance from line of contact. Donbas 2014-15, with Leer-3, Zhitel, R-934UM	practical	23
25 km	BM-27 Uragan Distance from front line, observed, one launcher hit by FPV drone, reported c. 2024-07	practical	5
about 15 km from enemy positions	2S4 Tyulpan Employment distance, reported MoD video, Ukraine, Nov 2022	practical (claim)	9
5-10 km behind forward edge	Tor Deployment distance, Tor-M1 in defence, first defensive position; 5-10 km between vehicles	practical	15

Table 4. Mobility and time into action

SYSTEM	CODE	WEIGHT	FORDING	INTO ACTION	OUT OF ACTION	PAGE
1B75 Penicillin	WH-AT	INA	INA	5 min	INA	20
1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16)	TR-A	14.8 t	Amphibious	INA	INA	33
2A36 Giatsint-B 152 mm towed gun	TW	9,760 kg	INA	4 min	4 min	10
2A65 Msta-B	TW	7,000 kg	INA	2 to 2.5 min	2 min	10
2S19 Msta-S	TR	42 t	1.2 m unprepared; 5 m prepared	1.5 min	1.5 min	8
2S35 Koalitsiya-SV	TR	48-55 t	2.1 m	1.5 min	under 1 min	8
2S4 Tyulpan	TR	27.5 t	1.0 m	25 min	25 min	9
2S5 Giatsint-S	TR	28.2 t	1.05 m	less than 2 min	less than 2 min	7
2S7 Pion / 2S7M Malka	TR	46,500 kg	1.2 m	5-6 min	3-5 min	9
96L6 all-altitude acquisition radar	WH-AT	INA	1.3 m	5 min	5 min	18
ARK-1M Rys	TR	15.7 t	6 km/h	5 min	5 min	19
Barnaul-T	TR-A	11,000 kg	5 km/h	5 min	7 min	35
BM-21 Grad 122 mm MRL	WH-AT	13.7 t	3 min	3 min	2 min	4
BM-27 Uragan	WH-AT	20 t	1.2 m	3 min	3 min	5
BM-30 Smerch	WH-AT	43.7 t	1.1 m	3 min	3 min	6
BMP-1KSh command and staff vehicle	TR	13 t	7 km/h	INA	INA	33
Borisoglebsk-2	TR-A	11,000 kg	5 km/h	15 min or less	15 min or less	23
Buk-M1 / Buk-M2 / Buk-M3	TR	34 t	1 m	5 min	5 min	14
IMR-2 / IMR-2M / IMR-3M combat engineer vehicle	TR	44.3 t	1.2 m unprepared; 5.0 m with equipment	6 min	INA	30
Infaua	WH-AT-A	13,614 kg	9 km/h	INA	INA	27
ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer	WH-AT	INA	1.8 m	INA	INA	31
Iskander-M	WH-AT	42.85 t	1.4 m	16 min to first launch from the march; 4 min from highest readiness	INA	11
Krasukha-2 / Krasukha-4	WH-AT	39,000 kg	1.75 m	less than 20 min	less than 20 min	21
Leer-2	WH-AT	7,800 kg	1.3 m	INA	INA	26
Leer-3	WH-AT	INA	INA	30 min	INA	26
Moskva-1	WH-AT	12,450 kg	1.75 m	45 min	INA	22
Murmansk-BN	WH-AT	up to 35 t	INA	72 h	INA	22
Nebo-M	WH-AT	INA	INA	12 m	15 min	18
Palantin	WH-AT	INA	INA	about 30 min	INA	25
Pantsir-S1	WH-AT	14.85 t	1.8 m	3 min	INA	16
PMP heavy folding pontoon bridge park	WH-AT	INA	INA	7 m of bridge per minute	a few minutes	28
Pole-21	NA	INA	INA	INA	INA	27

SYSTEM	CODE	WEIGHT	FORDING	INTO ACTION	OUT OF ACTION	PAGE
Polyana-D4	WH-AT	13.2 t	1.2 m	35 min	35 min	34
PRP-4A Argus / PRP-4M Deytery	TR-A	13.8 t	7 km/h	2 min	INA	34
PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry	TR-A	33.1 t	INA	5 min	INA	29
PU-12M7	WH-AT-A	13.6 t	10 km/h	3 min	INA	35
R-145BM Chayka command and staff vehicle	WH-AT-A	10.3 t	10 km/h	INA	INA	32
R-149 command and staff vehicles	WH-AT-A	15,000-15,050 kg	10 km/h	10 min on the move / 30 min halted	INA	32
R-330B Mandat-B and R-934BMV	WH-AT	11,000 kg	INA	30 min	INA	24
R-330Zh Zhitel	WH-AT	INA	INA	40 min or less	40 min or less	24
Repellent-1	WH-AT	23,650 kg	1.0 m	15 min	INA	27
S-300V4	TR	50 t	1.2 m	5 min	5 min	13
S-350 Vityaz	WH-AT	22 t	1.2 m	5 min	5 min	13
S-400 Triumf	WH-AT	INA	INA	5-10 min	INA	12
SNAR-10 / SNAR-10M1 Leopard	TR	12.7 t	INA	5 min	INA	20
Strela-10 and Osa short-range SAM	TR	12.1 t	1.2 m	3 to 5 min	3 to 5 min	17
TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge	WH-AT	19,500 kg	1.0 m	72 min	8 min	29
Tor	TR	32 t	1.0 m	less than 3 min	less than 5 min	15
Tornado-G	WH-AT	14 t	1.75 m	2 min	3 min	4
TOS-1A Solntsepek	TR	44.3 t	1.2 m	90 s from halt to fire	INA	7
Tunguska	TR	34 t	1.0 m	not more than 5 min	not more than 5 min	17
UR-77 Meteorit	TR-A	15.5 t	5 km/h	INA	30 to 40 min	30
Uragan-1M	WH-AT	40 t	1.4 m	3 min	3 min	5
Zoopark-1 / Zoopark-1M	TR	38.1 t	INA	5 min	5 min	19

Table 5. Source register

Codes are printed at the foot of each entry. Web sources are coded by publisher and numbered: APA Air Power Australia, AR Army Recognition, ARMADA Armada International, BCAT Bellingcat, BMPD bmpd blog, CSIS CSIS Missile Threat, FAS Federation of American Scientists, GS GlobalSecurity, MIL Military, MISS missiery.info, ORYX Oryx, RDP Russian Defense Policy, RUSI RUSI web, TASS TASS, TOPWAR Topwar, TWZ The War Zone, VPK VPK, VRF Vooruzhenie Rossii, WEG WEG online (ODIN), WMC Wikimedia Commons, WP Wikipedia (language after the hyphen), WSPOT WarSpotting. Other codes name a single document.

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